Healthcare Industry: Snr Systems Engineer(Azure/InTune/Defender)/DevOps/Data Engineer /SharePoint Consultant /DBA
Responsibilities:

- Administer, support, maintain and monitor Windows Server fleet (VMWare/AWS/Azure) systems, and in addition provide web server system administration and technical support as required for the Digital Information Office (DIO) Business.
- Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
- **Azure IaaS Management**: Deployed, managed, and maintained Azure IaaS solutions, encompassing compute (Virtual Machines, Scale Sets, AKS), storage (Blob, Queue, Disk), networking (VNet, Load Balancer, VPN Gateway), and security services (Azure AD, Security Center, Key Vault) to support mission-critical applications.
- **Azure Cloud Optimization**: Optimized Azure cloud environments for cost efficiency, performance, and security, utilizing tools like Azure Cost Management + Billing, Azure Advisor, Azure Monitor, Azure Automation, and Azure Security Center. Achieved a 20% cost reduction and a 30% performance improvement.
- **Network Engineering**: Led design and deployment of robust network solutions, leveraging deep knowledge of CIDR, TCP/IP stack, OSPF, RIP, and BGP for optimal performance & security across diverse WAN & hybrid cloud environments.
- **SD-WAN** (Software-Defined Wide Area Network): Implemented an SD-WAN solution using Cisco Meraki, VeloCloud across 10 branch offices, resulting in a 30% reduction in WAN costs and improved application performance.
- Leveraged Intune Mobile Device Management (MDM) and Mobile Application Management (MAM) to effectively manage and secure mobile devices and applications within the MS Endpoint Manager admin center, including PowerShell script
- Integrated Intune MDM with Microsoft Endpoint Manager to deliver a unified platform for managing and securing devices across the organization, streamlining workflows and improving overall security posture.
- Proficiently managed Microsoft Windows 2012 server services, including Active Directory, both cloud and on-premise, RADIUS, trusts, group policies, remote access, DHCP, Enterprise CA, DNS, and network device enrolment services.
- Managed IT infrastructure & EDW data backup using Commvault Data Protection /Veeam Backup for Microsoft 365
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Windows Server 2012 OS administration, providing support for services like DNS and DHCP, and contributing to network service maintenance. Contributed to the maintenance of enterprise CA and other network services
- Offered 3rd level technical support through various channels, including phone, email, and in-person interactions, resolving hardware and software issues, while also installing and configuring software and hardware.
- Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Snr Systems Engineer(Azure/InTune/Defender)/DevOps/Data Engineer /SharePoint Consultant
Responsibilities:

- Provides technical support and system administration for the day-to-day operations of the organisation's ICT systems.
- Demonstrated expertise in managing, supporting, and troubleshooting SharePoint 2013, Dynamics 365 CRM, and SQL Server 2012 on Windows Server 2012, with a strong focus on system and procedure documentation.
- Provided essential day-to-day ICT support for AD, AAD, Microsoft 365, MFA/SSO, and Azure, ensuring smooth operations.
- Implemented and Administered modern authentication technologies and InTune for efficient mobile management
- Provided 3rd level desktop, systems, and network support, diagnosing and resolving complex technical issues.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 Manulife Financial Corporation (MLJ), Tokyo
Finance/Insurance Domain: Senior Systems Engineer /ICT System Administrator (AD/SCCM/SCOM)/DBA
Responsibilities:

- Conducted the installation, configuration, and troubleshooting of hardware, software, and networking equipment.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate "zone1", "WWN1","WWN2"; Create a zoneset configuration by #cfgcreate "test_cfg", "zone1", "zone2"; Add a zone to a zone configuration via cfgadd, etc.

- Installed and configured a WebLogic domain with a frontend Apache Web Server acting as a reverse proxy. The backend integrates Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008 /BDD2007 Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components as well as enterprise file transfer and web analytics tools, such as AD FS, SSO, DNS, DFS, IIS7.0~10, etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Snr Systems Engineer /ICT System Administrator (AD/SCCM/SCOM) /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Installed and configured a WebLogic domain with a frontend Apache Web Server acting as a reverse proxy. The backend integrates Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Senior Systems Engineer /ICT Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patching for Operating Systems and Applications
- Planning, design and management of Active Directory with high level support to AD Migrations - within & between AD forests
- Proficiently handled hardware, software, and networking equipment installation and configuration. Provided effective support for Office Online Specialist functions, swiftly resolving technical issues while managing Active Directory and assisting in AD migrations between forests.
- Conducted installation, configuration, and testing of hardware and software solutions as well as routine maintenance tasks
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers & F5 load balancers v9
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java/Kotlin, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, PrivateGPT-X, AI Agents, Azure Open AI ML Services, Copilot, Google GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; MuleSoft Anypoint Studio and Mule 4, BizTalk Server 1.0~2020, RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata,
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,

- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure/InTune, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, Cisco Meraki, VeloCloud, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks. SSL-VPN, IPSec-VPN, Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

D365 Application Support Consultant –Remote - Dynamo Recruitment – 26/1/2025 HO in Vic.-Closed on 23/2/25 No resp.
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
D365 Application Support Consultant/Developer/Dynamics Technical Solution Architect/Senior DevOps/Data Engineer/ Software Engineer /Java/.NET/Python/AI/DL/ML/BI/SharePoint/D365 Developer /EDW.-Japan(7.5yrs) & Australia(25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: D365 Application Support Consultant/Snr DevOps/Data Engineer/.NET/Python/AI/ML/SP/D365 Developer
Responsibilities:

- Architected, developed, and implemented a 3-tiered Dynamics 365 Finance & Operations environment featuring: Presentation Tier: Web browsers and mobile devices for user interface; Application Tier: Application Object Server (AOS) running Dynamics 365 F&O services and Azure-hosted Windows server containers for cloud deployments.
- Highly experienced Dynamics 365 Architect with deep expertise in both CE and F&O, specializing in X++ development, system integrations, and upgrades.
- Develop and document functional specifications for system users, ensuring approval and knowledge sharing.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2022/Dynamics 365/Power BI/Power Automate & Power Apps)
- Provide day to day technical support and manage enhancements across the line of applications, both ERP & CRM systems.
- Developed custom web services and plugins in D365 using C#.NET with the Xrm.Sdk library in Visual Studio 2022. These solutions processed requests, performed various operations on D365 data, and populated response objects to fulfill business requirements. Utilized Power BI to extract customer data from Dynamics 365 F&O & load it into a SQL Server table
- Developed custom web service plugins in Dynamics 365 using C#.NET with Xrm.Sdk in Visual Studio 2022 to process requests, perform operations, and populate response objects for business users.
- Leveraged X++ frameworks such as the Trace Framework for performance analysis and debugging. Analyzed event handlers to identify and resolve conflicts with standard functionality. Effectively debugged and corrected X++ code, ensuring data integrity and system stability through rigorous testing and validation
- Successfully led the migration of multiple on-premises systems to the cloud, including the replacement of legacy CRM with Dynamics 365, encompassing both CE and F&O implementations.
- Developed custom components using PCF Controls to build model-driven apps & reusable APIs, integrated into Dynamics 365 & Power Apps canvases. Automated data retrieval from backend SQL 2022 Database and SharePoint 2022 Document Lists to display frontend Dynamics 365 activity records, enhancing business workflow processes and customer services.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 & its latest 2024 Release Wave 1 enhancements to develop Dynamics 365 applications, using customization & extension components to streamline data flows and integrate with SAP HR, SAP FinOps, S/4 HANA systems. Liquid for templating & rendering dynamic content within web applications.
- Undertake system tuning and scaling for both performance and security, while managing security profiles across the application in accordance with IT security principles.
- **Reduced development effort by 50% by leveraging OpenDevin** to eliminate the need for custom HTML, JavaScript, and canvas manipulation when customizing Dynamics 365, SharePoint 2022, & Power Apps using C#.NET Core in VS2022.
- Design, develop & customise SharePoint Online using Power Automate, SharePoint Framework (SPFx) or other M365 tools
- Developed Power Pages websites that seamlessly integrate and display Power BI Pro Paginated Reports, providing real-time insights into key performance indicators for Azure Synapse Analytics platform users.
- PowerBI Dashboards Development: Develop complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Take ownership of customer issues, follow problems through to resolution, and deliver technical troubleshooting in line with SLA agreements. Collaborate with application vendors to ensure high customer satisfaction through prompt ticket handling.
- Provide 2nd level support for all ERP and CRM system users (D365 F&O, D365 CE)
- **Automated repetitive tasks for Business Analysts and Data Analysts** by developing secure, offline local AI systems (PrivateGPT-X, Ollama3.2, Autogen Studio 2.0, etc.) for data analysis, replacing inefficient tools like Jira or Confluence.

- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
    - Azure: Azure DevOps
    - AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
    - GCP: Cloud Build/Cloud Deploy
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Designed and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python with ARM templates, Terraform, or Ansible. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration in Azure, & equivalent container orchestration services in AWS and GCP (e.g., ECS Fargate, GKE).
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Develop, test, and deploy AI models and algorithms, with a focus on Natural Language Processing (NLP) and text summarization, to develop a variety of AI Agents that automate as many repetitive business processes as possible.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: D365 Application Support Consultant/Developer/Snr DevOps/Data Engineer/Py/SP/BI/D365 Developer
Responsibilities:

- Designed and implemented Dynamics 365 F&O solutions utilizing a 3-tier architecture comprising: a Presentation Tier accessed through web browsers and mobile devices; an Application Tier leveraging the Application Object Server (AOS) and cloud-native technologies such as Azure-hosted Windows server containers for enhanced scalability and flexibility.
- Highly experienced Dynamics 365 Architect with a strong foundation in both CE and F&O, specializing in X++ development, system integrations, and upgrades. Proven success in designing, developing, and implementing robust and secure solutions within complex and regulated environments.
- Enhanced system performance and stability by leveraging X++ frameworks, such as the Trace Framework, for in-depth performance analysis and debugging. Proactively identified and resolved conflicts with standard functionality by analyzing event handlers. Demonstrated strong debugging and code correction skills, ensuring data integrity and system stability through rigorous testing and validation.
- Experienced in designing, implementing, and supporting integrated business solutions leveraging Microsoft Dynamics 365 (CE & F&O), Azure (including Azure AD), SharePoint, Power Platform, Logic Apps within the Azure iPaaS ecosystem, Service Bus, and PowerShell scripting to enhance operational efficiency and achieve business objectives.
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- Power BI Dashboards Development: Develop complex DAX formulas in Power BI to manipulate data.
- Create Power BI tiles and pin them to dashboards. Pin slicers or report pages to dashboards in Power BI Desktop/Service.
- Take ownership of customer issues, follow problems through to resolution, and deliver technical troubleshooting in line with SLA agreements. Collaborate with application vendors to ensure high customer satisfaction through prompt ticket handling.
- Provide 2nd level support for all ERP and CRM system users (D365 F&O, D365 CE)
- **Designed and implemented Medallion Architecture with Data Lake in Azure:**
- **Batch Laye**r: Utilize Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2017 to develop Dynamics CRM2016 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- **Design, develop, and maintain automated data pipelines and API orchestration using either:**
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.

- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Designed and implemented a scalable cloud-based solution on Microsoft Azure utilizing serverless App Services and Functions. Leveraged Azure SQL Database for structured data and Blob Storage for unstructured data. Possess strong knowledge of Oracle and SQL, along with familiarity with newer technologies like Dataverse, Databricks, and NoSQL.
- **Automated frequent staff onboarding and off boarding** of Privileged Account Access Management using CyberArk Privileged Session Manager (PSM v10.8). Implemented a proactive security monitoring solution powered by an AI-driven system (e.g., Llama 3.1 and Reflection) that continuously monitors for suspicious activity, triggers alerts for the Security team, and provides 24x7 proactive detection and response to security threats. The system is configured with multiple autonomous agents to ensure comprehensive coverage, including devices brought by employees (BYOD/BYOC). The AI is trained on a large dataset of normal and anomalous network traffic to enable accurate detection of security threats.
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- **Implemented cloud-agnostic application security for web applications and microservices** by scanning using Python, Bash, and PowerShell scripts, and integrating container scanning tools like Snyk and Aqua Security. This resulted in a 50% reduction in critical vulnerabilities, such as SQL injection, cross-site scripting (XSS), and remote code execution, ensuring compliance with industry best practices like OWASP Top 10.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Dynamics (CRM 2013/AX 7) Application Support Consultant/DevOps /Data Engineer/SP/K2 Developer
Responsibilities:

- Led the design, development, and deployment of a 3-tier Microsoft Dynamics AX environment utilizing a cloud-based architecture. This included:
  - Web and mobile interfaces for user accessibility.
  - Azure-hosted Application Object Server (AOS) for enhanced scalability and performance.
  - Azure SQL Database for secure and reliable data storage.
  - Integration with Azure services such as Azure Active Directory for identity management and Logic Apps for streamlined business processes.
- Highly skilled Microsoft Dynamics (AX 7 & CE) professional with expertise in integrating and automating business processes using Azure (including AAD), SharePoint, Power Platform, Logic Apps within Azure iPaaS ecosystem, Service Bus for reliable message queuing and real-time messaging, and PowerShell scripting to deliver impactful solutions.
- Experienced Dynamics AX 7 Developer with proven expertise in designing and developing robust solutions leveraging sysOperation Framework, Data Contracts Framework, Jobs Framework, Services Framework, UI Builder Framework, Workflow Framework, Batch Framework, Query Framework, Forms Framework, and Reports Framework within a fast-paced environment.
- Designed and implemented custom solutions using Power Apps and Microsoft Flow to enhance usability and efficiency within Dynamics AX 7 & SharePoint 2016 environments. Leveraged Mulesoft Anypoint API for scalable integrations and automation, while also utilizing Microsoft Flow to streamline business processes.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
  1. Azure: Azure DevOps
  2. AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
  3. GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM 2016/Microsoft Flow)
- Lead the development & support of existing solutions using Mulesoft Studio 3.10 & Mule ESB 3.10 through Agile practices.
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- Leverage SharePoint for process integration and Drupal/HTML with JavaScript for form development, along with backend SQL Server 2014 database for efficient data retrieval and customer service operations in Dynamics CRM 2016.
- Skilled in leveraging AI-powered tools to streamline development processes and enhance overall testing efficiency.
- Developed and maintained AI /ML models to enhance customer engagement and sales.
- Provide day to day technical support and manage enhancements across the line of applications, both ERP & CRM systems.

- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Leverage Python programming with CyberArk SDK** to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management.
- # Retrieve students registered data by: students_registered_data = retrieve_students_registered_data()
- Utilized Azure DevOps to collaborate effectively with team members and manage Power Platform projects.
- Integrated Power Platform solutions with other Azure services using Azure DevOps and Azure Pipelines.
- Skilled in creating impactful visualizations using Tableau and Power BI.
- Used Agile methodology to plan and execute tasks and used Git to manage code
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Employed Power BI, Tableau, or QlikView for real-time financial analysis.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Dynamics (CRM 2013/NAV2013) Application Support Consultant/Developer /Software Test Engineer
Responsibilities:

- Highly skilled Dynamics NAV Technical Consultant with proven expertise in designing, developing, and implementing custom solutions. Proficient in C/AL, .NET, and SQL Server. Strong analytical and problem-solving skills with a focus on delivering high-quality, efficient, and user-friendly NAV solutions.
- Expertise in system integrations, customizations, report development, and troubleshooting.
- Demonstrated experience/skills in developing Automation frameworks, test scripts, using applications and tools such as: GitLAB, OpenDevin, Cucumber, Gattling and Selenium for comprehensive testing and performance analysis.
- Provide day to day technical support and manage enhancements across the line of applications, both ERP & CRM systems.
- Proficient in HTML, JavaScript, and Selenium WebDriver for web testing automation.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented automated DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Dynamics (CRM2011/NAV2012) **Application Support Consultant/DevOps Engineer/CRM Developer**
Responsibilities:

- Highly skilled MS Dynamics NAV Technical Solution Developer with expertise in development, implementation, and support of NAV solutions. Proven ability to design, develop, and implement custom solutions, integrations, and customizations.
- Experienced MS Dynamics NAV Technical Solution Developer proficient in C/AL (Client/Server Application Language), .NET, and SQL Server. Strong analytical and problem-solving skills with a focus on delivering high-quality, efficient, and user-friendly NAV solutions. Expertise in system integrations, customizations, and report development.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**

Finance/Insurance Domain: Dynamics (CRM2010/NAV2010) **Application Support consultant/Senior Systems Engineer /DWH**
Responsibilities:

- Highly skilled MS Dynamics NAV Technical Solution Developer with expertise in development, implementation, and support of NAV solutions. Proven ability to design, develop, and implement custom solutions, integrations, and customizations.
- Experienced MS Dynamics NAV Technical Solution Developer proficient in C/AL (Client/Server Application Language), .NET, and SQL Server. Strong analytical and problem-solving skills with a focus on delivering high-quality, efficient, and user-friendly NAV solutions. Expertise in system integrations, customizations, and report development.
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Dynamics (CRM 4.0/NAV2009) **Application Support Consultant /BI Developer**
Responsibilities:

- Highly skilled Dynamics NAV Technical Consultant with proven expertise in designing, developing, and implementing custom solutions. Proficient in C/AL, .NET, and SQL Server. Strong analytical and problem-solving skills with a focus on delivering high-quality, efficient, and user-friendly NAV solutions.
- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer/ Dynamics (CRM 4.0/NAV 5.0) **Application Support Consultant /BI Developer/DWH**
Responsibilities:

- Highly skilled MS Dynamics NAV Technical Solution Developer with expertise in development, implementation, and support of NAV solutions. Proven ability to design, develop, and implement custom solutions, integrations, and customizations.
- Experienced MS Dynamics NAV Technical Solution Developer proficient in C/AL (Client/Server Application Language), .NET, and SQL Server. Strong analytical and problem-solving skills with a focus on delivering high-quality, efficient, and user-friendly NAV solutions. Expertise in system integrations, customizations, and report development.
- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate process

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc

- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
  Clouds: MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum
- **CMS:** SharePoint 2007~2022, Dynamics CRM3.0~D365(CE & F&O), X++ and associated frameworks, Dynamics AX 4.0/2009~2016, Dynamics NAV 2004~2008,Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Servers**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Application for Dynamics Technical Solution Architect-Developer role –Hitachi Construction Pty Ltd – 5/Jan/2025

of advertised Dynamics Technical Solution Architect/Developer position

Hiring Manager,

Hitachi Construction Machinery Pty Ltd

Dear Hiring Manager,

I am writing to express my keen interest in the Dynamics Technical Solution Architect / Developer position advertised on [platform where you saw the job ad] and to showcase my extensive experience and qualifications that align perfectly with your requirements.

Key Accomplishments

Throughout my 25-year career in Australia and 7.5 years in Japan, I have honed my expertise in Dynamics 365 F&O, with a proven track record of designing, developing, and implementing impactful enhancements. My in-depth knowledge of X++ development, system integrations, and upgrades positions me to make a significant contribution to your team's efforts in optimizing the D365 F&O platform.

In my previous role at Aged Care Quality Safety and Commission (ACQSC), I spearheaded the architecture, development, and deployment of a successful 3-tier Dynamics 365 F&O environment. This project involved a web and mobile-friendly presentation tier, a scalable Azure-hosted Application Object Server (AOS), and a secure Azure SQL Database for data storage.

Highlighting AI and MLOps Skills

I am particularly enthusiastic about your focus on leveraging AI and MLOps to continuously improve the D365 F&O platform. In my prior engagements, I have actively partnered with technical and functional consultants to design and implement solutions that incorporate these advanced technologies. For instance, at the University of Canberra, I collaborated with our Microsoft Partner to drive the adoption of AI and MLOps for enhancing product functionalities within Dynamics 365.

Beyond Technical Expertise

Beyond my technical skill set, I possess excellent communication and teamwork abilities. I am a strong believer in fostering a collaborative work environment and sharing my knowledge with colleagues. My proven ability to effectively troubleshoot issues and deliver exceptional user experiences ensures a smooth transition into your team.

I am confident that my qualifications and experience make me a strong candidate for this role. I am eager to learn more about your vision for optimizing the D365 F&O platform and how I can contribute to its success. Thank you for your time and consideration. I look forward to hearing from you soon.

Sincerely,

Zhi Ming Zhang

Mobile: 0402 997 679

件名: Dynamics Technical Solution Architect/Developer 職種応募

採用ご担当者

この度、貴社の求人情報サイトを拝見し、Dynamics Technical Solution Architect/Developer のポジションに応募させていただきます。張 志明と申します。

主な実績

オーストラリアでの 25 年と日本での 7.5 年のキャリアを通じて、私は Dynamics 365 F&O の専門知識を磨き、インパクトのある機能強化の設計、開発、実装の実績を築いてきました。 X++開発、システム統合、アップグレードに関する私の深い知識は、貴社のチームが D365 F&O プラットフォームを最適化する取り組みに大きく貢献できると確信しております。

Aged Care Quality Safety and Commission (ACQSC)での役割においては、成功裏に 3 層の Dynamics 365 F&O 環境のアーキテクチャ設計、開発、およびデプロイを主導しました。 このプロジェクトには、ウェブおよびモバイル対応のプレゼンテーション層、スケーラブルな Azure ホスト型アプリケーションオブジェクトサーバー(AOS)、および安全な Azure SQL データベースが含まれていました。

AI と MLOps スキルの強調

貴社が D365 F&O プラットフォームを継続的に改善するために AI と MLOps を活用することに特に注目しております。 以前の業務においては、技術コンサルタントおよび機能コンサルタントと積極的に連携し、これらの先進技術を組み込んだソリューションの設計と実装に取り組んでまいりました。 例えば、キャンベラ大学では、マイクロソフトパートナーと協力して、Dynamics 365 内の製品機能を強化するために AI と MLOps の導入を推進しました。

技術的専門知識を超えて

技術的なスキルセットに加えて、私は優れたコミュニケーション能力とチームワーク能力を有しています。 私は、協調的な職場環境を促進し、同僚と知識を共有することを強く信じています。 問題を効果的にトラブルシューティングし、優れたユーザーエクスペリエンスを提供する能力により、貴社のチームにスムーズに統合できると確信しております。

熱意と行動喚起

私の資格と経験が、このポジションに最適な候補者であると確信しております。 D365 F&O プラットフォームを最適化するという貴社のビジョンと、私がどのようにしてその成功に貢献できるかについて、より深く理解したいと考えております。 お時間をいただき、ご検討いただきありがとうございます。 どうぞよろしくお願いいたします。

Zhi Ming ZHANG - Australian Citizen File Ref: ZMZITILJP003
1303 / 29 Hunter St.| Parramatta 2150 | NSW Mobile: 0402-997-679 | mingjp@gmail.com

敬具

張 志明

Dynamics Technical Solution Architect/Developer – Parramatta – Hitachi – 5/Jan/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Dynamics Technical Solution Architect/Developer/Integration Specialist /Software Engineer /Senior DevOps/Data Engineer/ Java/.NET/Python/AI/DL/ML/BI/SharePoint/D365 Developer /EDW/ Clouds/Linux Admin.-Japan(7.5yrs) & Australia(25yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Dynamics Technical Solution Architect/Developer/Snr DevOps/Data Engineer/.NET/AI/ML/SP/D365 Developer
Responsibilities:

- Architected, developed, and implemented a 3-tiered Dynamics 365 Finance & Operations environment featuring: Presentation Tier: Web browsers and mobile devices for user interface; Application Tier: Application Object Server (AOS) running Dynamics 365 F&O services and Azure-hosted Windows server containers for cloud deployments.
- Highly experienced Dynamics 365 Architect with deep expertise in both CE and F&O, specializing in X++ development, system integrations, and upgrades. Proven success in designing, developing, and implementing complex solutions, seeking a challenging role at Hitachi Construction Machinery to contribute to their technology roadmap and drive business growth.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2022/Dynamics 365/Power BI/Power Automate & Power Apps)
- Developed custom web services and plugins in D365 using C#.NET with the Xrm.Sdk library in Visual Studio 2022. These solutions processed requests, performed various operations on D365 data, and populated response objects to fulfill business requirements. Utilized Power BI to extract customer data from Dynamics 365 F&O & load it into a SQL Server table
- Developed custom web service plugins in Dynamics 365 using C#.NET with Xrm.Sdk in Visual Studio 2022 to process requests, perform operations, and populate response objects for business users.
- Leveraged X++ frameworks such as the Trace Framework for performance analysis and debugging. Analyzed event handlers to identify and resolve conflicts with standard functionality. Effectively debugged and corrected X++ code, ensuring data integrity and system stability through rigorous testing and validation
- Successfully led the migration of multiple on-premises systems to the cloud, including the replacement of legacy CRM with Dynamics 365, encompassing both CE and F&O implementations.
- Developed custom components using PCF Controls to build model-driven apps & reusable APIs, integrated into Dynamics 365 & Power Apps canvases. Automated data retrieval from backend SQL 2022 Database and SharePoint 2022 Document Lists to display frontend Dynamics 365 activity records, enhancing business workflow processes and customer services.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 & its latest 2024 Release Wave 1 enhancements to develop Dynamics 365 applications, using customization & extension components to streamline data flows and integrate with SAP HR, SAP FinOps, S/4 HANA systems. Liquid for templating & rendering dynamic content within web applications.
- Design and develop custom Dynamics 365 customizations and extensions to streamline business processes, using Common Data Service, Power Automate, and AI Chatbot (formerly Power Virtual Agents)
- **Reduced development effort by 50% by leveraging OpenDevin** to eliminate the need for custom HTML, JavaScript, and canvas manipulation when customizing Dynamics 365, SharePoint 2022, & Power Apps using C#.NET Core in VS2022.
- Design, develop & customise SharePoint Online using Power Automate, SharePoint Framework (SPFx) or other M365 tools
- Developed Power Pages websites that seamlessly integrate and display Power BI Pro Paginated Reports, providing real-time insights into key performance indicators for Azure Synapse Analytics platform users.
- PowerBI Dashboards Development: Develop complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Partner with skilled technical and functional consultants to design, develop, and implement impactful enhancements, leveraging AI and MLOps to continuously improve our D365 F&O platform.
- **Automated repetitive tasks for Business Analysts and Data Analysts** by developing secure, offline local AI systems (PrivateGPT-X, Ollama3.2, Autogen Studio 2.0, etc.) for data analysis, replacing inefficient tools like Jira or Confluence.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
  - Azure: Azure DevOps
  - AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
  - GCP: Cloud Build/Cloud Deploy
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Designed and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.

- Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python with ARM templates, Terraform, or Ansible. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration in Azure, & equivalent container orchestration services in AWS and GCP (e.g., ECS Fargate, GKE).
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- **Implemented a Q-Learning-powered intrusion detection system** that reduced false positive rates by 30% and detected 95% of simulated Cyber-attacks within 24 hours. Deployed OpenAI's ChatGPT o3 model to assess the security implications of proposed actions before proceeding, ensuring that decisions align with established security policies and best practices.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Dynamics Technical Solution Architect/Developer/Snr DevOps/Data Engineer/Py/SP/BI/D365 Developer
Responsibilities:

- Designed and implemented Dynamics 365 F&O solutions utilizing a 3-tier architecture comprising: a Presentation Tier accessed through web browsers and mobile devices; an Application Tier leveraging the Application Object Server (AOS) and cloud-native technologies such as Azure-hosted Windows server containers for enhanced scalability and flexibility.
- Highly experienced Dynamics 365 Architect with a strong foundation in both CE and F&O, specializing in X++ development, system integrations, and upgrades. Proven success in designing, developing, and implementing robust and secure solutions within complex and regulated environments.
- Enhanced system performance and stability by leveraging X++ frameworks, such as the Trace Framework, for in-depth performance analysis and debugging. Proactively identified and resolved conflicts with standard functionality by analyzing event handlers. Demonstrated strong debugging and code correction skills, ensuring data integrity and system stability through rigorous testing and validation.
- Experienced in designing, implementing, and supporting integrated business solutions leveraging Microsoft Dynamics 365 (CE & F&O), Azure (including Azure AD), SharePoint, Power Platform, Logic Apps within the Azure iPaaS ecosystem, Service Bus, and PowerShell scripting to enhance operational efficiency and achieve business objectives.
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- Power BI Dashboards Development: Develop complex DAX formulas in Power BI to manipulate data.
- Create Power BI tiles and pin them to dashboards. Pin slicers or report pages to dashboards in Power BI Desktop/Service.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch. Integrated M365 environments (O365, MS Teams, SharePoint 2016, Dynamics 365, Power App, Microsoft Flow) into the Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- **Designed and implemented Medallion Architecture with Data Lake in Azure:**
- **Batch Layer**: Utilize Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2017 to develop Dynamics CRM2016 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- **Design, develop, and maintain automated data pipelines and API orchestration using either:**
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Designed and implemented a scalable cloud-based solution on Microsoft Azure utilizing serverless App Services and Functions. Leveraged Azure SQL Database for structured data and Blob Storage for unstructured data. Possess strong knowledge of Oracle and SQL, along with familiarity with newer technologies like Dataverse, Databricks, and NoSQL.

- **Automated frequent staff onboarding and off boarding** of Privileged Account Access Management using CyberArk Privileged Session Manager (PSM v10.8). Implemented a proactive security monitoring solution powered by an AI-driven system (e.g., Llama 3.1 and Reflection) that continuously monitors for suspicious activity, triggers alerts for the Security team, and provides 24x7 proactive detection and response to security threats. The system is configured with multiple autonomous agents to ensure comprehensive coverage, including devices brought by employees (BYOD/BYOC). The AI is trained on a large dataset of normal and anomalous network traffic to enable accurate detection of security threats.
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- **Implemented cloud-agnostic application security for web applications and microservices** by scanning using Python, Bash, and PowerShell scripts, and integrating container scanning tools like Snyk and Aqua Security. This resulted in a 50% reduction in critical vulnerabilities, such as SQL injection, cross-site scripting (XSS), and remote code execution, ensuring compliance with industry best practices like OWASP Top 10.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Dynamics Technical Solution Architect/Developer /Snr DevOps/Data Engineer /ML/SP/D365 /K2 Developer
Responsibilities:

- Led the design, development, and deployment of a 3-tier Microsoft Dynamics AX environment utilizing a cloud-based architecture. This included:
  - Web and mobile interfaces for user accessibility.
  - Azure-hosted Application Object Server (AOS) for enhanced scalability and performance.
  - Azure SQL Database for secure and reliable data storage.
  - Integration with Azure services such as Azure Active Directory for identity management and Logic Apps for streamlined business processes.
- Highly skilled Microsoft Dynamics 365 (CE & F&O) professional with expertise in integrating and automating business processes using Azure (including AAD), SharePoint, Power Platform, Logic Apps within Azure iPaaS ecosystem, Service Bus for reliable message queuing and real-time messaging, and PowerShell scripting to deliver impactful solutions.
- Experienced Dynamics 365 F&O Developer with proven expertise in designing and developing robust solutions leveraging sysOperation Framework (Mandatory) and X++ within a fast-paced environment.
- Designed and implemented custom solutions using Power Apps and Microsoft Flow to enhance usability and efficiency within Dynamics 365 & SharePoint 2016 environments. Leveraged Mulesoft Anypoint API for scalable integrations and automation, while also utilizing Microsoft Flow to streamline business processes.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
  1. Azure: Azure DevOps
  2. AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
  3. GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM 2016/Microsoft Flow)
- Lead the development & support of existing solutions using Mulesoft Studio 3.10 & Mule ESB 3.10 through Agile practices.
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- Leverage SharePoint for process integration and Drupal/HTML with JavaScript for form development, along with backend SQL Server 2014 database for efficient data retrieval and customer service operations in Dynamics CRM 2016.
- Skilled in leveraging AI-powered tools to streamline development processes and enhance overall testing efficiency.
- Developed and maintained AI /ML models to enhance customer engagement and sales.
- Drive collaboration with our Microsoft Partner to coordinate backlog of product enhancements by adopting AI and MLOps.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Leverage Python programming with CyberArk SDK** to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management.
- # Retrieve students registered data by: students_registered_data = retrieve_students_registered_data()
- Utilized Azure DevOps to collaborate effectively with team members and manage Power Platform projects.
- Integrated Power Platform solutions with other Azure services using Azure DevOps and Azure Pipelines.

- Skilled in creating impactful visualizations using Tableau and Power BI.
- Used Agile methodology to plan and execute tasks and used Git to manage code
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Employed Power BI, Tableau, or QlikView for real-time financial analysis.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Dynamics Technical Solution Developer /DevOps Engineer/Software Test Engineer/DBA
Responsibilities:

- Highly skilled Dynamics NAV Technical Consultant with proven expertise in designing, developing, and implementing custom solutions. Proficient in C/AL, .NET, and SQL Server. Strong analytical and problem-solving skills with a focus on delivering high-quality, efficient, and user-friendly NAV solutions.
- Expertise in system integrations, customizations, report development, and troubleshooting.
- Demonstrated experience/skills in developing Automation frameworks, test scripts, using applications and tools such as: GitLAB, OpenDevin, Cucumber, Gattling and Selenium for comprehensive testing and performance analysis.
- Proficient in HTML, JavaScript, and Selenium WebDriver for web testing automation.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented automated DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Dynamics Technical Solution Developer/DevOps Engineer/.NET/SharePoint Developer
Responsibilities:

- Highly skilled MS Dynamics NAV Technical Solution Developer with expertise in development, implementation, and support of NAV solutions. Proven ability to design, develop, and implement custom solutions, integrations, and customizations.
- Experienced MS Dynamics NAV Technical Solution Developer proficient in C/AL (Client/Server Application Language), .NET, and SQL Server. Strong analytical and problem-solving skills with a focus on delivering high-quality, efficient, and user-friendly NAV solutions. Expertise in system integrations, customizations, and report development.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Highly skilled MS Dynamics NAV Technical Solution Developer with expertise in development, implementation, and support of NAV solutions. Proven ability to design, develop, and implement custom solutions, integrations, and customizations.

- Experienced MS Dynamics NAV Technical Solution Developer proficient in C/AL (Client/Server Application Language), .NET, and SQL Server. Strong analytical and problem-solving skills with a focus on delivering high-quality, efficient, and user-friendly NAV solutions. Expertise in system integrations, customizations, and report development.
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Highly skilled Dynamics NAV Technical Consultant with proven expertise in designing, developing, and implementing custom solutions. Proficient in C/AL, .NET, and SQL Server. Strong analytical and problem-solving skills with a focus on delivering high-quality, efficient, and user-friendly NAV solutions.
- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /MS Dynamics NAV Technical Solution Developer/Snr Systems Engineer /DWH
Responsibilities:

- Highly skilled MS Dynamics NAV Technical Solution Developer with expertise in development, implementation, and support of NAV solutions. Proven ability to design, develop, and implement custom solutions, integrations, and customizations.
- Experienced MS Dynamics NAV Technical Solution Developer proficient in C/AL (Client/Server Application Language), .NET, and SQL Server. Strong analytical and problem-solving skills with a focus on delivering high-quality, efficient, and user-friendly NAV solutions. Expertise in system integrations, customizations, and report development.
- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate process

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc

- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

- Highly skilled Microsoft Dynamics 365 (CE & F&O) professional with expertise in integrating and automating business processes using Azure (including AAD), SharePoint, Power Platform, Logic Apps within Azure iPaaS ecosystem, Service Bus for reliable message queuing and real-time messaging, and PowerShell scripting to deliver impactful solutions.

- Highly experienced Dynamics 365 F&O Architect with proven success in designing, developing, and implementing complex solutions, specializing in X++ development, system integrations, and upgrades, seeking a challenging role at Hitachi Construction Machinery to contribute to their technology roadmap and drive business growth.

DCS

- Experienced in designing, implementing, and supporting integrated business solutions leveraging Microsoft Dynamics 365 (CE & F&O), Azure (including Azure AD), SharePoint, Power Platform, Logic Apps within the Azure iPaaS ecosystem, Service Bus, and PowerShell scripting to enhance operational efficiency and achieve business objectives.

- Full-stack developer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, and Azure cloud. Skilled in developing and consuming enterprise APIs using Azure API Gateway, REST API, or Mulesoft Anypoint Platform, with experience in integrating with privileged access management solutions like CyberArk
- Led the development of tailored Mulesoft solutions that align with business requirements for scalability, security, and efficiency. Guide teams in adhering to standards and governance frameworks, leveraging technical expertise in server-side languages (Java, Node.js, etc.) and front-end frameworks (React JS/Next JS, Mulesoft Anypoint API, etc.).

- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.

AWS

- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.

- Leveraged Infrastructure as Code (IaC) principles to automate the provisioning and configuration of virtual private clouds (VPCs), subnets, security groups using Terraform with Python (Boto3). Adopted ECS Fargate for automated container orchestration, enabling scalable and efficient deployment and management of containerized applications.

- Design, develop test and deploy security focused and AI-Powered ML applications, deep learning (DL) models and systems

- **Applied DevOps Best Practices to MLOps**: **Effectively manage ML models using proven DevOps practices, helping your organization stay ahead in a rapidly evolving tech landscape.**
- **Centralized Model Management**
- **Version Control Simplification**
- **Leverage Existing Tools**
- **Unified ML Tools Integration**

- **Enhanced Security Protocols**

Developed custom web services and plugins in Dynamics 365 using C#.NET with the Xrm.Sdk library in Visual Studio 2022. These solutions processed requests via Event-Handler to trigger different workflows, ReceivePurchaseOrder, ShipProducts,etc.), performed various operations on Dynamics 365 data, and populated response objects to fulfill business requirements.

- Leveraged LangChain to create versatile AI agents capable of data summarization, classification, and prediction. Utilized Pydantic for data validation to ensure data quality and consistency. These agents assisted Business Analysts and Data Analysts in accelerating data exploration, generating insights, and automating routine tasks.
- Leveraged MuleSoft Anypoint Platform's hybrid deployment capabilities to design and deploy APIs that seamlessly integrate AWS Lambda functions with Azure Logic Apps, enabling real-time data synchronization between the two platforms.

Senior DevOps Engineer v2– GBST Sydney – 1/1/2025
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer (Hybrid) /Integration Solution Architect /Software Engineer/Senior Data Engineer /Java/Salesforce/Mulesoft Developer /Python AI/ML/DL/BI/.NET/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DevOps Engineer/Snr Data Engineer/Python/Go/.NET /AI/ML/DL/SharePoint/D365 Developer /EDW

Responsibilities:

- Designed and implemented a secure hybrid cloud architecture integrating AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted site-to-site connections using Cloud VPN Gateways across all platforms. Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python (Boto3) with CloudFormation, Terraform, or Ansible. Utilized ECS Fargate for efficient and scalable container orchestration.
- Architect and implement scalable AWS infrastructures using Python (Boto3) & CloudFormation to meet business objectives.
- Ensured robust security and compliance of AWS environments through the effective use of CloudTrail and AWS Config.
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., `dfx force:source:pull`) from Salesforce orgs.
- Built secure and scalable AWS Landing Zones using Infrastructure as Code (IaC) with AWS CloudFormation templates or Terraform. Leveraged AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch.
- Design and lead the migration of complex applications from Rancher/Docker on EC2 to ECS Fargate, optimizing performance and cost efficiency. Used CloudWatch Contributor Insights to analyze High-Cardinality Data & Traffic Trends.
- Built and managed ECS clusters and task definitions, ensuring seamless application deployment and management.
- Implemented robust CI/CD pipelines for automated builds, testing, and deployment to ECS Fargate.
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- **CloudWatch Observability Expertise:** Implemented comprehensive CloudWatch observability solutions, including metrics, logs, and dashboards. Leveraged CloudWatch Insights for advanced log analysis and troubleshooting.
- Implemented custom CloudWatch metrics using Boto3 to track critical application performance indicators, enabling data-driven decision-making and proactive issue resolution: cloudwatch = boto3.client('cloudwatch')
- Python or Go programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Automation**:Utilized AWS CodeBuild, Git Actions, GitLab CI, or Jenkins for automated build, test, & deployment processes. Used Ansible, Chef, or Puppet for configuration management & infrastructure provisioning in cloud-agnostic environments.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
    - **Azure: Azure DevOps**
    - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Proven ability to develop high-quality Salesforce solutions in the financial services sector, leveraging a robust technology stack including client-side LWC (Lightning Web Components), JavaScript, CSS, and HTML5 for a seamless user experience, and Apex programming for server-side logic, data manipulation, and integration with external systems.
- Automated Business Analysts and Data Analysts daily repeated tasks by developing secure,offline local AI systems (PrivateGPT-X, Ollama-L, Autogen Studio 3.0, etc.) for data analysis. Leveraged LangChain to create versatile AI agents capable of data summarization, classification, and prediction. Utilized Pydantic for datavalidation to ensure data quality and consistency. These agents assisted Business Analysts and Data Analysts in accelerating data exploration, generating insights, and automating routine tasks.
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, RNN, ANN, GAN) based on dataset type, size, and desired result interpretability.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov Central Bank: Senior DevOps Engineer/Senior Data Engineer/Java/Python /Go/AI/ML/BI/SP Developer /EDW
Responsibilities:

- **Built and implemented Medallion Architecture with Data Lake in GCP:**
- **Batch Layer:** Utilized Google Cloud Dataproc, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, & unrefined data.

- **Serving Layer:** Implemented BigQuery with its built-in Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer:** Leveraged Cloud Pub/Sub as the unified service layer for real-time data ingestion and BigQuery for querying the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
  Utilized BigQuery for data querying, loading, exporting to Pandas DataFrame, and executing parameterized queries within the Python programming environment. query_job = client.query(query, job_config=job_config)
- Designed and developed Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/ J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
- Full-stack engineer skilled in React/TypeScript for frontend development, Java/APIs for backend development, and AWS cloud. Proficient in integration tools like MuleSoft and J2EE to ensure seamless data flow using RESTful web services.
- Proven ability to develop high-quality Salesforce solutions in the financial services sector, leveraging a robust technology stack including client-side LWC, JavaScript, CSS, and HTML5 for a seamless user experience, and Apex programming for server-side logic, data manipulation, and integration with external systems.
- Expand capabilities across the entire data pipeline lifecycle, encompassing the building, running, testing, and operating of data pipelines that ingest, filter, sort, join, and/or transform datasets on both GCP and SAP Data Services by utilization of Cloud Build for pipeline automation, GitOps for infrastructure-as-code management, and Argo CD for continuous delivery.
- **Automation and CI/CD**: Leveraged industry-standard tools like Helm and Kubespray to streamline & customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.
- Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.
- Developed and deployed ML models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision): model = torch.hub.load('pytorch/vision:v0.6.0', 'mobilenet_v2', pretrained=True)
- Expertise in implementing and maintaining a robust suite of cloud security controls, including Identity and Access Management (IAM), data encryption, network security, vulnerability management, and data loss prevention (DLP). Ensured compliance with industry standards and regulations such as CIS, NIST, GDPR, and HIPAA.
- Designed, developed, and maintained a robust ETL/ELT pipeline/API orchestration using either:
  1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
  3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
- Proficient in Linux system administration, including kernel optimization (sysctl, vmstat), performance tuning (iostat, sar), and security hardening (iptables, fail2ban).
- Developed and deployed ML models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision: model = models.detection.fasterrcnn_resnet50_fpn(pretrained=True)
- Applied technical expertise in monitoring, observability, performance management, risk management, troubleshooting, and service recoverability. Executed and reported on non-functional testing, contributing to system improvements.
- Migrated on-premise applications to the Google Cloud Platform (GCP) cloud using Velostrata, Cloud Migrate for Compute Engine, Migrate for Anthos, and CloudEndure Migration to GKE environment.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Worked with enterprise CI/CD capabilities and tools such as GoCD, GitLab, Code Build, and Cloud Formation to provide guidance on best practices and help enable major programs to deliver business value.
- Deploy and manage microservices-based applications using Docker CRE and Kubernetes on GCP GKE clusters.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior DevOps Engineer/Integration Solution Architect/Java/.NET/Python/Go/Mulesoft/SP/K2 Developer
Responsibilities:

- Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications. Build and maintain all non-production and production environments.

- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, Dynamics 365 CRM, and Nintet K2 Five workflows. Additionally, the platform integrates with data visualization tools like Power BI and Tableau for further data analysis.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Automated Salesforce Development Workflow:
- Utilized Salesforce DX tools (`dfx force:source:pull`) to retrieve and manage code (metadata) from development orgs.
- Implemented unit testing frameworks like Jest to ensure code quality on retrieved Salesforce code.
- Configured CI/CD pipelines to automate code retrieval, testing, & deployment to staging orgs using `dfx force:source:deploy`. (Optional: additional deployment steps like code analysis and MuleSoft application deployments if applicable)
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., `dfx force:source:pull`) from Salesforce orgs.
- **Automated Deployments with SFDX:** Minimized manual errors and accelerated releases.
- **Enhanced Code Quality**: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.
- **MuleSoft Integration**: Built seamless Salesforce-MuleSoft integrations within CI/CD for end-to-end automation.
- Designed and maintained robust CI/CD pipelines using platform-specific tools:
    1. **Azure: Azure DevOps**
    2. **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    3. **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- **Automated Kubernetes Deployments:** Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.
- Proven ability to design & implement highly scalable, resilient, & cost-effective cloud-native applications on the Azure platform, leveraging technologies such as Azure Service Bus, Azure Event Hubs, Azure Container Service (ACS), and Azure DevOps.
- **Secured API Access for Student Enrollment**: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of students' data.
- **CI/CD and DevSecOps Automation:** Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or even on-premises environment.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Cloud/DevOps Engineer/Data Engineer /Software Test Engineer/Linux Administrator
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed Amazon Redshift with its built-in data processing capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.

- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- Developed and maintained automation scripts using Python, Go, Bash to streamline development & operational workflows.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.
- Development of MS Dynamics CRM by Apex Visualforce in Salesforce Cloud, including integration with Sharepoint 2010
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within AWS cloud architecture.
- **Security**: Secured secrets with HashiCorp Vault or AWS Secrets Manager and implemented network segmentation and firewalls. Integrated security scanning tools throughout development.
- **Leverage AWS services for cloud-native design patterns**:
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- **Optimized CI/CD pipelines with DevOps tools**: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon Elastic Container Registry (ECR) and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Successfully leveraged ITIL 4.0 frameworks, including Incident Management, Problem Management, and Continual Improvement, to optimize incident response processes and drive service excellence.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Linux Server Provisioning: Utilize Ansible playbooks (e.g., ec2_instance.yml, configure_router.yml, network_config.yml, install_packages.yml, deploy_app.yml) for creating EC2 instances, configuring networks, installing packages, and deploying applications.
- Windows Server Configuration: Employ Chef, initiated with knife bootstrap windows_server_ip chef-server_url, for managing Active Directory, group policies, and software installations.
- Database Provisioning: Leverage Puppet to automate database creation (e.g., mysql_database { 'mydatabase': ensure => present, }) and user management.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Cloud/DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using AWS API Gateway
- Deploy and maintain web applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Sora, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Phi 3, Copilot, Google GEMINI, Metaverse Llama 3.1, Mistral SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Dynamics 365 & Azure Integration Specialist /Software Engineer /Senior DevOps/Data Engineer/Clouds/Linux Admin./ Java/.NET/Python/AI Automation/DL/ML/BI/SharePoint/D365 Developer /EDW-Japan(7.5yrs) & Australia(24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Dynamics 365 & Azure Integration Specialist/Snr DevOps/Data Engineer/.NET/PyAI/ML/SP/D365 Developer
Responsibilities:

- Highly skilled Microsoft Dynamics 365 (CE & F&O) professional with expertise in integrating and automating business processes using Azure (including AAD), SharePoint, Power Platform, Logic Apps within Azure iPaaS ecosystem, Service Bus for reliable message queuing and real-time messaging, and PowerShell scripting to deliver impactful solutions.
- Successfully migrated and integrated all on-prem systems to the cloud as well as replaced legacy CRM with Dynamics 365.
- Developed custom components using PCF Controls to build model-driven apps & reusable APIs, integrated into Dynamics 365 & Power Apps canvases. Automated data retrieval from backend SQL 2022 Database and SharePoint 2022 Document Lists to display frontend Dynamics 365 activity records, enhancing business workflow processes and customer services.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 & its latest 2024 Release Wave 1 enhancements to develop Dynamics 365 applications, using customization & extension components to streamline data flows and integrate with SAP HR, SAP FinOps, S/4 HANA systems. Liquid for templating & rendering dynamic content within web applications.
- Design and develop custom Dynamics 365 customizations and extensions to streamline business processes, using Common Data Service, Power Automate, and AI Chatbot (formerly Power Virtual Agents)
- Design, develop & customise SharePoint Online using Power Automate, SharePoint Framework (SPFx) or other M365 tools
- Developed Power Pages websites that seamlessly integrate and display Power BI Pro Paginated Reports, providing real-time insights into key performance indicators for Azure Synapse Analytics platform users.
- PowerBI Dashboards Development: Develop complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Drive collaboration with our Microsoft Partner to coordinate backlog of product enhancements by adopting AI and MLOps.
- **Automated repetitive tasks for Business Analysts and Data Analysts** by developing secure, offline local AI systems (PrivateGPT-X, Ollama3.2, Autogen Studio 2.0, etc.) for data analysis, replacing inefficient tools like Jira or Confluence.
- Leveraged LangChain to create versatile AI agents capable of data summarization, classification, and prediction. Utilized Pydantic for data validation to ensure data quality and consistency. These agents assisted Business Analysts and Data Analysts in accelerating data exploration, generating insights, and automating routine tasks.
- **Reduced development effort by 50% by leveraging OpenDevin** to eliminate the need for custom HTML, JavaScript, and canvas manipulation when customizing Dynamics 365, SharePoint 2022, & Power Apps using C#.NET Core in VS2022.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
    - Azure: Azure DevOps
    - AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
    - GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Designed and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python with ARM templates, Terraform, or Ansible. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration in Azure, & equivalent container orchestration services in AWS and GCP (e.g., ECS Fargate, GKE).
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2022/Dynamics 365/Power BI/Power Automate & Power Apps)
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Design, develop test and deploy security focused and AI-Powered ML applications, deep learning (DL) models and systems
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- **Implemented a Q-Learning-powered intrusion detection system** that reduced false positive rates by 30% and detected 95% of simulated Cyber-attacks within 24 hours. Deployed OpenAI's ChatGPT o1 model to assess the security implications of proposed actions before proceeding, ensuring that decisions align with established security policies and best practices.
- **Implemented cloud-agnostic application security for web applications and microservices** by scanning using Python, Bash, and PowerShell scripts, and integrating container scanning tools like Snyk and Aqua Security. This resulted in a 50%

reduction in critical vulnerabilities, such as SQL injection, cross-site scripting (XSS), and remote code execution, ensuring compliance with industry best practices like OWASP Top 10.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: D365 & Azure Automation Engineer /Snr DevOps/Data Engineer/Python/ML/SP/Power BI/D365 Developer
Responsibilities:

- Experienced in designing, implementing, and supporting integrated business solutions leveraging Microsoft Dynamics 365 (CE & F&O), Azure (including Azure AD), SharePoint, Power Platform, Logic Apps within the Azure iPaaS ecosystem, Service Bus, and PowerShell scripting to enhance operational efficiency and achieve business objectives.
- Design and develop applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).
- Full-stack developer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, and Azure cloud. Skilled in developing and consuming enterprise APIs using Azure API Gateway, REST API, or Mulesoft Anypoint Platform, with experience in integrating with privileged access management solutions like CyberArk
- Lead the development of tailored Mulesoft solutions that align with business requirements for scalability, security, and efficiency. Guide teams in adhering to standards and governance frameworks, leveraging technical expertise in server-side languages (Java, Node.js, etc.) and front-end frameworks (React JS/Next JS, Mulesoft Anypoint API, etc.).
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- Power BI Dashboards Development: Develop complex DAX formulas in Power BI to manipulate data.
- Create Power BI tiles and pin them to dashboards. Pin slicers or report pages to dashboards in Power BI Desktop/Service.
- Apply nested row context within calculations and conditional formatting with gradients.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch. Integrated M365 environments (O365, MS Teams, SharePoint 2016, Dynamics 365, Power App, Microsoft Flow) into the Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- **Designed and implemented Medallion Architecture with Data Lake in Azure:**
- **Batch Laye**r: Utilize Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Leveraged Infrastructure as Code (IaC) principles to automate the provisioning and configuration of virtual private clouds (VPCs), subnets, security groups using Terraform with Python (Boto3). Adopted ECS Fargate for automated container orchestration, enabling scalable and efficient deployment and management of containerized applications.
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2017 to develop Dynamics CRM2016 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- **Design, develop, and maintain automated data pipelines and API orchestration using either:**
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Designed and implemented a scalable cloud-based solution on Microsoft Azure utilizing serverless App Services and Functions. Leveraged Azure SQL Database for structured data and Blob Storage for unstructured data. Possess strong knowledge of Oracle and SQL, along with familiarity with newer technologies like Dataverse, Databricks, and NoSQL.
- **Automated frequent staff onboarding and off boarding** of Privileged Account Access Management using CyberArk Privileged Session Manager (PSM v10.8). Implemented a proactive security monitoring solution powered by an AI-driven

system (e.g., Llama 3.1 and Reflection) that continuously monitors for suspicious activity, triggers alerts for the Security team, and provides 24x7 proactive detection and response to security threats. The system is configured with multiple autonomous agents to ensure comprehensive coverage, including devices brought by employees (BYOD/BYOC). The AI is trained on a large dataset of normal and anomalous network traffic to enable accurate detection of security threats.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Dynamics 365 & Azure Integration Architect /Snr DevOps/Data Engineer /AI/ML/SP/D365 /K2 Developer
Responsibilities:

- Highly skilled Microsoft Dynamics 365 (CE & F&O) professional with expertise in integrating and automating business processes using Azure (including AAD), SharePoint, Power Platform, Logic Apps within Azure iPaaS ecosystem, Service Bus for reliable message queuing and real-time messaging, and PowerShell scripting to deliver impactful solutions.
- Designed and implemented custom solutions using Power Apps and Microsoft Flow to enhance usability and efficiency within Dynamics 365 & SharePoint 2016 environments. Leveraged Mulesoft Anypoint API for scalable integrations and automation, while also utilizing Microsoft Flow to streamline business processes.
- Leveraged MuleSoft Anypoint Platform's hybrid deployment capabilities to design and deploy APIs that seamlessly integrate AWS Lambda functions with Azure Logic Apps, enabling real-time data synchronization between the two platforms.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
    1. Azure: Azure DevOps
    2. AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
    3. GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM 2016/Microsoft Flow)
- Lead the development & support of existing solutions using Mulesoft Studio 3.10 & Mule ESB 3.10 through Agile practices.
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- Leverage SharePoint for process integration and Drupal/HTML with JavaScript for form development, along with backend SQL Server 2014 database for efficient data retrieval and customer service operations in Dynamics CRM 2016.
- Skilled in leveraging AI-powered tools to streamline development processes and enhance overall testing efficiency.
- Developed and maintained AI /ML models to enhance customer engagement and sales.
- Drive collaboration with our Microsoft Partner to coordinate backlog of product enhancements by adopting AI and MLOps.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Leverage Python programming with CyberArk SDK** to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management.
- # Retrieve students registered data by: students_registered_data = retrieve_students_registered_data()
- Utilized Azure DevOps to collaborate effectively with team members and manage Power Platform projects.
- Integrated Power Platform solutions with other Azure services using Azure DevOps and Azure Pipelines.
- Ensured consistent quality & efficient delivery of Power Platform projects through using Azure DevOps & Azure Pipelines.
- Proficient in modern data analytics and engineering tools like dbt and Dataform.
- Skilled in creating impactful visualizations using Tableau and Power BI.
- Used Agile methodology to plan and execute tasks and used Git to manage code
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Employed Power BI, Tableau, or QlikView for real-time financial analysis.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: D365 & Azure Solutions Engineer /DevOps Engineer/Software Test Engineer/Clouds/Linux Admin./DBA
Responsibilities:

- Developing, customizing, and implementing data integration solutions for MS Dynamics CRM2013 using KingswaySoft SSIS Integration Toolkit for Dynamics CRM v8.0. Leverage Visual Studio Online / VS Code, TFS, or Git for version control.
- Work closely with CRM Consultants on providing Clients with customised solutions: Requirement Analysis, Gap Analysis, Solution Development, Data Migration, Testing, Implementation and Support of Microsoft Dynamics NAV and CRM solutions
- Dynamics CRM2013 development either using Using KingswaySoft XrmSvcToolkit or MS Dynamics CRM Developer Toolkit
- Demonstrated experience/skills in developing Automation frameworks, test scripts, using applications and tools such as: GitLAB, OpenDevin, Cucumber, Gattling and Selenium for comprehensive testing and performance analysis.
- Proficient in HTML, JavaScript, and Selenium WebDriver for web testing automation.
- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Collaborate with internal project teams to gather requirements and develop solutions that boost user experience and functionality (sites, sub-sites, pages, libraries, data or document management, layout, custom views, etc)
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented automated DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: D365 & Azure Solutions Engineer /DevOps Engineer/Software Test Engineer/.NET/SPoint Developer
Responsibilities:

- Led the development, implementation, and maintenance of integrations between diverse systems and applications.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Lead the development and support of existing solutions using MS Power Platform suite of tools through Agile practices.
- Developing, customizing, and implementing data integration solutions for MS Dynamics CRM2011 using KingswaySoft SSIS Integration Toolkit for Dynamics CRM. Leverage Visual Studio 2012 / VS Code, TFS, or Git for version control.
- Collaborate with the business to deliver scalable, secure, and efficient Power Platform solutions aligned with business needs. Mentor teams in best practices and governance to ensure solution quality
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.

- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Led a project to replace an existing insurance web application, resulting in a significant reduction in IT costs and improved operational efficiency. The new application seamlessly integrated with Sales Force Automation (SFA) and project management tools like Jira or Confluence, streamlining workflows and enhancing collaboration.
- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate process

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,

- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Outline of Principle Data Engineer(AWS Glue-Ab Initio) -CBA –iGreenData – 23/12/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Principle Data Engineer (AWS Glue/Ab Initio) /Integration Solution Architect/Snr DevOps Engineer/Software Engineer/ Python/R Developer /Mulesoft /AI/ML/BI/SharePoint/Dynamics 365 Developer /EDW/DBA –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Principle Data Engineer (AWS Glue/Ab Initio) /Snr DevOps Engineer/Pyhon/AI/ML/SP/D365 Developer /EDW
Responsibilities:

- Built and implemented **Medallion Architecture with Data Lake in AWS**:
- **Batch Layer:** Utilize Amazon EMR, a scalable managed Hadoop framework service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Amazon Kinesis Data Streams as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying Silver & potentially Gold layer of Data Lake for advanced analytics and reporting.
- Leveraged AWS Glue Studio for ETL, Apache Spark for data processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Utilized Enterprise Ab Initio's powerful ETL tools (GDE, ADC, Express-IT, Control Center) & advanced techniques (Meta Programming, Co>Relate) to design, develop, & deploy efficient & scalable data integration solutions for on-premises data pipelines, leveraging its mature technology, GUI-based development, parallel processing, & robust data quality features.
- Leveraged AWS Glue Studio, Boto3, and Kinesis Data Streams to build a complex real-time data pipeline, ingesting data from a Python script-based generator, processing it using Glue Streaming ETL, and storing the results in Amazon S3.
- Apache Spark: Utilize Spark SQL and DataFrame APIs to perform complex data transformations and aggregations.
- AWS Glue: Leveraged AWS Glue Studio to create and maintain ETL jobs for data pipelines.
- Data Access and Visualization: Developed complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Tableau: Use Tableau to create interactive dashboards and visualizations to provide insights to business users.
- Amazon QuickSight: Leverage QuickSight for self-service BI and ad-hoc analysis.

- Delivered data products & services leveraging Power BI, ML, and R to automate data insights, informing business decision-making and achieving strategic goals through data modeling and statistical analysis.
- Automated Business Analysts and Data Analysts daily repeated tasks by developing secure,offline local AI systems (PrivateGPT-X, Ollama-L, Autogen Studio 3.0, etc.) for data analysis. Leveraged LangChain to create versatile AI agents capable of data summarization, classification, and prediction. Utilized Pydantic for datavalidation to ensure data quality and consistency. These agents assisted Business Analysts and Data Analysts in accelerating data exploration, generating insights, and automating routine tasks.
- Collaborate with clients & stakeholders to architect, develop, & manage databases, ETL pipelines, & management systems, ensuring end-to-end solutions align with business and user needs while meeting functional and non-functional requirements.
- **Leveraged Kafka (MSK) Topics, MSK, producers, consumers, Kafka Connect, and Flink to build robust real-time data pipelines, enabling efficient ingestion, transformation, and processing of streaming data.**
- **Design, develop, and maintain a robust ETL pipeline/API orchestration using either**:
    - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, RNN, ANN, GAN) based on dataset type, size, and desired result interpretability.
- Data development, optimization, and manipulation using PySpark libraries (PySpark Core, PySpark SQL, PySpark MLlib etc.) or SQL, including PySpark scripting.
- Automation and Orchestration: Experience in automating workflows and scheduling jobs using tools like AutoSys to improve efficiency and reliability.
- Data Governance and Quality: Passionate about ensuring data quality and adherence to data governance standards.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Principle Data Engineer (AWS Glue/Ab Initio) /Snr DevOps Engineer /Py/R/Java/AI/ML/SP/BI Developer
Responsibilities:

- **Architected and implemented a modern 3-layer lambda architecture on GCP**:
- **Batch layer**: Utilized Google Cloud Dataproc, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed BigQuery with its built-in Spark capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Cloud Pub/Sub as the unified service layer for real-time data ingestion and BigQuery for querying the data warehouse for advanced analytics and reporting.
- Leveraged a diverse technology stack including AWS Glue Studio, Apache Spark, AWS S3, AWS Redshift, and Mulesoft Anypoint Platform for data engineering and integration tasks, alongside experience with traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Utilized Enterprise Ab Initio's powerful ETL tools (GDE, ADC, Express-IT, Control Center) & advanced techniques (Meta Programming, Co>Relate) to design, develop, & deploy efficient & scalable data integration solutions for on-premises data pipelines, leveraging its mature technology, GUI-based development, parallel processing, & robust data quality features.
- Adapted at modeling and normalizing complex data structures using tools like Cloud Spanner and Cloud SQL for optimal performance and scalability within the GCP environment.
- Proven ability to manage and analyze intricate, high-volume, high-dimensional data from various sources within a serverless GCP Lambda architecture, leveraging BigQuery and Cloud Dataflow for efficient data processing and analysis.
- Data Processing and Transformation:
- Apache Spark: Utilize Spark SQL and DataFrame APIs to perform complex data transformations and aggregations.
- AWS Glue: Leveraged AWS Glue Studio to create and maintain ETL jobs for data pipelines.
- Data Access and Visualization: Developed complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Tableau: Use Tableau to create interactive dashboards and visualizations to provide insights to business users.
- Amazon QuickSight: Leverage QuickSight for self-service BI and ad-hoc analysis.

- Delivered data products & services leveraging Power BI, ML, and R to automate data insights, informing business decision-making and achieving strategic goals through data modeling and statistical analysis.
- Built and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms. Leveraged infrastructure-as-code tools like Terraform or Ansible for automated provisioning and configuration, ensuring consistency and efficiency.
- Design and develop Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
- Data Processing and Transformation:
- Apache Spark: Utilize Spark SQL and DataFrame APIs to perform complex data transformations and aggregations.
- AWS Glue: Leverage Glue ETL jobs to create and maintain data pipelines.
- Data Access and Visualization: Developed complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Tableau: Use Tableau to create interactive dashboards and visualizations to provide insights to business users.
- Amazon QuickSight: Leverage QuickSight for self-service BI and ad-hoc analysis.
- Delivered data products & services leveraging Power BI, ML, and R to automate data insights, informing business decision-making and achieving strategic goals through data modeling and statistical analysis.
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- Used MuleSoft Anypoint Platform (including API, Exchange, and DataWeave) for complex structured data transformations among disparate systems, while leveraging DBT for standardization and basic transformations.
- Possess a strong understanding of data pipelines, utilizing tools like Cloud Dataflow for ETL processes within the GCP ecosystem, feeding data into BigQuery for in-depth analytics and Cloud Data Studio for creating insightful reports.
- Well-versed in data wrangling techniques for data acquisition from diverse sources, employing tools like Cloud Storage and Cloud Functions for data ingestion, and leveraging tools like Dataflow and BigQuery for data cleaning, transformation, and preparation for analysis within the GCP environment.
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Develop statistical models for ad reporting using R programming with core R packages, modeling packages from tidyverse, and reporting and visualization packages including report, R Shiny, ggplot2, Tableau, Power BI, MicroStrategy & SAP BO.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Principle Data Engineer (AWS Glue/Ab Initio) /Snr DevOps Engineer Engineer/AI/ML/SP/K2 Developer
Responsibilities:

- Designed and implemented a scalable Snowflake EDW (Terraform) for the hybrid Azure and AWS environment. Data from various sources (including on-premises databases) is ingested using Azure Data Factory (ADF). SSIS and Spark Notebooks handle data transformations, with Spark Notebooks leveraging Apache Spark for complex big data scenarios. This unified platform integrates with Machine Learning tools from both clouds, enabling advanced analytics and large-scale ML.
- Leveraged SPARX for data modeling to support efficient pipeline development within a comprehensive enterprise architecture framework. This data model can then be used for integration with Snowflake, AWS, SAS, and Python in either a Salesforce or Mulesoft development environment.
- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Utilized Enterprise Ab Initio's powerful ETL tools (GDE, ADC, Express-IT, Control Center) & advanced techniques (Meta Programming, Co>Relate) to design, develop, & deploy efficient & scalable data integration solutions for on-premises data pipelines, leveraging its mature technology, GUI-based development, parallel processing, & robust data quality features.
- Architected and delivered high-performance data pipelines using the Microsoft data stack (SSIS, Azure Synapse, HDInsight, ADF, Databricks) to ensure data accuracy and efficiency, empowering real-time analytics (Azure Stream Analytics) and insightful visualizations (Power BI) for improved decision-making.
- Deriving business solutions through creative analysis and insights, whilst ensuring solutions are robust and fit for purpose
- Blended advanced analytics with popular BI tools: crafted interactive R visuals in Tableau, leveraged Rserve for seamless data exchange, analyzed & prepped data within Power BI's model using R scripts, built custom R visualizations, and mastered data interoperability between R, SAS, and SPSS (Statistical Package for Social Science).

- **Optimized CI/CD pipelines with DevOps tools**: Used tools like Jenkins, GitHub Action or GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages
- Integrated R to SQL Server & working with SQL Databases using RODBC and dplyr to querying SQL Databases in RStudio.
- Designed ETL processes using source-to-target mappings and design documents that incorporated security, performance tuning, and other best practices, including minimizing data input, implementing incremental data updates, maximizing data quality, and automating tasks.
- Implemented parallel processing, secured the data preparation area, and adhered to DevSecOps principles for secure development practices when building ETL data pipelines.
- Deploy & test code from lower environments to production. Document technical specifications, data models, process flows etc. Perform comprehensive ETL process and other technical components to ensure data quality and functionality.
- Optimize data delivery through automation and scalable infrastructure, evaluating process improvements.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis.
- Designed data warehouse (DWH) concepts based on both Bill Inmon and Ralph Kimball methodologies, built data models using Tableau or Toad Data Modeler 6.x, developed OLAP cubes using SSAS or Cognos TM1, and visualized data using Power BI, Tableau 10, or QlikView.
- Designed and implemented CI/CD pipelines across Azure DevOps, GitHub Actions, and Jenkins, utilizing scripting languages (Python, Ansible, etc.) for automated tasks, to ensure continuous delivery of applications.
- Utilized Azure Monitor, logging, partitioning, indexing, and optimizing the ADF pipeline structure to analyze system and application performance, providing insights for business case development to enhance service delivery and reduce issues.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies, and Azure Security Center.
- Monitored and troubleshooted Azure services using Splunk, Data Dog and Azure Network Monitor
- **Docker Images:** Proficient in building and running Docker containers in AWS and Azure Hybrid Cloud using Docker commands and managed services like Amazon ECR and Azure Container Registry (ACR) : `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- Designed and developed K2 workflows, forms, and reports on a SharePoint platform, integrating data from multiple line-of-business systems (SharePoint, CRM, etc.) using SmartObjects in K2 Designer, to facilitate data consolidation and decision-making.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Developed and deployed machine learning models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision).

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Senior Data Engineer (AWS Glue)/DevOps Engineer/Test Engineer /Clouds/Linux Administrator/DBA
Responsibilities:

- Built and implemented **Medallion Architecture with Data Lake in AWS**:
- **Batch Layer**: Utilize Amazon EMR (Elastic MapReduce) with its Spark capabilities for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Amazon Redshift with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Amazon Kinesis Data Streams and Amazon Redshift Spectrum for real-time analysis through Amazon Kinesis Data Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Utilized Matillion experience and ERwin data modeling expertise to design and implement robust ETL pipelines, ensuring data quality and alignment with business needs.
- Leveraged Amazon Redshift for high-performance analytics on large datasets and Amazon Athena for ad-hoc SQL queries on AWS S3 data. Utilized Splunk, ELK Stack, and Prometheus for monitoring and troubleshooting data pipelines.
- Building data pipelines using Amazon EMR with PySpark to extract data from various sources (Oracle/Excel), store it in a Data Lake, and then transform and model it for analytics and reporting purposes.
- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.

- Utilized Enterprise Ab Initio's powerful ETL tools (GDE, ADC, Express-IT, Control Center) & advanced techniques (Meta Programming, Co>Relate) to design, develop, & deploy efficient & scalable data integration solutions for on-premises data pipelines, leveraging its mature technology, GUI-based development, parallel processing, & robust data quality features.
- Completed design, data models, loading approaches and source to target mapping documents.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a mapping document that defines how the source data fields correspond to the target data fields.
- Implemented the mapping logic based on the mapping document, including writing code or using tools to map the source data to the target data according to the defined mapping rules.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Using either Python or Java developed custom ETL/API orchestrations which can apply to any cloud or on-prem application.
- Implemented automated testing frameworks (Selenium, JUnit) & CI/CD pipelines (Azure DevOps) for efficient testing across applications. Collaborated with developers to ensure timely and high-quality testing.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Developed and deployed cloud-native applications using Azure Functions, Azure Service Fabric and Azure Service Bus
- Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Data Engineer /DevOps Engineer /Software Test Engineer/SharePoint Consultant /DBA
Responsibilities:

- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Utilized Enterprise Ab Initio's powerful ETL tools (GDE, ADC, Express-IT, Control Center) & advanced techniques (Meta Programming, Co>Relate) to design, develop, & deploy efficient & scalable data integration solutions for on-premises data pipelines, leveraging its mature technology, GUI-based development, parallel processing, & robust data quality features.
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Data Engineer/Software Test Engineer/Senior Systems Engineer /DBA
Responsibilities:

- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Utilized Enterprise Ab Initio's powerful ETL tools (GDE, ADC, Express-IT, Control Center) & advanced techniques (Meta Programming, Co>Relate) to design, develop, & deploy efficient & scalable data integration solutions for on-premises data pipelines, leveraging its mature technology, GUI-based development, parallel processing, & robust data quality features.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Leveraged Apache Spark for ETL processing, AWS S3 for data storage, AWS Redshift for data warehousing, and Mulesoft Anypoint Platform for integration, along with experience in traditional technologies like Teradata, Hadoop, Tableau, and SAS, and AWS-native services like EMR and Redshift.
- Utilized Enterprise Ab Initio's powerful ETL tools (GDE, ADC, Express-IT, Control Center) & advanced techniques (Meta Programming, Co>Relate) to design, develop, & deploy efficient & scalable data integration solutions for on-premises data pipelines, leveraging its mature technology, GUI-based development, parallel processing, & robust data quality features.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Data Analyst /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, AWS Glue, Ab Initio, Apache Spark, PySpark, AWS Athena, Redshift, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Software Engineer – Syd, not SFO, Starup co. – SustainRecruit – 22/12/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Software Engineer/Integration Solutions Architect/Senior DevOps Engineer /Senior Data Engineer / Clouds Admin/ Full Stack Developer(Java/.Net/React/Py)/AI/ML/BI/MuleSoft/SharePoint/D365 Developer/EDW –Japan (7.5yrs) & Aust.(24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Software Engineer/Solutions Architect/Snr DevOps/Data Engineer/Java/.Net/Py/AI/ML/SP/D365 Developer
Responsibilities:

- Senior Software Engineer with Expertise in Event-Driven Architecture (Kafka) for Scalable SaaS Backend Systems
- Developed an extendable SDK that supported multiple pluggable channels (Kafka, SNS, etc.) to generate events. Each application/service generates events to a Kafka topic, which can then trigger Lambda functions or other downstream processes (services or applications that consume the events produced by the Kafka topic.)
- Full-stack developer proficient in client-side Next.js/React and server-side Django/Python, Flask/Python, and AWS (AWS API Gateway, REST APIs). Experienced with Java-based Mulesoft Anypoint Platform for API integration.
- Utilized Django as the backend to handle data, business logic, and AI/LLM model interactions to automate scheduled credit & payment for approved providers, while using Next.js for the frontend to provide a rich user interface & dynamic content.
- **Client-Side Rendering**: Leveraging Next.js to render AI/LLM-powered components directly on the client-side, providing a seamless user experience. **Server-Side Rendering (SSR)**: Utilizing Next.js to render pages on the server, improving SEO and initial page load times for AI-driven content for Real-time Payments and NPP (New Payments Platform)
- **Server-side logic**: Expertise in handling event generation, interacting with external services (Kafka, SNS), and managing communication channels using Go, Java, Python, C#, or JavaScript.
- Skilled in building microservices applications using Java & JavaScript with frameworks like Spring Boot & Node.js. Expertise in consuming RESTful web services to integrate with ACQSC's enterprise services layer, databases, and host systems.
- **API Integration**: Leveraging Next.js's built-in data fetching capabilities or Mulesoft DataWeave to consume APIs created with Django, Flask, REST, or Mulesoft Anypoint Platform to fetch AI/LLM-generated content from real-time web applications.
- **Greenfield Projects**: Collaborate with talented engineers from various domains, utilizing the latest AI/LLM models on offline OLLAMA 3.2 to simulate predictive results. This process is conducted before building & shaping large systems from scratch.
- Adapting a combined approach using Large Conceptual Models (LCMs) and Large Action Models (LAMs) could potentially automate the identification and correction of mispayments, reducing the need for human intervention.
- **Pattern Recognition**: Leverage well trained LCMs on vast payment data to identify patterns and anomalies indicating mispayments, such as unusual amounts, incorrect recipients, or timing discrepancies.
- **Natural Language Processing**: Use LCMs to analyze payment descriptions, invoices, and other textual data to detect errors or inconsistencies that might signal a mispayment.
- **Rule-Based Systems**: Programmed LCMs with specific rules to flag potential mispayments, such as payments exceeding certain thresholds or those made to unauthorized recipients.
- **Automated Corrections**: Train LAMs to automatically correct simple mispayments, such as those involving incorrect account numbers or amounts. **Triggering Human Intervention**: For more complex cases, LAMs can flag the issue and trigger a workflow involving human review and approval.
- Adapting MIT B2B Liquid AI with 1000x energy efficiency over ChatGPT, leveraging organoid intelligence (OI) to accelerate brain organoid development for neurological disease research, and developing efficient AI solutions at all scales.
- Automated Business Analysts and Data Analysts daily repeated tasks by developing secure,offline local AI systems (PrivateGPT-X, Ollama-L, Autogen Studio 3.0, etc.) for data analysis. Leveraged LangChain to create versatile AI agents capable of data summarization, classification, and prediction. Utilized Pydantic for datavalidation to ensure data quality and consistency. These agents assisted Business Analysts and Data Analysts in accelerating data exploration, generating insights, and automating routine tasks. Strong proficiency in database design, optimization, and SQL.
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- Implemented a Q-Learning-powered intrusion detection system that reduced false positive rates by 30% and detected 95% of simulated cyber attacks within 24 hours. Deployed OpenAI's ChatGPT o1-o3 models to assess the security implications of proposed actions before proceeding, ensuring that decisions align with established security policies and best practices.
- Architected and implemented a hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Configured encrypted tunnels for data transfer using IPsec via Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Leveraged Infrastructure as Code (IaC) principles with Terraform and Python (Boto3) to automate the provisioning and configuration of VPCs, subnets, and Network Security Groups (NSGs). Utilized ECS Fargate for serverless container orchestration, enabling efficient and scalable deployment of containerized applications.
- Designed, developed and maintained automated CI/CD pipelines using platform-specific tools:
    - Azure: Azure DevOps, Azure Pipeline
    - AWS: AWS CodePipeline/CodeBuild/CodeDeploy
    - GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to clients' specific needs.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Senoir Software Engineer/Snr DevOps/Data Engineer/Java/.Net/Python/AI/ML/SP/D365 Developer/EDW
Responsibilities:

- Designed and developed applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).
- Full-stack Developer Proficient in React/TypeScript, Python/Flask/Django, Azure Cloud, and API development (AWS API Gateway, REST, Mulesoft Anypoint Platform). Experienced in integrating with PAM solutions like CyberArk.
- Employed Django to construct a robust backend system, incorporating AI/LLM models to streamline payment automation for major financial institutions. Developed an intuitive Next.js frontend to facilitate user interaction and dynamic content delivery. Improved the performance and accessibility of AI-driven Real-time Payments and NPP (New Payments Platform) content by utilizing Next.js for both client-side and server-side rendering.
- Integrated AI/LLM-generated content from real-time web applications using Next.js data fetching or Mulesoft DataWeave to consume APIs built with Django, Flask, REST, or Mulesoft Anypoint Platform.
- Integration Solutions Architect specializing in designing frameworks to integrate client's modern cloud-based software solutions with legacy applications (SAP BusinessObjects, SAP FinOps, and SAP S/4HANA), utilizing cloud-native integration with AWS Lambda, AWS Step Functions, & AWS API Gateway for RBA financial services workflow automation.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch. Integrated M365 environments (O365, MS Teams, SharePoint 2016, Dynamics 365, Power App, Microsoft Flow) into the Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- **CI/CD automation**: Either utilizing GitHub Actions with Bash scripts or using Mulesoft Anypoint Platform to automate the build, test, and deployment processes, streamlining the software delivery pipeline.
- Python or GoLang programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- Power BI Dashboards Development: Develop complex DAX formulas in Power BI to manipulate data.
- Collaborated with a world-class engineering team to develop, scale, and own key platform features, ensuring their success.
- **Designed and implemented Medallion Architecture with Data Lake in Azure:**
- Leveraged Infrastructure as Code (IaC) principles to automate the provisioning and configuration of virtual private clouds (VPCs), subnets, security groups using Terraform with Python (Boto3). Adopted ECS Fargate for automated container orchestration, enabling scalable and efficient deployment and management of containerized applications.
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana) and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Developed and deployed digital solutions (integrations, APIs, business process automations, and cloud integrations) on AWS, Azure, GCP, or on-premise environments, leveraging container orchestration services such as ECS, EKS, AKS, GKE, or Kubernetes. Strong proficiency in database design, optimization, and SQL.
- **Design, develop, and maintain automated data pipelines and API orchestration using either:**
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- **Implemented cloud-agnostic application security for web applications and microservices** by scanning using Python, Bash, and PowerShell scripts, and integrating container scanning tools like Snyk and Aqua Security. This resulted in a 50% reduction in critical vulnerabilities, such as SQL injection, cross-site scripting (XSS), and remote code execution, ensuring compliance with industry best practices like OWASP Top 10.
- Proficient in creating DevOps pipelines & related frameworks, such as Agile, Lean, ITIL4, & DevSecOps, using tools like Git, Jenkins, Docker, Kubernetes, & Ansible, while leading solution design discussions at various levels within an organization.
- Automated frequent onboarding and off boarding of Privileged Account Access Management using CyberArk Privileged Session Manager (PSM v10.8). Implemented a proactive security monitoring solution powered by an AI-driven system (e.g., Llama 3.1 and Reflection) that continuously monitors for suspicious activity, triggers alerts for the Security team, and provides 24x7 proactive detection and response to security threats. The system is configured with multiple autonomous agents to ensure comprehensive coverage, including devices brought by employees (BYOD/BYOC). The AI is trained on a large dataset of normal and anomalous network traffic to enable accurate detection of security threats.
- Immersed experience adopting ISO27001, ACC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.

- Provisioning & managing cloud resources, virtual networks, & secrets, adhering to governance standards & best practices.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Software Engineer/Snr DevOps/Data Engineer/Java/.Net/Py/AI/ML/SP/D365/K2 Developer/EDW
Responsibilities:

- Integration Solutions Specialist focused on designing frameworks to integrate client's modern cloud-based software solutions with legacy applications (SAP HR, SAP FinOps, and SAP S/4HANA), utilizing cloud-native integration with AWS Lambda, AWS Step Functions, and AWS API Gateway for streamlining student enrollment processes.
- Results-oriented full-stack developer with a strong foundation in React, TypeScript, Python, AWS, and API integration. Experienced in enhancing application security with PAM solutions.
- Optimized credit and received payment processes for enrolled students by building a scalable Django backend integrated with AI/LLM models. Designed a user-centric Next.js frontend to enhance operational efficiency.
- Leveraged React Router and the Fetch API to seamlessly integrate AI/LLM-generated content from APIs built with Django, Flask, REST, or Mulesoft Anypoint Platform into real-time web applications.
- Employed Gatsby or Remix to build performant, data-driven real-time web applications, seamlessly integrating AI/LLM-generated content from APIs built with Django, Flask, REST, or Mulesoft Anypoint Platform.
- Developed and deployed ML models for image classification using PyTorch, leveraging pre-trained models like ResNet-50 (e.g., implemented a simple model by modifying the output layer: model.fc= torch.nn.Linear(model.fc.in_features, 10)).
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Develop statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse ; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.
- Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- Developed and deployed digital solutions (integrations, APIs, business process automations, and cloud integrations) either on AWS ECS or EKS.
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
- Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.
- Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Software Engineer/DevOps/Data Engineer/Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- Automated cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Proven track record in designing, developing, and deploying scalable, high-performance backend systems.
- Deep expertise in Python, Django, and related frameworks.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Contributed to the design and implementation of integration solutions, meeting both business and technical requirements.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.

- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Software Engineer /DevOps Engineer /Software Test Engineer /C#.NET/SharPoint Developer
Responsibilities:

- Led the development, implementation, and maintenance of integrations between diverse systems and applications.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins, BitBucket, and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Systems Integration Developer /DevOps Engineer/Senior Systems Engineer /DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, TFS) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET Core (Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1~o3, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, Mulesoft Studio 7 Mule 4, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, TIBCO Cloud Integration, TIBCO BW CE 6.x ~ CE2.92, Tibco Mashery API Gateway, Mulesoft Anypoint Exchange/DataWeave, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum.
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Cloud Infrastructure Engineer/Integration Solution Architect/Snr DevOps/Data Engineer/Snr Network Engineer /Java/.Net /Python /AI/ML/BI/MuleSoft/BizTalk/SharePoint/D365 Developer/EDW –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Cloud Infrastructure Engineer/Snr Network/DevOps/Data Engineer/Py/AI/ML/SP/D365 Developer/EDW
Responsibilities:

- Architected and implemented a hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Configured encrypted tunnels for data transfer using IPsec via Cloud VPN Gateways across all three platforms.
- Designed and implemented automated VPN solutions using WireGuard with Perfect Forward Secrecy (PFS) to secure remote access and inter-site communication, ensuring confidentiality and integrity of sensitive data.
- Leveraged Infrastructure as Code (IaC) principles with Terraform and Python (Boto3) to automate the provisioning and configuration of VPCs, subnets, and Network Security Groups (NSGs). Utilized ECS Fargate for serverless container orchestration, enabling efficient and scalable deployment of containerized applications in Docker.
- Designed & implemented secure Azure landing zones using Azure Blueprints & VPNs to integrate Windows-based cloud workloads within M365 environments (O365, MS Teams, SharePoint 2019/2022, Dynamics 365, Power BI, Power Automate, etc.). Employed advanced networking & security best practices to safeguard data confidentiality, integrity, & availability.
- Leveraged Intune Mobile Device Management (MDM) and Mobile Application Management (MAM) to effectively manage and secure mobile devices and applications within the Microsoft Endpoint Manager admin center.
- Proven track record in administering Windows Server, with a focus on DNS, DHCP, Active Directory, and Group Policy.
- Provided technical support for critical business applications. Managed and optimized Azure, AWS and GCP infrastructures.
- Designed, implemented, and maintained Azure solutions. Troubleshooted and resolved Azure/AWS related issues.
- Developed and implemented CI/CD pipelines using Azure Pipelines/AWS Code Pipeline/GCP Cloud Build, GitHub Actions and YAML scripting to automate the deployment of business applications, streamlining development & improving efficiency.
- Strong understanding and practical experience in managing large-scale, multifaceted Azure environments composed of multiple intertwined subscriptions. Proven ability to identify, diagnose, and resolve complex technical issues
- **Cloud Infrastructure Engineering: Azure Expertise**: Proficient in designing, implementing, and managing Azure infrastructure, including VNets, subnets, NSGs, firewall rules and intrusion prevention systems, SSO, FortiGate Firewalls, FortiClient VPN with MFA, and Azure Entra ID (formerly known as AAD)-based access control for Azure Key Vault.
- **Network Engineering**: Led design and deployment of robust network solutions, leveraging deep knowledge of CIDR, TCP/IP stack, OSPF, RIP, and BGP for optimal performance & security across diverse WAN & hybrid cloud environments.
- **SD-WAN**: Experience with SD-WAN solutions to optimize WAN performance and security.
- **Automation**: Skilled in automating infrastructure tasks using Python and Ansible, including IaC practices with Azure ARM. **Security**: Proficient in implementing security best practices, including NSGs, Azure Firewall, and VPN configurations.
- Managed and configured Microsoft Azure cloud services. Designed, built, and deployed Azure environments and workloads.
- Supporting and troubleshooting cloud resources to ensure availability, resiliency and costing efficiency.
- Designed, developed and maintained automated CI/CD pipelines using platform-specific tools:
    - Azure: Azure DevOps, Azure Pipeline
    - AWS: AWS CodePipeline/CodeBuild/CodeDeploy
    - GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to clients' specific needs.
- Provisioning & managing cloud resources, virtual networks, & secrets, adhering to governance standards & best practices.
- Extensive experience in implementing and managing Azure Key Vault to securely store, retrieve, and manage cryptographic keys, secrets, and certificates. Proven ability to design and deploy Azure Virtual Networks (VNet) to create isolated and secure network environments for Azure resources.
- Utilized Azure Monitor/AWS CloudWatch to monitor, troubleshoot, and optimize Kubernetes clusters for peak performance and reliability, implement security best practices, network policies, secrets management and develop automation scripts to manage infrastructure components, ensuring consistent environments across clusters.
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- Implemented a Q-Learning-powered intrusion detection system that reduced false positive rates by 30% and detected 95% of simulated cyber attacks within 24 hours. Deployed OpenAI's ChatGPT o1 model to assess the security implications of proposed actions before proceeding, ensuring that decisions align with established security policies and best practices.
- **Automated frequent staff onboarding and off boarding** Privileged Account Access Management using CyberArk Privileged Session Manager (PSM v10.8). Implemented a proactive security monitoring solution powered by an AI-driven system (e.g., Llama 3.2 and Reflection) that continuously monitors for suspicious activity, triggers alerts for the Security team, and provides 24x7 proactive detection and response to security threats. The system is configured with multiple autonomous agents to ensure comprehensive coverage, including devices brought by employees (BYOD/BYOC). The AI is trained on a large dataset of normal and anomalous network traffic to enable accurate detection of security threats.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Snr Cloud Infrastructure Engineer/Snr Network/DevOps/Data Engineer/Java/Py/AI/ML/SP/BI Developer
Responsibilities:

- Designed and implemented robust application and platform solutions using Mulesoft and cloud-native CI/CD tools (e.g., Azure DevOps, AWS DevOps, GCP Cloud Build/Cloud Deploy). Ensured alignment with RBA architectural standards to support critical organizational systems and services, integrating across AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Built secure and scalable AWS Landing Zones using Infrastructure as Code (IaC) with AWS CloudFormation templates or Terraform. Leveraged AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch. Oversee and manage installations to ensure timely, cost-effective, and high-quality completion, thereby minimizing operational and project risks and controlling costs.
- Utilized Terraform to provision the infrastructure (VMs, networks, ARM, etc.), and then used PowerShell Desired State Configuration (DSC) to configure the operating systems and applications on those machines. This hybrid approach allows you to manage both infrastructure and configuration in a consistent and automated way, aligning well with IaC principles.
- Implemented robust CI/CD pipelines with Mulesoft to automate application delivery and reduce manual intervention. Leveraged Terraform to efficiently provision and manage cloud infrastructure resources (Azure, AWS, GCP) for Mulesoft Anypoint API deployments connecting to major financial institutions 24x7.
- **Networking Proficiency**: Extensive experience with Cisco networking technologies.
- **Cloud Infrastructure Engineering**: **Azure Expertise**: Proven proficiency in designing, implementing, and managing Azure infrastructure, including VNets, subnets, NSGs, SSO, FortiGate Firewalls, FortiClient VPN with MFA, and Azure AD-based access control for Azure Key Vault.
- **Network Engineering**: Engineered and optimized intricate network topologies, demonstrating expertise in CIDR, TCP/IP, OSPF, RIP, BGP, ISIS, BFD, LDP, and L2TP protocols to deliver reliable, scalable WAN & hybrid cloud solutions.
- Deployed and managed IP and MPLS networks connectivities between RBA and all major financial institutions
- **SD-WAN**: Experienced in leveraging SD-WAN solutions to optimize WAN performance and security.
- **Cloud Operations**: Managed & configured Azure cloud services, ensuring high availability, resiliency, & cost-efficiency. Supported development & test environments and associated DevOps tools (Git, Ansible, Jenkins, Docker, etc.).
- Designed, developed, and maintained automated data pipelines and API orchestration using either:
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- Optimized data pipelines utilizing Airflow and Luigi, reducing processing time by 30% and improving resource utilization.
- Automated model versioning and rollbacks with MLflow, enabling seamless rollbacks and mitigating production issues: e.g. mlflow.sklearn.log_model(model, "linear_model") & rollback by client.transition_model_version_stage()
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- **Integration Solution Design with Azure DevOps CI/CD**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).
- Implementing Integrated **DevSecOps** practices with IaC security principles, utilizing ARM templates, Azure PowerShell, Azure CLI, and AWS CLI to automate security configurations and ensure a consistent security posture across deployments.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Developed and implemented robust CI/CD pipelines using GitHub Actions to streamline Mulesoft application deployment. Leveraged YAML scripting and automation to enhance RBA development efficiency and reduce risk.
- **CloudWatch Observability Expertise:** Implemented comprehensive CloudWatch observability solutions, including metrics, logs, and dashboards. Leveraged CloudWatch Insights for advanced log analysis and troubleshooting.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Cloud Infrastructure Engineer/Snr Network/DevOps/Data Engineer/Python/ML/SPoint/K2 Developer
Responsibilities:

- Cloud Integration Solution Architect with expertise in designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow

- Managed and configured Microsoft Azure cloud services. Designed, built, and deployed Azure environments and workloads.
- Provided technical supporting and troubleshooting cloud resources to ensure availability, resiliency and costing efficiency.
- Strategic planning regarding optimizations for the improvement of the core infrastructure.
- **Networking Proficiency**: Extensive experience with Cisco networking technologies
- **Cloud Infrastructure Engineering**: Azure expert, proficient in designing, implementing, and managing Azure infrastructure, including VNets, subnets, NSGs, firewall rules, SSO, FortiGate, FortiClient VPN with MFA, and Azure AD-based access control for Azure Key Vault.
- Demonstrated expertise in integrating Terraform and Mulesoft to automate complex API deployments and infrastructure provisioning.Successfully designed and implemented CI/CD pipelines using Terraform and Mulesoft, optimizing application lifecycle management.Applied Terraform to provision and manage cloud infrastructure resources (Azure, AWS) with a focus on efficiency, security, and scalability.
- Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
- Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
- Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
- Successfully installed, configured, and tested new and existing hardware, software, and WAN/LAN technologies.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
- Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Cloud Infrastructure Engineer/Senior Network/DevOps Engineer/Clouds/Linux Administrator
Responsibilities:

- Skilled Cloud Integration Architect with expertise in designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow
- **Network Engineering**: Strong understanding of the interplay between CIDR, the TCP/IP stack, and routing protocols (IGPs: OSPF, RIP; EGP: BGP) in WAN and hybrid cloud environments.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python with ARM templates, Terraform, or Ansible. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration in Azure, & equivalent container orchestration services in AWS and GCP (e.g., ECS Fargate, GKE).
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Proficient in Linux system administration, including kernel optimization (sysctl, vmstat), performance tuning (iostat, sar), and security hardening (iptables, fail2ban). Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all EML's data pipelines.
- Developed, reviewed, and maintained cloud support documentation and best practices
- Escalated and resolved performance issues with cloud providers and vendors
- **Change Management**: Planned and managed change implementation for cloud systems and applications.
- Managed the integration of system solutions across various platforms and technologies, supporting DevOps and Cloud.

- Optimized the pipeline for performance, cost, and efficiency by applying best practices such as resource allocation, scaling, health checks, and logging in addition to partitioning, indexing, and optimizing the ADF pipeline structure.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Streamlined DevOps processes by implementing robust version control practices with Git, including branching strategies, code reviews, and merge requests. Utilized configuration management tools to consolidate settings and streamline infrastructure management, ensuring consistent provisioning.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Cloud Infrastructure Engineer/Snr Network/DevOps Engineer/SharPoint Developer
Responsibilities:

- Proven track record of designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- Streamlined cloud operations by automating resource provisioning and management using infrastructure as code (IaC). Proficient in configuring and managing hybrid cloud environments (Azure, AWS, GCP) across development, testing, and production, leveraging tools like Ansible, CloudFormation, boto3, and Pulumi.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: System Administrator /Senior Systems Engineer /Administrator (AD/SCCM/SCOM)/DBA
Responsibilities:

- Managed and maintained Azure AD, D365 CRM, SharePoint, and M365 for a large organization, supporting over 1000 users.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Implemented JIRA project management tool, streamlining issue tracking and collaboration across cross-functional teams.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate “zone1”, “WWN1”,”WWN2”; Create a zoneset configuration by #cfgcreate “test_cfg”, “zone1”, “zone2”; Add a zone to a zone configuration via cfgadd, etc.
- Create Device Collection to define source files and distribution points through UNC Network path; Assign membership rules for the collection; Configure install & uninstall program scripts; Create deployments by selected software to deploy in root dir.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Network Administrator /Systems Engineer /Wintel Systems Administrator (AD/SCCM/SCOM) /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.

- Systems Planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Designed and maintained a robust and secure network infrastructure, supporting data and voice communication for 1200+ employees.
- Implemented advanced firewall configurations and intrusion prevention systems, safeguarding against cyber threats.
- Conducted regular network performance monitoring and optimization, enhancing network efficiency by 20%.
- Designed and managed network infrasctructure and security measures.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Senior Network Engineer/Senior Systems Engineer/Integration Developer /Data Engineer**
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, and infrastructure products on HP G3-G6 servers, HP and Dell Blade Servers, Dell and Citrix Desktops, & HP/NetApp/Dell storage solutions.
- Led data center infrastructure planning and management, ensuring the availability and reliability of critical business applications. Provided network design for BAU operations and technical support to the US Head Office SOC Team.
- Deployed and managed IP and MPLS network connectivity between MS global data centers and branches.
- Worked in the NOC team to perform large-scale Windows Server auto-installation using PXE scripts on Active Directory-linked Altiris imaging servers and auto-install Windows desktop (XP/Vista) OS using MDT 2008/BDD 2007 Update 2 deployment best practices
- Implemented SCCM for automated software deployments and patch management, reducing manual efforts by 40%.
- Provided technical guidance to junior team members and collaborated with cross-functional teams on IT projects.
- Designing and architecting SCCM2007R2 for enterprise deployment, patch and update management solution
- Plan, design and manage Active Directory with high level support to AD Migrations - within and between AD forests
- **Networking Proficiency**: Extensive experience with Cisco networking technologies
- **Network Engineering**: Understood and configured complex network environments, leveraging knowledge of CIDR, the TCP/IP stack, and routing protocols (IGPs: OSPF, RIP; EGP: BGP) in both WAN and hybrid cloud scenarios.
- Design, implement, configure and administer Infrastructure solutions for IaaS, Saas, PaaS etc.
- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Assisted in managing IT infrastructure and ensuring optimal performance.
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers& F5 load balancers v9,
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Administration of Windows Server2003~2008R2, SQL Server2005~2008, AD, DNS, DHCP, Exchange2007, MOSS2007
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities. Promoting & improving incident & problem management, change / release management policies & procedures by SCOM

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.

- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
- **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senoir Solutions Designer Developer – Sydney Water – 15/10/2024 – Interviewed on 7 /11/ 2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senoir Solutions Designer Developer/Integration Specialist /Software Engineer /Snr DevOps Engineer /Snr Data Engineer /Java/.Net/Python Developer /AI/ML/BI/TIBCO/MuleSoft/SharePoint/D365 Developer/EDW –Japan (7.5yrs) & Aust.(24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Solutions Designer Developer/Snr DevOps/Data Engineer/TIBCO/Java/Py/AI/ML/ML/SP/D365 Developer
Responsibilities:

- Integration Specialist specializing in designing and implementing robust integration frameworks to connect modern cloud-based software solutions with legacy on-premises applications. Leveraging cloud-native integration platforms like Azure Logic Apps, AWS Lambda, and TIBCO Cloud Integration, as well as traditional integration tools such as Tibco BusinessWorks and Tibco Mashery API Gateway, to automate approval processes for aged care approved providers.
- Designed, built, and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, Azure VNets, and TIBCO Cloud Integration Platform. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all four platforms. Leveraged infrastructure-as-code (IaC) tools like Ansible for automated provisioning and configuration of networks, and utilized AWS ECS Fargate for efficient and scalable container orchestration.
- Developed digital solutions (integrations, APIs, business process automations, cloud integration) on the TIBCO Container Platform, ensuring data quality, availability, security (with appropriate authentication, authorization, non-repudiation, and data encryption), and service stability.
- Proficient in TIBCO BW CE 2.92, enabling clients to build cloud-native applications with an API-first approach, designing and implementing robust APIs that serve as the foundation for their microservices architecture. Deploy these applications to container-based PaaS platforms such as Cloud Foundry™, Kubernetes, and OpenShift, or to similar Docker-supported cloud environments, leveraging the power of containerization to ensure rapid deployment, scalability, and portability.
- Designed and implemented integration solutions using either Mulesoft Anypoint Exchange/DataWeave or TIBCO BusinessWorks CE 2.92 to automate data exchange between disparate systems, improving data processing speed by 25%.
- Designed and implemented automated infrastructure using TIBCO BW Container Instances or AWS ECS Fargate, demonstrating strong understanding of architecture, cloud-native integration, containerization, patterns, and software development. Successfully managed container orchestration, provisioning, scaling, and deployment, eliminating manual infrastructure management, & focusing on application development and deployment for improved efficiency and scalability.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
    - Azure: Azure DevOps, Azure Pipeline
    - AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
    - GCP: Cloud Build/Cloud Deploy
    - TIBOC: TIBCO Cloud Integration, TIBCO ActiveMatrix BW, TIBCO Cloud Mashery API, TIBCO BW CE2.92
- **Developed and deployed digital solutions (integrations, APIs, business process automations, and cloud integrations) either on the TIBCO Container Platform or on the Docker platform with Kubernetes.**
- **Develop Integration Flows**: Design flows in TIBCO BusinessWorks to act as intermediaries between systems.

- Utilize Connectors: Leverage pre-built connectors from TIBCO Cloud Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, Dynamics 365, and Power Automate custom connectors.
- **Configure Connections**: Provide access credentials, URLs, and other details for each system within the connectors.
- **Design Integration Logic**: Transform data, handle errors, & implement security measures in TIBCO ActiveMatrix BW.
- Design and implement automated business process workflows primarily using TIBCO Workflow, in conjunction with Microsoft Power Automate, Power Apps, AWS/Azure DevOps, Python/PySpark, and Azure Data Factory for automated data ETL/ELT processes. Deploy applications to the TIBCO Cloud and use TIBCO EMS to track performance.
- **Integration Solution Design with Azure DevOps CI/CD**: Design Integration Flow: Define data flow, transformations, and choose the orchestration platform (TIBCO BusinessWorks, Azure Logic Apps, or Hybrid).
- **Develop & Build**: Use Azure DevOps pipeline with source control (Git, BitBucket) to trigger builds for TIBCO BW CE, Azure IPaaS, BizTalk. Monitor and troubleshoot integration processes using ELK Stack (Elasticsearch, Logstash, and Kibana).
- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., TIBCO CLI for TIBCO BusinessWorks, ARM templates for IPaaS/BizTalk).
- **Optional Post-Deployment**: Include tasks for notifications, scripts, or additional actions after deployments.
- Automated Business Analysts and Data Analysts daily repeated tasks by developing secure,offline local AI systems (PrivateGPT-X, Ollama-L, Autogen Studio 3.0, etc.) for data analysis. Leveraged LangChain to create versatile AI agents capable of data summarization, classification, and prediction. Utilized Pydantic for datavalidation to ensure data quality and consistency. These agents assisted Business Analysts and Data Analysts in accelerating data exploration, generating insights, and automating routine tasks.
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- Implemented a Q-Learning-powered intrusion detection system that reduced false positive rates by 30% and detected 95% of simulated cyber attacks within 24 hours. Deployed OpenAI's ChatGPT o1 model to assess the security implications of proposed actions before proceeding, ensuring that decisions align with established security policies and best practices.
- Technical support for aged care business applications covering SharePoint 2019/2022/Dynamics 365, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Solutions Designer Developer/Snr DevOps/Data Engineer/TIBCO/Java/AI/SP/BI/D365 Developer
Responsibilities:

- Integration Solutions Architect specializing in designing frameworks to integrate client's modern cloud-based software solutions with legacy applications (SAP BusinessObjects, SAP FinOps, and SAP S/4HANA), utilizing cloud-native integration with Azure Functions, AWS Step Functions, and Tibco Cloud Integration, Tibco BusinessWorks, and Tibco Mashery API Gateway for RBA financial services workflow automation.
- Designed, built, and secured a hybrid cloud architecture connecting AWS VPCs, GCP Cloud VPCs, Azure VNets, and TIBCO Cloud Integration Platform. Established encrypted tunnels for data transfer using Cloud VPN Gateways across all four platforms. Leveraged infrastructure-as-code (IaC) tools like Ansible for automated provisioning and configuration of networks, and utilized **TIBCO Spotfire** for data visualization and analytics.
- Designed and developed applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using TIBCO BusinessWorks, TIBCO EMS, Java8/9/J2EE, or IBPy (Python).
- Full-stack developer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, TIBCO BusinessWorks, TIBCO EMS, and Azure cloud. Skilled in developing and consuming enterprise APIs using Azure API Gateway, REST API, or TIBCO API Gateway, with expertise in XML constructs, schemas, SOAP, and REST protocols.
- Proficient in creating DevOps pipelines & related frameworks, such as Agile, Lean, ITIL4, & DevSecOps, using tools like Git, Jenkins, Docker, Kubernetes, & Ansible, while leading solution design discussions at various levels within an organization.
- Leveraged either Mulesoft Anypoint Platform or TIBCO Cloud Integration's hybrid deployment capabilities to create flexible APIs that seamlessly connected GCP/AWS/Azure API platforms through their API Gateways and Azure IPaaS environments. By adopting a 'write-once, deploy-anywhere' approach, we eliminated the need for complex infrastructure management tools like Terraform, resulting in increased efficiency and reduced development time.
- Automated data workflows between diverse systems using Mulesoft Anypoint Exchange/DataWeave or TIBCO BW CE 7.x.
- Skilled in integration technologies, including TIBCO BW CE, TIBCO Messaging (EMS and Kafka),Azure, and cloud-native microservices, with expertise in BPM, solution design, development, deployment, SOA governance, and API management.
- Designed and implemented robust, scalable infrastructure solutions using container orchestration platforms like TIBCO BusinessWorks Container Instances or AWS ECS Fargate. Successfully managed container provisioning, scaling, and deployment, ensuring high availability and performance. Contributed to improving operational efficiency and reducing manual overhead while maintaining strict security and compliance standards.
- Design, develop, and maintain automated data pipelines and API orchestration using either:
  - **TIBCO Cloud Integration** or **TIBCO Data Virtualization** for data processing, integration, and API management, along with **TIBCO Cloud Enterprise** for deployment and orchestration.
  - **Azure Synapse Analytics** with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  - **AWS Glue** for data processing and integration with other AWS services to load data into target Aurora databases.

- **GCP Cloud Dataflow** or **Cloud Data Fusion** for data processing and integration, and Google BigQuery for data warehousing and analytics.

- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Implemented scalable digital solutions, including integrations, APIs, business process automations, and cloud integrations, utilizing cutting-edge technologies like TIBCO Container Platform or Docker with Kubernetes.
- **Integration Solution Design with Azure DevOps CI/CD**: Design Integration Flow: Define data flow, transformations, and choose the orchestration platform (TIBCO BusinessWorks, Azure Logic Apps, or Hybrid).
- **Develop & Build**: Use Azure DevOps pipeline with source control (Git, BitBucket) to trigger builds for TIBCO BW CE, Azure IPaaS, BizTalk. Monitor and troubleshoot integration processes using ELK Stack.
- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., TIBCO CLI for TIBCO BusinessWorks, ARM templates for IPaaS/BizTalk).
- **Optional Post-Deployment**: Include tasks for notifications, scripts, or additional actions after deployments.
- Automated frequent onboarding and off boarding of Privileged Account Access Management using CyberArk Privileged Session Manager (PSM v10.8). Implemented a proactive security monitoring solution powered by an AI-driven system (e.g., Llama 3.1 and Reflection) that continuously monitors for suspicious activity, triggers alerts for the Security team, and provides 24x7 proactive detection and response to security threats. The system is configured with multiple autonomous agents to ensure comprehensive coverage, including devices brought by employees (BYOD/BYOC). The AI is trained on a large dataset of normal and anomalous network traffic to enable accurate detection of security threats.
- Immersed experience adopting ISO27001, ACC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Migrated IBM TM1 OLAP DB, PostgreSQL and Oracle databases to Azure SQL through Power BI Pro Get Data connectors.
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Solutions Designer Developer/Snr DevOps/Data Engineer/TIBCO/Java/Py/ML/SP/BI/D365/K2 Developer
Responsibilities:

- Integration Solutions Specialist focused on designing frameworks to integrate client's modern cloud-based software solutions with legacy applications (SAP HR, SAP FinOps, and SAP S/4HANA), utilizing cloud-native integration with Azure Functions, Azure Logic Apps, AWS API, and Tibco Cloud Integration, Tibco BusinessWorks, and Tibco Mashery API Gateway for streamlining student enrollment processes.
- Developed robust integration solutions using Mulesoft Anypoint Exchange/DataWeave or TIBCO BusinessWorks to streamline data exchange between various systems, boosting data processing efficiency by 25%.
- Implemented automated infrastructure solutions using TIBCO BusinessWorks Container Instances or Azure App Service for container orchestration. Effectively managed container provisioning, scaling, and deployment, streamlining operations and reducing manual overhead. Focused on application development and deployment, boosting efficiency and scalability.
- Skilled Full-Stack Engineer leveraging React/TypeScript for frontend, Java/APIs for backend development and expertise in AWS cloud. Proficient in integration tools like MuleSoft and J2EE for seamless data flow using RESTful web services.
- Designed and implemented automated infrastructure solutions using TIBCO BusinessWorks Container Instances or Azure Kubernetes Service(AKS). Successfully managed container orchestration, provisioning, scaling, and deployment, eliminating the need for manual infrastructure management. Focused on application development and deployment, resulting in improved efficiency and scalability.
- Developed and deployed digital solutions (integrations, APIs, business process automations, and cloud integrations) either on the TIBCO Container Platform or on the Docker platform with Kubernetes.
- Python or GoLang programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, Azure Import/Export and Azure SQL.
- Overseeing the design and implementation of knowledge transfer programs to equip operational support teams with the necessary skills and expertise to maintain the solution's reliability and minimize operational risks.
- Develop statistical models for ad reporting using R programming with Core R packages, modeling packages of Tidyverse ; Reporting and Visualization packages of report, R-Shiny, ggplot2, Tableau, Power BI, Microstrategy, SAP Business Objects.
- Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
- Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
- Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins

- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
- Access Tableau data from R with `reticulate`- Tableau Server Client (TSC), create interactive visuals like Tableau using `r2d3`, and generate Tableau-compatible charts with `plotly`.
- Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Solutions Designer Developer/DevOps Engineer/Software Test Engineer/Cloud/Linux Administrator
Responsibilities:

- Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Contributed to the design and implementation of integration solutions, meeting both business and technical requirements.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented robust DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Systems Integration Developer/DevOps Engineer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Led the development, implementation, and maintenance of integrations between diverse systems and applications.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.

- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins, BitBucket, and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Systems Integration Developer /DevOps Engineer/Senior Systems Engineer /DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, TFS) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.

- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET Core (Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, Mulesoft Studio 7 Mule 4, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, TIBCO Cloud Integration, TIBCO BW CE 6.x ~ CE2.92, Tibco Mashery API Gateway, Mulesoft Anypoint Exchange/DataWeave, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum.
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior D365 & Power Platform Consultant – Fully remote to ACT – The Acqurium – 6/10/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior D365 & Power Platform Consultant/Software Engineer /Senior DevOps/Data Engineer/Clouds/Linux Administrator/ Java/.NET/Python/AI Automation/DL/ML/BI/SharePoint/D365 Developer /EDW/DBA-Japan(7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior D365 & Power Platform Consultant /Snr DevOps/Data Engineer/.NET/Python/AI/ML/SP/D365 Developer
Responsibilities:

- Design, develop, and implement D365 and Power Platform solutions using Microsoft Power Platform (Power Apps, Power Automate, Power BI, AI Chatbot etc.) for a variety of government clients.(Aged Care, Department of Health etc.)
- Developed custom components using PCF Controls to build model-driven apps & reusable APIs, integrated into Dynamics 365 & Power Apps canvases. Automated data retrieval from backend SQL 2022 Database and SharePoint 2022 Document Lists to display frontend Dynamics 365 activity records, enhancing business workflow processes and customer services.

- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 & its latest 2024 Release Wave 1 enhancements to develop Dynamics 365 applications, using customization & extension components to streamline data flows and integrate with SAP HR, SAP FinOps, S/4 HANA systems. Liquid for templating & rendering dynamic content within web applications.
- Work closely with clients to understand their business needs and translate them into effective technical solutions.
- Design and develop custom Dynamics 365 customizations and extensions to streamline business processes, using Common Data Service, Power Automate, AI Chatbot and TIBCO ActiveMatrix BusinessWorks™ 5.14.0
- Design, develop & customise SharePoint Online using Power Automate, SharePoint Framework (SPFx) or other M365 tools
- Manage all aspects of the project lifecycle, from initial requirements gathering to deployment and ongoing support.
- Developed Power Pages websites that seamlessly integrate and display Power BI Pro Paginated Reports, providing real-time insights into key performance indicators for Azure Synapse Analytics platform users.
- PowerBI Dashboards Development: Develop complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Provide ongoing technical expertise guidance to clients, ensuring they maximize their return on investment in MS technology
- **Automated repetitive tasks for Business Analysts and Data Analysts** by developing secure, offline local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis, replacing inefficient tools like Jira or Confluence.
- Leveraged LangChain to create versatile AI agents capable of data summarization, classification, and prediction. Utilized Pydantic for data validation to ensure data quality and consistency. These agents assisted Business Analysts and Data Analysts in accelerating data exploration, generating insights, and automating routine tasks.
- **Reduced development effort by 50% using OpenAI Canvas, OpenDevin** to eliminate the need for custom HTML, JavaScript, & canvas manipulation when customizing D365, SharePoint 2022, & Power Apps using C#.NET Core in VS2022
- **Automated CI/CD pipelines** for data and machine learning models using AWS and Azure DevOps/Pipelines, Kustomize, Kubespray, Terraform for IaC, Docker for containerized applications, Kubernetes for scalability, Ansible, and GitLab CI/CD.
- **Automated frequent staff onboarding and off boarding** of Privileged Account Access Management using CyberArk Privileged Session Manager (PSM v10.8). Implemented a proactive security monitoring solution powered by an AI-driven system (e.g., Llama 3.2 and Reflection) that continuously monitors for suspicious activity, triggers alerts for the Security team, and provides 24x7 proactive detection and response to security threats. The system is configured with multiple autonomous agents to ensure comprehensive coverage, including devices brought by employees (BYOD/BYOC). The AI is trained on a large dataset of normal and anomalous network traffic to enable accurate detection of security threats.
- Designed and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms.
- Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python with ARM templates, Terraform, or Ansible. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration in Azure, & equivalent container orchestration services in AWS and GCP (e.g., ECS Fargate, GKE).
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2022/Dynamics 365/Power BI/Power Automate & Power Apps)
- Technical support for aged care business applications covering SharePoint 2022/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- **Unleash insights from diverse data**: Applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, RNN, ANN, GAN) based on dataset type, size, and desired result interpretability.
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- **Implemented a Q-Learning-powered intrusion detection system** that reduced false positive rates by 30% and detected 95% of simulated Cyber-attacks within 24 hours. Deployed OpenAI's ChatGPT o1 model to assess the security implications of proposed actions before proceeding, ensuring that decisions align with established security policies and best practices.
- **Implemented cloud-agnostic application security for web applications and microservices** by scanning using Python, Bash, and PowerShell scripts, and integrating container scanning tools like Snyk and Aqua Security. This resulted in a 50% reduction in critical vulnerabilities, such as SQL injection, cross-site scripting (XSS), and remote code execution, ensuring compliance with industry best practices like OWASP Top 10.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr D365 & MS Flow Consultant/Snr DevOps/Data Engineer/Java/AI/ML/Power BI/D365 Developer
Responsibilities:

- Design and develop applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).
- Full-stack developer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, and Azure cloud. Skilled in developing and consuming enterprise APIs using Azure API Gateway, REST API, or Mulesoft Anypoint Platform, with experience in integrating with privileged access management solutions like CyberArk
- Lead the development of tailored Mulesoft solutions that align with business requirements for scalability, security, and efficiency. Guide teams in adhering to standards and governance frameworks, leveraging technical expertise in server-side languages (Java, Node.js, etc.) and front-end frameworks (React JS/Next JS, Mulesoft Anypoint API, etc.).

- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- Power BI Dashboards Development: Develop complex DAX formulas in Power BI to manipulate data.
- Create Power BI tiles and pin them to dashboards. Pin slicers or report pages to dashboards in Power BI Desktop/Service.
- Apply nested row context within calculations and conditional formatting with gradients.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch. Integrated M365 environments (O365, MS Teams, SharePoint 2016, Dynamics 365, Power App, Microsoft Flow) into the Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- **Designed and implemented Medallion Architecture with Data Lake in Azure:**
- **Batch Laye**r: Utilize Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Leveraged Infrastructure as Code (IaC) principles to automate the provisioning and configuration of virtual private clouds (VPCs), subnets, security groups using Terraform with Python (Boto3). Adopted ECS Fargate for automated container orchestration, enabling scalable and efficient deployment and management of containerized applications.
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana) and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2017 to develop Dynamics CRM2016 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- **Design, develop, and maintain automated data pipelines and API orchestration using either:**
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Designed and implemented a scalable cloud-based solution on Microsoft Azure utilizing serverless App Services and Functions. Leveraged Azure SQL Database for structured data and Blob Storage for unstructured data. Possess strong knowledge of Oracle and SQL, along with familiarity with newer technologies like Dataverse, Databricks, and NoSQL.
- **Implemented cloud-agnostic application security for web applications and microservices** by scanning using Python, Bash, and PowerShell scripts, and integrating container scanning tools like Snyk and Aqua Security. This resulted in a 50% reduction in critical vulnerabilities, such as SQL injection, cross-site scripting (XSS), and remote code execution, ensuring compliance with industry best practices like OWASP Top 10.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Dynamics CRM2016 & MS Flow Consultant/Snr DevOps/Data Engineer/AI/ML/SP/D365/K2 Developer
Responsibilities:

- Lead the development & support of existing solutions using Mulesoft Studio 3.10 & Mule ESB 3.10 through Agile practices.
- Software Integration Architect who successfully designed and implemented Mulesoft extensions to integrate the client's modern cloud-based HR platform with an on-site payroll system, automating employee onboarding & offboarding processes.
- Designed and implemented custom solutions using Power Apps and Microsoft Flow to enhance usability and efficiency within Dynamics 365 & SharePoint 2016 environments. Leveraged Mulesoft Anypoint API for scalable integrations and automation, while also utilizing Microsoft Flow to streamline business processes.
- Leveraged MuleSoft Anypoint Platform's hybrid deployment capabilities to design and deploy APIs that seamlessly integrate AWS Lambda functions with Azure Logic Apps, enabling real-time data synchronization between the two platforms.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
    1. Azure: Azure DevOps
    2. AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)

3. GCP: Cloud Build/Cloud Deploy

- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM 2016/Microsoft Flow)
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- Leverage SharePoint for process integration and Drupal/HTML with JavaScript for form development, along with backend SQL Server 2014 database for efficient data retrieval and customer service operations in Dynamics CRM 2016.
- Skilled in leveraging AI-powered tools to streamline development processes and enhance overall testing efficiency.
- Developed and maintained AI /ML models to enhance customer engagement and sales.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Leverage Python programming with CyberArk SDK** to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management.
- # Retrieve students registered data by: students_registered_data = retrieve_students_registered_data()
- Utilized Azure DevOps to collaborate effectively with team members and manage Power Platform projects.
- Integrated Power Platform solutions with other Azure services using Azure DevOps and Azure Pipelines.
- Ensured consistent quality & efficient delivery of Power Platform projects through using Azure DevOps & Azure Pipelines.
- Proficient in modern data analytics and engineering tools like dbt and Dataform.
- Skilled in creating impactful visualizations using Tableau and Power BI.
- Used Agile methodology to plan and execute tasks and used Git to manage code
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Employed Power BI, Tableau, or QlikView for real-time financial analysis.
- Design and development of K2 workflows, forms & reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare: Dynamics CRM2013/MS Flow Consultant/DevOps Engineer/Software Test Engineer/Clouds/Linux Admin
Responsibilities:

- Lead the development and support of existing solutions using MS Power Platform suite of tools through Agile practices.
- Developing, customizing, and implementing data integration solutions for MS Dynamics CRM2013 using KingswaySoft SSIS Integration Toolkit for Dynamics CRM v8.0. Leverage Visual Studio Online / VS Code, TFS, or Git for version control.
- Collaborate with the business to deliver scalable, secure, and efficient Power Platform solutions aligned with business needs. Mentor teams in best practices and governance to ensure solution quality.
- Work closely with CRM Consultants on providing clients with customized solutions: Requirement analysis, Gap analysis, Solution development, Data migration, Testing, Implementation & Support of Microsoft Dynamics NAV and CRM solutions. Dynamics CRM 2013 development either using KingswaySoft XrmSvcToolkit or MS Dynamics CRM Developer Toolkit.
- Demonstrated experience/skills in developing Automation frameworks, test scripts, using applications and tools such as: GitLAB, OpenDevin, Cucumber, Gattling and Selenium for comprehensive testing and performance analysis.
- Proficient in HTML, JavaScript, and Selenium WebDriver for web testing automation.
- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Collaborate with internal project teams to gather requirements and develop solutions that boost user experience and functionality (sites, sub-sites, pages, libraries, data or document management, layout, custom views, etc)
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented automated DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.

- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases.
- Multitasking with Unix/Linux servers' administration and global firewall management, including install, configure & support
- VoIP IP Phones and Video conference meeting froom setup fro meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Dynamics CRM 2013 Consultant/DevOps Engineer/Software Test Engineer/.NET/SharPt Developer
Responsibilities:

- Led the development, implementation, and maintenance of integrations between diverse systems and applications.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Lead the development and support of existing solutions using MS Power Platform suite of tools through Agile practices.
- Developing, customizing, and implementing data integration solutions for MS Dynamics CRM2011 using KingswaySoft SSIS Integration Toolkit for Dynamics CRM. Leverage Visual Studio 2012 / VS Code, TFS, or Git for version control.
- Collaborate with the business to deliver scalable, secure, and efficient Power Platform solutions aligned with business needs. Mentor teams in best practices and governance to ensure solution quality
- **Skilled in Agile QA**: Planning, coordinating, and executing tests within Aigile workflows.
- **Broad Testing Experince**: Handles functional testing, database testing, performance testing, and security testing.
- Utilized Apache Tomcat, NGINX/NGINX Plus and JBoss for application/web servers.
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery.
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA

- Led a project to replace an existing insurance web application, resulting in a significant reduction in IT costs and improved operational efficiency. The new application seamlessly integrated with Sales Force Automation (SFA) and project management tools like Jira or Confluence, streamlining workflows and enhancing collaboration.
- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Led a project to replace an existing insurance web application, resulting in a significant reduction in IT costs and improved operational efficiency. The new application seamlessly integrated with Sales Force Automation (SFA) and project management tools like Jira or Confluence, streamlining workflows and enhancing collaboration.

- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: **Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH**
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and BIML script.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate process

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Digital Power Platform Senior Developer – DCS called back by team members – 30/9/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Application Developer/Software Engineer /Senior DevOps/Data Engineer/Clouds/Linux Administrator/ Java/.NET/Python/AI Automation/DL/ML/BI/SharePoint/D365 Developer /EDW/DBA-Japan(7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Digital Power Platform Senior Developer/Snr DevOps/Data Engineer/.NET/Python/AI/ML/SP/D365 Developer
Responsibilities:

- Senior Application Developer with 5+ years of experience in designing, developing, and implementing scalable, secure, and high-performance applications using Microsoft Power Platform (Power Apps, Power Automate, Power BI, AI Chatbot etc.).
- Lead the development and support of existing solutions using MS Power Platform suite of tools through Agile practices.
- Developed custom components using PCF Controls to build model-driven apps & reusable APIs, integrated into Dynamics 365 & Power Apps canvases. Automated data retrieval from backend SQL 2022 Database and SharePoint 2022 Document Lists to display frontend Dynamics 365 activity records, enhancing business workflow processes and customer services.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 & its latest 2024 Release Wave 1 enhancements to develop Dynamics 365 applications, using customization & extension components to streamline data flows and integrate with SAP HR, SAP FinOps, S/4 HANA systems. Liquid for templating & rendering dynamic content within web applications.
- Design and develop custom Dynamics 365 customizations and extensions to streamline business processes, using Common Data Service, Power Automate, and AI Chatbot (formerly Power Virtual Agents)
- Design, develop & customise SharePoint Online using Power Automate, SharePoint Framework (SPFx) or other M365 tools
- Developed Power Pages websites that seamlessly integrate and display Power BI Pro Paginated Reports, providing real-time insights into key performance indicators for Azure Synapse Analytics platform users.
- PowerBI Dashboards Development: Develop complex DAX formulas (in Power BI or Azure DAX Studio) to manipulate data.
- Drive collaboration with our Microsoft Partner to coordinate backlog of product enhancements by adopting AI and MLOps.
- **Automated repetitive tasks for Business Analysts and Data Analysts** by developing secure, offline local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis, replacing inefficient tools like Jira or Confluence.
- Leveraged LangChain to create versatile AI agents capable of data summarization, classification, and prediction. Utilized Pydantic for data validation to ensure data quality and consistency. These agents assisted Business Analysts and Data Analysts in accelerating data exploration, generating insights, and automating routine tasks.
- **Reduced development effort by 50% by leveraging OpenDevin** to eliminate the need for custom HTML, JavaScript, and canvas manipulation when customizing Dynamics 365, SharePoint 2022, & Power Apps using C#.NET Core in VS2022.
- **Automated CI/CD pipelines** for data and machine learning models using AWS and Azure DevOps/Pipelines, Kustomize, Kubespray, Terraform for IaC, Docker for containerized applications, Kubernetes for scalability, Ansible, and GitLab CI/CD.
- **Automated frequent staff onboarding and off boarding** of Privileged Account Access Management using CyberArk Privileged Session Manager (PSM v10.8). Implemented a proactive security monitoring solution powered by an AI-driven system (e.g., Llama 3.1 and Reflection) that continuously monitors for suspicious activity, triggers alerts for the Security team, and provides 24x7 proactive detection and response to security threats. The system is configured with multiple autonomous agents to ensure comprehensive coverage, including devices brought by employees (BYOD/BYOC). The AI is trained on a large dataset of normal and anomalous network traffic to enable accurate detection of security threats.
- Designed and implemented hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python with ARM templates, Terraform, or Ansible. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration in Azure, & equivalent container orchestration services in AWS and GCP (e.g., ECS Fargate, GKE).
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2022/Dynamics 365/Power BI/Power Automate & Power Apps)
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, RNN, ANN, GAN) based on dataset type, size, and desired result interpretability.
- Design, develop test and deploy security focused and AI-Powered ML applications, deep learning (DL) models and systems
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.

- **Implemented a Q-Learning-powered intrusion detection system** that reduced false positive rates by 30% and detected 95% of simulated Cyber-attacks within 24 hours. Deployed OpenAI's ChatGPT o1 model to assess the security implications of proposed actions before proceeding, ensuring that decisions align with established security policies and best practices.
- **Implemented cloud-agnostic application security for web applications and microservices** by scanning using Python, Bash, and PowerShell scripts, and integrating container scanning tools like Snyk and Aqua Security. This resulted in a 50% reduction in critical vulnerabilities, such as SQL injection, cross-site scripting (XSS), and remote code execution, ensuring compliance with industry best practices like OWASP Top 10.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Senior Application Developer(Java)/Snr DevOps/Data Engineer/Python/ML/SP/Power BI/D365 Developer
Responsibilities:

- Design and develop applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).
- Full-stack developer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, and Azure cloud. Skilled in developing and consuming enterprise APIs using Azure API Gateway, REST API, or Mulesoft Anypoint Platform, with experience in integrating with privileged access management solutions like CyberArk
- Lead the development of tailored Mulesoft solutions that align with business requirements for scalability, security, and efficiency. Guide teams in adhering to standards and governance frameworks, leveraging technical expertise in server-side languages (Java, Node.js, etc.) and front-end frameworks (React JS/Next JS, Mulesoft Anypoint API, etc.).
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- PowerBI Paginated reporting Development using Report Builder which directly connects to various data sources (SQL Server, Azure Synapse, etc.) within the report itself. No underlying data model or semantic layer.
- Power BI Dashboards Development: Develop complex DAX formulas in Power BI to manipulate data.
- Create Power BI tiles and pin them to dashboards. Pin slicers or report pages to dashboards in Power BI Desktop/Service.
- Apply nested row context within calculations and conditional formatting with gradients.
- Built secure and scalable AWS Landing Zones using IaC with AWS CloudFormation or Terraform. Used AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch. Integrated M365 environments (O365, MS Teams, SharePoint 2016, Dynamics 365, Power App, Microsoft Flow) into the Landing Zones for a comprehensive and secure platform for Microsoft workloads.
- **Designed and implemented Medallion Architecture with Data Lake in Azure:**
- **Batch Laye**r: Utilize Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- **Serving Layer**: Implement Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer**: Leverage Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Leveraged Infrastructure as Code (IaC) principles to automate the provisioning and configuration of virtual private clouds (VPCs), subnets, security groups using Terraform with Python (Boto3). Adopted ECS Fargate for automated container orchestration, enabling scalable and efficient deployment and management of containerized applications.
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2017 to develop Dynamics CRM2016 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- **Design, develop, and maintain automated data pipelines and API orchestration using either:**
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across hybrid clouds & on-premises deployments without downtime
- Designed and implemented a scalable cloud-based solution on Microsoft Azure utilizing serverless App Services and Functions. Leveraged Azure SQL Database for structured data and Blob Storage for unstructured data. Possess strong knowledge of Oracle and SQL, along with familiarity with newer technologies like Dataverse, Databricks, and NoSQL.

- **Implemented cloud-agnostic application security for web applications and microservices** by scanning using Python, Bash, and PowerShell scripts, and integrating container scanning tools like Snyk and Aqua Security. This resulted in a 50% reduction in critical vulnerabilities, such as SQL injection, cross-site scripting (XSS), and remote code execution, ensuring compliance with industry best practices like OWASP Top 10.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Senior Application Developer(Java/.NET)/Snr DevOps/Data Engineer /AI/ML/SP/D365 /K2 Developer
Responsibilities:

- Software Integration Architect who successfully designed and implemented Mulesoft extensions to integrate the client's modern cloud-based HR platform with an on-site payroll system, automating employee onboarding & offboarding processes.
- Designed and implemented custom solutions using Power Apps and Microsoft Flow to enhance usability and efficiency within Dynamics 365 & SharePoint 2016 environments. Leveraged Mulesoft Anypoint API for scalable integrations and automation, while also utilizing Microsoft Flow to streamline business processes.
- Leveraged MuleSoft Anypoint Platform's hybrid deployment capabilities to design and deploy APIs that seamlessly integrate AWS Lambda functions with Azure Logic Apps, enabling real-time data synchronization between the two platforms.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
    1. Azure: Azure DevOps
    2. AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
    3. GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2016/Dynamics CRM 2016/Microsoft Flow)
- Lead the development & support of existing solutions using Mulesoft Studio 3.10 & Mule ESB 3.10 through Agile practices.
- Developed custom components using Mulesoft Anypoint API to create interactive and reusable user interface elements that can be integrated into Power Apps canvases or Microsoft Flow workflows.
- Utilized KingswaySoft SSIS Integration Toolkit for MS Dynamics 365 and MS Dynamics 365 developer toolkit in VS 2015 to develop Dynamics CRM2013 applications, using customization & extension components to Streamlin data flows & integrate with SAP FinOps, S/4 HANA systems & using Liquid for templating and rendering dynamic content within web applications.
- Leverage SharePoint for process integration and Drupal/HTML with JavaScript for form development, along with backend SQL Server 2014 database for efficient data retrieval and customer service operations in Dynamics CRM 2016.
- Skilled in leveraging AI-powered tools to streamline development processes and enhance overall testing efficiency.
- Developed and maintained AI /ML models to enhance customer engagement and sales.
- Drive collaboration with our Microsoft Partner to coordinate backlog of product enhancements by adopting AI and MLOps.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Leverage Python programming with CyberArk SDK** to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management.
- # Retrieve students registered data by: students_registered_data = retrieve_students_registered_data()
- Utilized Azure DevOps to collaborate effectively with team members and manage Power Platform projects.
- Integrated Power Platform solutions with other Azure services using Azure DevOps and Azure Pipelines.
- Ensured consistent quality & efficient delivery of Power Platform projects through using Azure DevOps & Azure Pipelines.
- Proficient in modern data analytics and engineering tools like dbt and Dataform.
- Skilled in creating impactful visualizations using Tableau and Power BI.
- Used Agile methodology to plan and execute tasks and used Git to manage code
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Employed Power BI, Tableau, or QlikView for real-time financial analysis.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**

Healthcare Sector: Application Developer/DevOps Engineer/Software Test Engineer/Clouds/Linux Administrator/DBA
Responsibilities:

- Lead the development and support of existing solutions using MS Power Platform suite of tools through Agile practices.
- Developing, customizing, and implementing data integration solutions for MS Dynamics CRM2013 using KingswaySoft SSIS Integration Toolkit for Dynamics CRM v8.0. Leverage Visual Studio Online / VS Code, TFS, or Git for version control.
- Collaborate with the business to deliver scalable, secure, and efficient Power Platform solutions aligned with business needs. Mentor teams in best practices and governance to ensure solution quality
- Work closely with CRM Consultants on providing Clients with customised solutions: Requirement Analysis, Gap Analysis, Solution Development, Data Migration, Testing, Implementation and Support of Microsoft Dynamics NAV and CRM solutions
- Dynamics CRM2013 development either using Using KingswaySoft XrmSvcToolkit or MS Dynamics CRM Developer Toolkit
- Demonstrated experience/skills in developing Automation frameworks, test scripts, using applications and tools such as: GitLAB, OpenDevin, Cucumber, Gattling and Selenium for comprehensive testing and performance analysis.
- Proficient in HTML, JavaScript, and Selenium WebDriver for web testing automation.
- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Collaborate with internal project teams to gather requirements and develop solutions that boost user experience and functionality (sites, sub-sites, pages, libraries, data or document management, layout, custom views, etc)
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented automated DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Application Developer/DevOps Engineer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Led the development, implementation, and maintenance of integrations between diverse systems and applications.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Lead the development and support of existing solutions using MS Power Platform suite of tools through Agile practices.
- Developing, customizing, and implementing data integration solutions for MS Dynamics CRM2011 using KingswaySoft SSIS Integration Toolkit for Dynamics CRM. Leverage Visual Studio 2012 / VS Code, TFS, or Git for version control.
- Collaborate with the business to deliver scalable, secure, and efficient Power Platform solutions aligned with business needs. Mentor teams in best practices and governance to ensure solution quality
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Led a project to replace an existing insurance web application, resulting in a significant reduction in IT costs and improved operational efficiency. The new application seamlessly integrated with Sales Force Automation (SFA) and project management tools like Jira or Confluence, streamlining workflows and enhancing collaboration.
- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate process

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc

- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Cloud Operational Support Engineer – NSWPF v2 – Olypic Park – Parra – 25/9/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cloud Operational Support Engineer/Integration Solution Architect/Snr DevOps Engineer/Snr Data Engineer /Java/.Net /Python /AI/ML/BI/MuleSoft/BizTalk/SharePoint/D365 Developer/EDW –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Cloud Operational Support Engineer/Snr DevOps/Data Engineer/Java/Py/AI/ML/SP/D365 Developer/EDW
Responsibilities:

- Integration Solution Architect specialized in designing and implementing complex integrations between modern cloud-based software (Azure, AWS, GCP, Mulesoft) and legacy on-site applications using IPasS and Biztalk.
- Design and implement application and platform solutions using Mulesoft and Azure DevOps/AWS DevOps/GCP Cloud Build/Cloud Deploy, ensuring alignment with the latest architecture standards to support organizational systems and services in compliance with ACQSC DTI Architecture, connected to AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Configured VPCs, subnets, security groups, and network gateways to establish encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms.
- Developed and implemented CI/CD pipelines using Azure Pipelines/AWS Code Pipeline/GCP Cloud Build, GitHub Actions and YAML scripting to automate the deployment of Mulesoft applications, streamlining development & improving efficiency.
- Proficient in integrating Terraform with Mulesoft Anypoint Platform to automate API deployments and infrastructure management. Developed & implemented CI/CD pipelines using Terraform and Mulesoft to streamline application lifecycle.
- Leveraged Terraform to provision and manage cloud infrastructure resources (Azure, AWS) for Mulesoft deployments.
- Identify and report on performance issues in assigned systems and applications to assist owners in creating business cases for necessary upgrades or decisions aimed at reducing issues and eliminating adverse impacts on service delivery.
- Configure and manage hybrid cloud environments (Azure, AWS, and GCP) across development, testing, and production stages using infrastructure as code (IaC) practices. Employ tools like Ansible, CloudFormation, boto 3 and Pulumi to automate the provisioning and management of resources, ensuring consistency, scalability, and reduced manual effort.
- Designed and implemented high-performance integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform. Leveraged Anypoint Exchange and DataWeave to automate data exchange between disparate systems, resulting in a 25% improvement in data processing speed and enhanced operational efficiency.

- Implemented robust version control practices using Git (GitHub Actions, GitHub Enterprise), including branching strategies, code reviews, and merge requests. Leveraged configuration management tools to consolidate settings from diverse sources, streamlining infrastructure management and ensuring consistent provisioning.
- Utilized Azure Monitor/AWS CloudWatch to monitor, troubleshoot, and optimize Kubernetes clusters for peak performance and reliability, implement security best practices, network policies, secrets management and develop automation scripts to manage infrastructure components, ensuring consistent environments across clusters.
- Coordinate with project, testing, and operations teams to ensure seamless deployments and deliverables.
- **Integration Solution Design with Azure DevOps CI/CD**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
- Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline. The pipeline will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
- Leverage Azure Monitor or AWS CloudWatch to track pipeline performance metrics, use Application Insights or AWS X-Ray to pinpoint bottlenecks within individual stages, and employ Amazon DevOps Services or Azure DevOps Services to implement continuous optimization strategies, ensuring efficient and reliable deployments.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- Implemented a Q-Learning-powered intrusion detection system that reduced false positive rates by 30% and detected 95% of simulated cyber attacks within 24 hours. Deployed OpenAI's ChatGPT o1 model to assess the security implications of proposed actions before proceeding, ensuring that decisions align with established security policies and best practices.
- Automated frequent onboarding and off boarding of Privileged Account Access Management using CyberArk Privileged Session Manager (PSM v10.8). Implemented a proactive security monitoring solution powered by an AI-driven system (e.g., Llama 3.1 and Reflection) that continuously monitors for suspicious activity, triggers alerts for the Security team, and provides 24x7 proactive detection and response to security threats. The system is configured with multiple autonomous agents to ensure comprehensive coverage, including devices brought by employees (BYOD/BYOC). The AI is trained on a large dataset of normal and anomalous network traffic to enable accurate detection of security threats.
- Designed and implemented secure VPCs using Azure Blueprints to support the integration of Windows-based cloud workloads within M365 environments (O365/MS Teams/SharePoint 2019/2022/Dynamics 365/Power BI/Power Automate, etc.). Leveraged advanced networking and security best practices to ensure data confidentiality, integrity, and availability
- Ensured effective access to data and information assets while maintaining security, privacy, confidentiality, and innovation by building a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT). Implement CI/CD pipelines (using tools like Airflow or Luigi) to guarantee resilience against outages.
- Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Cloud Operational Support Engineer/Snr DevOps/Data Engineer/Java/Py/AI/ML/ML/SP/BI Developer
Responsibilities:

- Software Integration Solution Architect specializing in designing frameworks to seamlessly integrate clients' modern cloud-based software solutions with legacy on-site applications. Oversee the integration of system solutions across diverse platforms and technologies within various environments, including the support of DevOps and cloud platforms.
- Designed and implemented robust application and platform solutions using Mulesoft and cloud-native CI/CD tools (e.g., Azure DevOps, AWS DevOps, GCP Cloud Build/Cloud Deploy). Ensured alignment with RBA architectural standards to support critical organizational systems and services, integrating across AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Streamlined API deployments and infrastructure management by integrating Terraform with Mulesoft Anypoint Platform.
- Implemented robust CI/CD piplelines utilizing Terraform and Mulesoft to automate application delivery and reduce manual intervention. Leveraged Terraform to efficiently provision and manage cloud infrastructure resources (Azure, AWS) for Mulesoft deployments.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, and Dynamics 365 CRM.
- **Configure Connections**: Provide access credentials, URLs, and other details for each system within the connectors.
- **Design Integration Logic**: Transform data, handle errors, and implement security measures within the Mule flow.
- **Deploy and Monitor**: Deploy application to Mule runtime environment & use Mulesoft's monitor tools to track performance.
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Led the API lifecycle management process for all internal APIs, ensuring consistency, security, and version control.
- **Integration Solution Design with Azure DevOps CI/CD**: **Design Integration Flow**: Define data flow, transformations, and choose the orchestration platform (MuleSoft, Azure Logic Apps, or Hybrid).
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (MuleSoft, Azure IPaaS, BizTalk).

- **Deployment Stages**: Configure separate deployment stages for each platform using appropriate tasks (e.g., Maven/Anypoint CLI for MuleSoft, ARM templates for IPaaS/BizTalk).
- **Optional Post-Deployment**: Include tasks for notifications, scripts, or additional actions after deployments.
- Managed and implemented large-scale or complex deployments and changes to the organization's information systems, ensuring alignment with operational requirements and successful solution delivery.
- Built secure and scalable AWS Landing Zones using Infrastructure as Code (IaC) with AWS CloudFormation templates or Terraform. Leveraged AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch. Oversee and manage installations to ensure timely, cost-effective, and high-quality completion, thereby minimizing operational and project risks and controlling costs.
- Implementing Integrated **DevSecOps** practices with IaC security principles, utilizing ARM templates, Azure PowerShell, Azure CLI, and AWS CLI to automate security configurations and ensure a consistent security posture across deployments.
- Evaluate and document the performance of assigned systems and applications to assist owners in creating business cases for necessary upgrades or decisions aimed at reducing issues and eliminating adverse impacts on service delivery.
- Successfully implemented innovative network solutions to address RBA security challenges. Configured VPCs, subnets, and security groups to establish highly secure, encrypted data transfer tunnels across multiple cloud environments.
- Developed and implemented robust CI/CD pipelines using Azure Pipelines, AWS Code Pipeline, GCP Cloud Build, or GitHub Actions to streamline Mulesoft application deployment. Leveraged YAML scripting and automation to enhance RBA development efficiency and reduce risk.
- Designing, implementing, and managing on-premises Kubernetes clusters to ensure high availability and performance, deploying and managing containerized applications on Kubernetes using internal tooling (e.g., OpenShift, Rancher, K3s), assisting in the design and implementation of CI/CD pipelines tailored to on-premises environments.
- **CloudWatch Observability Expertise:** Implemented comprehensive CloudWatch observability solutions, including metrics, logs, and dashboards. Leveraged CloudWatch Insights for advanced log analysis and troubleshooting.
- Design, develop, and maintain automated data pipelines and API orchestration using either:
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- Develop, review, and maintain comprehensive application and platform support documentation to assist others in restoring services and minimize the impact of unplanned outages. Proactively identify and escalate performance issues with internal and external providers to minimize the adverse impact of system availability on business continuity.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Cloud Operational Support Engineer/Snr DevOps/Data Engineer/Java/Python/ML/SPoint/K2 Developer
Responsibilities:

- Cloud Integration Solution Architect with expertise in designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow
- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments.
- Used BizTalk Server for legacy system integration by defining data structures (schemas), mapping data formats, utilizing adapters for communication protocols (e.g., FILE, REST), & orchestrating message flows (optional) for complex scenarios.
- Develop and implement comprehensive change management plans for applications and platforms, ensuring that changes are executed strategically to meet business objectives while minimizing risks to the institution and its data.
- Demonstrated expertise in integrating Terraform and Mulesoft to automate complex API deployments and infrastructure provisioning.Successfully designed and implemented CI/CD pipelines using Terraform and Mulesoft, optimizing application lifecycle management.Applied Terraform to provision and manage cloud infrastructure resources (Azure, AWS) with a focus on efficiency, security, and scalability.
- Provided 3rd level technical support, consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with standards and requirements.
- Collaborate with cross-functional teams to define integration requirements & ensure solutions align with business objectives.
- Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
- Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics

- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python
- Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
- Successfully installed, configured, and tested new and existing hardware, software, and WAN/LAN technologies.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
- Design and development of K2 workflows, forms and reports on a SharePoint platform & its backend SQL Server database.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Sector: Cloud Operational Support Engineer/DevOps Engineer/Software Test Engineer/Linux Administrator
Responsibilities:

- Skilled Cloud Integration Architect with expertise in designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python with ARM templates, Terraform, or Ansible. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration in Azure, & equivalent container orchestration services in AWS and GCP (e.g., ECS Fargate, GKE).
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Proficient in Linux system administration, including kernel optimization (sysctl, vmstat), performance tuning (iostat, sar), and security hardening (iptables, fail2ban). Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all EML's data pipelines.
- Developed, reviewed, and maintained cloud support documentation and best practices
- Escalated and resolved performance issues with cloud providers and vendors
- **Change Management**: Planned and managed change implementation for cloud systems and applications.
- Managed the integration of system solutions across various platforms and technologies, supporting DevOps and Cloud.
- Optimized the pipeline for performance, cost, and efficiency by applying best practices such as resource allocation, scaling, health checks, and logging in addition to partitioning, indexing, and optimizing the ADF pipeline structure.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Streamlined DevOps processes by implementing robust version control practices with Git, including branching strategies, code reviews, and merge requests. Utilized configuration management tools to consolidate settings and streamline infrastructure management, ensuring consistent provisioning.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Created a Synapse Analytics workspace in Azure Portal to provision SQL Data Warehouse for transformed data.
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.

Oct. 2011 – Sep. 2012 Sunshine Coast Council(SCC), QLD Australia
Government Council: Cloud Operational Support Engineer/DevOps Engineer/Software Test Engineer/SharPoint Developer
Responsibilities:

- Proven track record of designing and implementing seamless integrations between modern cloud-based software solutions and legacy on-site applications, ensuring optimal system interoperability and data flow.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.

- Streamlined cloud operations by automating resource provisioning and management using infrastructure as code (IaC). Proficient in configuring and managing hybrid cloud environments (Azure, AWS, GCP) across development, testing, and production, leveraging tools like Ansible, CloudFormation, boto3, and Pulumi.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Leveraged Intune Mobile Device Management (MDM) and Mobile Application Management (MAM) to effectively manage and secure mobile devices and applications within the Microsoft Endpoint Manager admin center.
- **Software deployment in a co-management environment**: Use Intune to deploy cloud-based applications (e.g., Office 365) and SCCM to deploy on-premises applications (e.g., legacy line-of-business apps).
- **Device configuration**: Use Intune for mobile device management and SCCM for on-premises device configuration.
- Intune Engineer Key Achievements (Hybrid Join environment)
- Implemented and managed Intune configurations and policies for both domain-joined and Azure AD joined devices.
- Use Endpoint analytics to help identify the policies or the hardware issues that slow down the devices and improve end user experience. Deploy Microsoft 365 apps to the devices using Microsoft Intune.
- Use Windows autopatch with Microsoft Intune to manage the patching and keep software up to date.
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Database backup (SQL, MongoDB, DynamoDB) via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Systems Integration Developer /DevOps Engineer/Senior Systems Engineer /DWH
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Administered and monitored Unix/Linux-based systems, ensuring smooth operation of critical applications.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Integration Developer /Senior Systems Engineer /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Installed and configured a WebLogic domain with integrated Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Executed ELT/ETL processes & advanced analytics using R & Python with various libraries for data enrichment & analysis..
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Systems Integration Developer /Data Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.

- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- **Develop and Build**: Use Azure DevOps pipeline with source code management (e.g., Git) to trigger builds for each integration platform (Azure IPaaS, BizTalk).
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Developed and maintained integration solutions using BizTalk Server to connect legacy systems with modern applications, enabling seamless data flow and modernization efforts.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
- **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Automation Developer – Seems for Services Aust? – Just Digital People and Private Advertiser – 17/9/2024 both no resp.
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Automation Developer /Software Engineer /Senior DevOps Engineer/Senior Data Engineer/Clouds/Linux Administrator/ Java/.NET/Python/AI/DL/ML/BI/SharePoint/D365 Developer /EDW/DBA-Japan(7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Automation Developer/Snr DevOps Engineer/Snr Data Engineer/.NET/Python/AI/ML/SP/D365 Developer/EDW
Responsibilities:

- Designed and implemented a hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms.
- Leveraged Infrastructure as Code (IaC) principles to automate the provisioning and configuration of virtual private clouds (VPCs), subnets, security groups using Terraform with Python (Boto3). Adopted ECS Fargate for automated container orchestration, enabling scalable and efficient deployment and management of containerized applications.
- Design and implement automated business process workflows using Microsoft Power Automate, Power Apps, AWS/Azure DevOps, and Python/PySpark for automated data ETL/ELT process via Azure Data Factory.
- Leverage SharePoint for process integration and Drupal/HTML with JavaScript for form development, along with a mirrored backend SQL Server 2022 database for efficient data retrieval and customer service operations in Dynamics 365.
- Automated Business Analysts and Data Analysts daily repeated tasks by developing secure, offline local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. Leveraged LangChain to create versatile AI agents capable of data summarization, classification, and prediction. Utilized Pydantic for data validation to ensure data quality and consistency. These agents assisted Business Analysts and Data Analysts in accelerating data exploration, generating insights, and automating routine tasks.
- Demonstrated experience/skills in developing Automation frameworks, test scripts, using applications and tools such as: GitLAB, OpenDevin, Cucumber, Gattling and Selenium for comprehensive testing and performance analysis.
- **Project Setup with proposed automation framework**: Create a GitLab project to store the automation-related files.
- **Define Test Cases in OpenDevin**: Create a new project in OpenDevin.
- **Visually design the test case flow**: Use OpenDevin's intuitive interface to map out the desired interactions with web elements, data inputs, and expected outcomes.
- Proficient in HTML, JavaScript, and Selenium WebDriver for web testing automation. Leverage AI tools to streamline initial code generation, ensuring efficient and robust test solutions.
- **Integrate with Selenium and Cucumber**:
- **Create a Selenium WebDriver instance**: Instantiate a WebDriver object to interact with the web browser.
- **Load the HTML page**: Use the WebDriver to load the HTML page created by OpenDevin.
- **Execute JavaScript code**: Employ the WebDriver's executeScript() method to execute the JavaScript code within the Selenium context.
- Utilized Harrison AI platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- **Designed and maintained automated AI/ML pipelines using platform-specific tools**:
  - Azure: Azure Machine Learning (AML), Azure OpenAI Service
  - AWS: AWS SageMaker
  - GCP: GCP Vertex AI
- Develop medical-tailored LLMs (BERT, GPT-4o, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- Utilized organoid intelligence (OI) to accelerate the development of brain organoids for neurological disease research.
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, RNN, ANN, GAN) based on dataset type, size, and desired result interpretability.
- Design, develop test and deploy security focused and AI-Powered ML applications, deep learning (DL) models and systems
- **Cloud Security Automation with AI**: Leveraged AI tools to detect & prevent threats, manage vulnerabilities, & secure data.
- Implemented a Q-Learning-powered intrusion detection system that reduced false positive rates by 30% and detected 95% of simulated cyber attacks within 24 hours. Deployed OpenAI's ChatGPT o1 model to assess the security implications of proposed actions before proceeding, ensuring that decisions align with established security policies and best practices.
- Automated frequent onboarding and off boarding of Privileged Account Access Management using CyberArk Privileged Session Manager (PSM v10.8). Implemented a proactive security monitoring solution powered by an AI-driven system (e.g., Llama 3.1 and Reflection) that continuously monitors for suspicious activity, triggers alerts for the Security team, and provides 24x7 proactive detection and response to security threats. The system is configured with multiple autonomous agents to ensure comprehensive coverage, including devices brought by employees (BYOD/BYOC). The AI is trained on a large dataset of normal and anomalous network traffic to enable accurate detection of security threats.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Automation Developer/Snr DevOps/Data Engineer/Java/Python/AI/ML /SP/Power BI Developer /EDW
Responsibilities:

- Integration Solution Architect specializing in designing frameworks to integrate client's modern cloud-based software solutions with legacy on-site applications.
- Utilized MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between GCP infrastructure (including Apigee) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- Automated the provisioning of microservices applications deployed using AWS ECS/EKS, Azure AKS & GCP's GKE to manage complex containerized environments. Successfully navigated and enhanced a 20-service ecosystem of evolving APIs and microservices using Amazon API Gateway, AWS AppSync, AWS Lambda, GCP Cloud Endpoints, Cloud Functions, or Mulesoft Anypoint Platform. Improved API performance by 20% and increased scalability by 50%.
- Built secure and scalable AWS Landing Zones using Infrastructure as Code (IaC) with AWS CloudFormation templates or Terraform. Leveraged AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch.
- Design and lead the migration of complex applications from Rancher/Docker on EC2 to ECS Fargate, optimizing performance and cost efficiency. Used CloudWatch Contributor Insights to analyze High-Cardinality Data & Traffic Trends.
- Infrastructure Automation Specialist with expertise in Ansible for multi-platform environments (Azure, AWS, GCP and on-premises) and DSC for Windows-specific configurations. (e.g. Configuring a Windows Service as MyService.dsc and run it in PowerShell: Start-DscConfiguration -Path ".¥MyService.dsc")
- Design and develop applications similar to Interactive Brokers (IB) OMS Trade WorkStation (TWS) for prime brokerage and trading in financial markets using Java8/9/J2EE or IBPy (Python).
- Full-stack developer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, and Azure cloud. Skilled in developing and consuming enterprise APIs using Azure API Gateway, REST API, or Mulesoft Anypoint Platform, with experience in integrating with privileged access management solutions like CyberArk
- Demonstrated proficiency in developing and implementing automated testing frameworks using industry-standard tools such as GitLab, Cucumber, Gatling, and Selenium. Skilled in leveraging OpenDevin for efficient test case design and execution.
- Possess a strong foundation in HTML, JavaScript, and Selenium WebDriver, with experience integrating these technologies to create robust and scalable test automation solutions. Proficient in utilizing AI-powered tools to streamline code generation and enhance overall testing efficiency.
- Employed IaC principles with Python to streamline network resource management using ARM, Terraform, or Ansible. Leveraged Azure Kubernetes Service (AKS) and equivalent container orchestration platforms for scalable and efficient application deployment.
- Design, develop, and maintain automated data pipelines and API orchestration using either:
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
    - Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- Work on NLP, computer vision, and recommender system projects using Python, TensorFlow, and scikit-learn.
- Utilized the AWS Deep Learning AMIs to build custom environments and workflows with MxNet as well as other frameworks including TensorFlow , PyTorch, Chainer, Keras, Caffe, Caffe2, and Microsoft Cognitive Toolkit.
- Designed and built AI-powered chatbots capable of handling complex financial queries and providing personalized recommendations to a large number of customers, resulting in a 20% reduction in customer support inquiries.
- Data Preprocessing: Clean, normalize, and transform the data to make it suitable for the AI training model.
- Feature Engineering: Select or create features that are most relevant to the problem.
- Split the Data: Divide the dataset into training and testing sets.
- Choose a Model: Select a machine learning or deep learning model based on the nature of the problem.
- Train the Model: Use the training dataset to train the chosen model.
- Model Evaluation: Assess the performance of the model using the testing dataset.
- Hyperparameter Tuning: Optimize the hyperparameters of the model for better performance.
- Deployment: If satisfied with the model, deploy it for real-world use.
- Deploy machine learning solutions for various clients using IBM's Watson Studio platform.
- Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Automation Developer/Snr DevOps/Data Engineer/Java/.NET/Python/AI/ML/SharePoint /K2 Developer
Responsibilities:

- Software Integration Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications.
- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments.
- **Designed and maintained automated CI/CD pipelines using platform-specific tools**:
  - Azure: Azure DevOps
  - AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)
  - GCP: Cloud Build/Cloud Deploy
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Skilled in designing, developing, and implementing automated testing solutions. Proficient in utilizing industry-standard tools such as GitLab, Cucumber, Gatling, and Selenium to create efficient and effective test frameworks. Possess a strong foundation in web technologies, including HTML, JavaScript, and Selenium WebDriver, and have experience integrating these tools to build robust and scalable test automation solutions.
- Skilled in leveraging AI-powered tools to streamline development processes and enhance overall testing efficiency.
- Developed and maintained AI /ML models to enhance customer engagement and sales.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Leverage Python programming with CyberArk SDK to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management.
- # Retrieve students registered data by: students_registered_data = retrieve_students_registered_data()
- **Develop custom portals** using modern frameworks like React, Angular, or Vue.js for specific departments, teams, or use cases that require highly tailored experiences. Integrate and consume these portals' enterprise APIs with CyberArk PSM infrastructure to enhance user satisfaction and productivity and offer a seamless user experience.
- Tailored effective presentations for diverse technical audiences, showcasing strong communication skills.
- Designed and built AI-powered Chatbots tools that can simulate human conversation for international students frequently asked Questions and Answers to automate customer services, and provide personalized recommendations.
- Proficient in technologies like React, GCP, FastAPI, Postgres, Kubernetes, and Next-js.
- Proficient in modern data analytics and engineering tools like dbt and Dataform.
- Skilled in creating impactful visualizations using Tableau and Power BI.
- Worked on projects involving web development (Spring Boot), data integration (Oracle Data Integrator) and business intelligence (Oracle BI Publisher)
- Used Agile methodology to plan and execute tasks and used Git to manage code
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Leveraging Infrastructure as Code tools, such as Amazon Elastic Kubernetes Service (EKS), Azure Kubernetes Service (AKS), and Google Kubernetes Engine (GKE), to define, provision, and orchestrate clusters, nodes, resources, and optimize application pipelines for efficient management and scaling.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Employed Power BI, Tableau, or QlikView for real-time financial analysis.
- Either using BMC Control-M v9.0 to manage, schedule multiple data transformation jobs or using SSIS with VS or SSDT
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: Automation/Integration Developer/DevOps Engineer/Software Test Engineer//Linux Administrator/DBA
Responsibilities:

- Designed and implemented hybrid cloud infrastructure architectures integrating AWS and Azure. Utilize Cloud VPN Gateways on both AWS and Azure to establish secure, encrypted tunnels for data transfer. Employ tools like Terraform or Ansible to automate infrastructure provisioning and management across both platforms.

- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Design and implement data pipelines utilising a standard toolset including dbt, Cloud Dataflow, Cloud Composer, Cloud Pub/Sub, and Confluent Cloud Kafka.
- Perform code release using CI/CD tools with BuildKite and GitHub: Define the CI/CD steps within the BuildKite pipeline: Fetching code from GitHub, including running tests, building the application, and deploying it to the target environment.
- Implement automated monitoring with Prometheus and Grafana to collect metrics for all Commission's data pipelines.
- Contributed to the design and implementation of integration solutions, meeting both business and technical requirements.
- Provided 3rd level technical support and troubleshooting for integration-related issues to maintain system reliability.
- Leveraged Docker to package applications and dependencies as standardized units, guaranteeing consistent and portable execution across diverse environments.
- Implemented automated DevOps pipelines automating SDLC (build, test, deploy, monitor) for various online applications, solving complex problems and pushing technical boundaries to deliver high-quality code.
- Proficient in CI/CD pipelines & tools (Jenkins, GitLab CI/CD, CircleCI, Azure DevOps) to integrate development and testing processes. Utilized Git & Plastic SCM for version control, Maven, Gradle, and npm for building, & JUnit, pytest, & Selenium for automated testing. Implemented GitOps deployment strategy for efficient infrastructure and application management.
- Analyzed data sources, types, and formats to define transformation, loading, and data model requirements.
- Extensive expertise in programming languages such as Java, Python, Go, C#, PHP and PowerShell scripting languages,
- Analyzed requirements, plans tests, and collaborates effectively to ensure efficient and timely testing.
- Created and executed high-quality automated and manual tests covering functional, non-functional, and regression aspects.
- Identified and recommended infrastructure improvements, upgrades, and repairs for IT hardware and communication systems, including SQL, Oracle, PostGres, MySQL, DynamoDB and MongoDB for databases
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Systems Integration Developer/DevOps Engineer/Software Test Engineer/.NET/SharPoint Developer
Responsibilities:

- Led the development, implementation, and maintenance of integrations between diverse systems and applications.
- Provided technical consulting and subject matter expertise to ensure alignment with organizational strategy and governance. Took technical responsibility for all stages of the integration development process, ensuring compliance with requirements.
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- **Skilled in Agile QA**: planning, coordinating, and executing tests within Agile workflows.
- **Broad Testing Experience**: Handles functional testing, database testing, performance testing, and security testing.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers
- Performed backup and recovery operations using Azure Backup and Azure Site Recovery
- High proficiency in Networking, including Cisco, Meraki SD-WAN, FortiGate Firewalls and tested in FortiClient VPN with Multifactor Authentication (MFA), to enhance network performance and protect against cyber threats.
- Designed and built cloud-based infrastructure solutions in AWS and Azure, optimizing performance and scalability.
- Utilized AWS DevOps tools like Terraform, Vault, and Artifactory within an agile environment, leveraging GitLab or a similar CI/CD platform for version control and build automation.
- Monitored and analyzed network performance using tools such as Nagios, Zabbix, Wireshark, and SolarWinds
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform

- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /BI Consultant /DWH
Responsibilities:

- Led a project to replace an existing insurance web application, optimizing IT costs and integrating with Sales Force Automation (SFA).
- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate process

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, Strawberry, OpenAI o1, LM-Studio, Azure Open AI ML Services, Copilot, Google GEMINI Pro 1.5 MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.

- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

Senior Cloud Engineer – Australian Retirement Trust – 15/9/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Cloud Engineer/Senior DevOps Engineer/Cyber Security Engineer/Senior Systems Engineer/Senior Data Engineer/ Linux/Unix Administrator/Java/.NET/Python/AI/ML/BI/SharePoint/D365 Developer/EDW -Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr Cloud Engineer/Snr DevOps/Cyber Security Engineer/.NET/Python/AI/ML/SharePt/D365 Developer/EDW
Responsibilities:

- Designed and implemented CyberArk Privileged Access Solution (PAS) across hybrid cloud architecture integrating AWS VPCs,GCP Cloud VPCs, and Azure VNets. Established encrypted site-to-site connections using Azure VPN Gateways & AWS/GCP VPN Gateways across all platforms,including installation, development, design and troubleshooting.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2019/Dynamics 365/Power BI/Power Automate & Power App)
- Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python with ARM templates, Terraform, or Ansible. Utilized Azure Kubernetes Service (AKS) for efficient and scalable container orchestration in Azure, & equivalent container orchestration services in AWS and GCP (e.g., ECS Fargate, GKE).
- Implemented a zero-trust architecture using EntraID, CyberArk PSM10.8, and Saviynt Forgerock to enhance identity and privileged access management. Leveraged cloud-native security controls & automation to ensure compliance & mitigate risks
- **Drive Identity and Privileged Access Management (IAM) Maturity**: Enhance IAM Solutions: Implement and optimize IAM tools such as EntraID, CyberArk, and Saviynt Forgerock to strengthen security & streamline identity management processes.
- Leverage CyberArk Dynamic Privilege Access (DPA) to centralize and secure all PAM accounts, eliminating the risk of compromised credentials and improving overall security posture.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
  - **Azure: Azure DevOps**
  - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
  - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- **Infrastructure Automation Specialist** with expertise in Ansible for multi-platform environments (Azure, AWS, GCP and on-premises) and DSC for Windows-specific configurations. (e.g. Configuring a Windows Service as MyService.dsc and run it in PowerShell: Start-DscConfiguration -Path ".¥MyService.dsc")
- **Linux Server Provisioning**: Utilized Ansible-Playbook to automate the creation of Azure Virtual Machines (VMs), network configuration, package installation, and application deployment by Bash command:
  ansible-playbook azure_vm_instance.yml,configure_router.yml,network_config.yml,install_packages.yml,deploy_app.yml
- **Integrate with Cloud IAM Services**: Seamlessly integrate with cloud-based IAM services like Microsoft Entra, AWS IAM, and GCP IAM to provide a unified and scalable identity management solution.
- **Cloud Security with AI**: Leveraged AI tools to detect and prevent threats, manage vulnerabilities, and secure data.
- Implemented a Q-Learning-powered intrusion detection system that reduced false positive rates by 30% and detected 95% of simulated cyber attacks within 24 hours. Deployed OpenAI's Strawberry model to assess the security implications of proposed actions before proceeding, ensuring that decisions align with established security policies and best practices.

- **Leveraged CyberArk Privileged Session Manager (PSM) to provide 24x7 proactive detection and response to security threats** by configuring multiple autonomous agents, powered by **offline** AI (Llama 3.1 and Reflection), to continuously monitor for suspicious activity and trigger alerts for the Security team to investigate and mitigate risks, especially for devices brought by employees (BYOD/BYOC), including managing third parties and vendors.
- Proficient in integration tools like MuleSoft and web frameworks (e.g., Flask, Django, FastAPI) to build robust, scalable, and efficient data pipelines using RESTful web services. Develop Python web application components, integrating with and consuming enterprise APIs. Skilled in Python, PowerShell, and Bash scripting for automating tasks and driving a culture of self-service and automation within the identity landscape.
- Leverage Python, Java or .Net to implement custom rate limiting based on the number of failed authentication attempts using SAML assertions to prevent brute-force attacks targeting user credentials.
- Enhanced communication security by deploying Telegram with end-to-end encryption, safeguarding data from interception and tampering by combining symmetric and asymmetric encryption using AES-256, DH-KEM, RSA etc.
- Implemented Saviynt's just-in-time (JIT) access for CyberArk accounts, significantly enhancing security by limiting privileged account activation to only when needed, minimized the risk of unauthorized access during idle periods.
- Successfully leveraged ITIL 4.0 frameworks, including Incident Management, Problem Management, and Continual Improvement, to optimize incident response processes and drive service excellence.

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Cloud Engineer/ Snr DevOps/Data/Cyber Security Engineer/ Java/Python/AI/ML/SP Developer/EDW
Responsibilities:

- Successfully delivered a CyberArk PAS solution that securely connects AWS, GCP, and Azure environments, enhancing privileged access management and mitigating security risks.
- Enabled RAB to achieve greater agility and innovation by designing and deploying secure cloud environments using Azure Blueprints. Integrated with Office 365 services such as Azure Active Directory, Exchange Online, SharePoint Online, and Teams to facilitate collaboration, improve information sharing, and support the organization's strategic goals.
- Implemented a robust zero-trust security framework using EntraID, CyberArk PSM10.8, and Saviynt Forgerock to enhance identity and privileged access management at RBA. This proactive approach significantly reduced the risk of unauthorized access and ensured compliance with industry regulations.
- Proactively drive RBA's Identity and Privileged Access Management (IAM) maturity by implementing and optimizing industry-leading tools like EntraID, CyberArk, and Saviynt Forgerock. This ensures robust security measures and strict adherence to regulatory standards.
- Streamline privileged access management processes and improve operational efficiency by leveraging CyberArk Dynamic Privilege Access (DPA) to centralize and secure all privileged accounts. This reduces the risk of unauthorized access and enhances RBA's overall security posture.
- Leverage Python programming with CyberArk SDK to automate the assignment of privileged accounts to institution traders based on traders online registered personal information data, ensuring efficient and secure access management:
  import cyberark_sdk # Connect to CyberArk PSM by:
  cyberark_client = cyberark_sdk.Client(username="your_username", password="your_password")
- Employed IaC principles with Python to streamline network resource management using ARM, Terraform, or Ansible. Leveraged Azure Kubernetes Service (AKS) and equivalent container orchestration platforms for scalable and efficient application deployment.
- Design, develop, and maintain robust data pipelines and API orchestration using either:
  1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
  2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
  3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Applied CyberArk to address regulatory requirements like GDPR, HIPAA, or PCI DSS for privileged access management.
- Establish a robust data governance framework by designing a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework like DBT. Leverage CyberArk Dynamic Privilege Access (DPA) to secure access to sensitive data within the platform, ensuring seamless data portability and compliance with industry regulations while safeguarding RBA's sensitive information.
- Infrastructure Automation Specialist with expertise in Ansible for multi-platform environments (Azure, AWS, GCP, and on-premises) and DSC for Windows-specific configurations. (e.g. Configuring a Windows Service using a DSC configuration script and running it in PowerShell: dscconfig MyService.dsc or Invoke-DscResource MyService)"
- **Cloud Security with AI**: Leveraged AI tools to detect and prevent threats, manage vulnerabilities, and secure data. Develop financial security-tailored LLMs (e.g., BLOOM-Fin, GPT-3-Finance) and integrated ML/DL algorithms for financial data analysis and risk assessment, enabling accurate, compliant, and personalized financial insights and decision support.

- **Network Security via SIEM:** Hybrid Cloud Security with Netskope SASE: Deploy Netskope Service Edge in data centers and cloud environments. Configure SASE functionalities including CASB, SWG, ZTNA, and DLP. Establish secure connections between on-premises and cloud (e.g., ExpressRoute, Direct Connect).
- **DevSecOps Security**: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
- **Industry Compliance** (RBA financial market services): Implemented 5 key security controls aligned with ACSC Essential 8 (e.g., Application Control, MFA), ISM 37 securijectty controls (physical access controls with mandatory security clearances), & relevant NIST CSF categories (IAM, Protective Actions) to address RBA's financial sector risks & regulations.
- **Compliance Adherence**: Ensured compliance with ISO27001/2's 114 security controls and NIST800-53 standards.
- **Utilized various security tools:** Splunk, McAfee, Secret Server, CrowdStrike, Mimecast, Qualys, Tenable AD, Fortinet.
- Full-stack developer skilled in React/TypeScript for frontend development, Python/Flask or Django for backend development, and Azure cloud. Skilled in developing and consuming enterprise APIs using Azure API Gateway, REST API, or Mulesoft Anypoint Platform, with experience in integrating with privileged access management solutions like CyberArk
- Proficient in Linux system administration, including kernel optimization (sysctl, vmstat), performance tuning (iostat, sar), and security hardening (iptables, fail2ban). Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr Cloud Engineer/ Snr DevOps/Data/Cyber Security Engineer /.NET/Python/AI/ML/SharePt /K2 Developer
Responsibilities:

- Implemented a secure, hybrid-cloud PAS solution using CyberArk, AWS, GCP, and Azure, including network configuration, VPN tunnel establishment, and troubleshooting.
- Defined granular policies for user authentication methods, such as SAML for Azure AD and LDAP for AWS IAM, to specify which authentication methods are required for different resources or user groups. These policies were implemented to grant secure access to designated resources through CyberArk PSM while ensuring a seamless user experience. Enabled comprehensive security features like cloud malware protection, data loss prevention (DLP), and web filtering across both cloud environments, creating a robust hybrid cloud security posture.
- Leverage Python programming with CyberArk SDK to automate the assignment of privileged accounts to international students based on their online registered personal information data, ensuring efficient and secure access management.
- # Retrieve students registered data by: students_registered_data = retrieve_students_registered_data()
- **Develop custom portals** using modern frameworks like React, Angular, or Vue.js for specific departments, teams, or use cases that require highly tailored experiences. Integrate and consume these portals' enterprise APIs with CyberArk PSM infrastructure to enhance user satisfaction and productivity and offer a seamless user experience.
- Leverage CyberArk PSM and established IAM platforms like Azure AD, Okta, or OneLogin to provide intuitive self-service portals with features such as password reset, MFA enrollment for international students, & group membership management.
- **Implement Identity Governance and Administration (IGA) tools** to automate identity lifecycle management tasks, such as user provisioning, deprovisioning, and access certification. Integrate IGA solutions with CyberArk like Saviynt or SailPoint to enforce role-based access control (RBAC), ensure compliance with security policies, and reduce security risks.
- Expert in designing and implementing Azure Logic Apps to orchestrate complex business processes and integrate with various enterprise APIs. Proficient in building event-driven applications using Azure Functions, triggered by API calls or other events. Skilled in utilizing Azure's serverless capabilities to develop and deploy scalable API-driven applications.
- Experienced in integrating enterprise APIs with Azure tools to create seamless data workflows and automate processes
- Skilled in utilizing Azure Service Bus to enable reliable and scalable messaging between applications and APIs
- Experienced in integrating enterprise APIs with Azure tools using RESTful APIs, SOAP, or other protocols.
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Threat modeling for proactive risk mitigation**: Proactive identification and assessment of potential threats and vulnerabilities through threat modeling techniques to implement preventive measures and reduce risk exposure.
- **Scan Regularly**: Regularly scan base images for vulnerabilities using dedicated tools like Snyk or Anchore. Set clear criteria for triggering patch upgrades based on severity and impact.
- **Automate Uplifts**: Integrate image building and deployment into the CI/CD pipeline. Automate patching for critical vulnerabilities based on established policy using tools like Skaffold.
- **Monitor and Report**: Visualize scan results and patching progress on a centralized dashboard. Set up alerts for critical vulnerabilities and successful/failed patching attempts.
- **Continuous Improvement**: Track patching time and continuously improve processes to minimize response time. Foster communication between security & development teams to address underlying code issues and prevent future vulnerabilities.
- **Cloud Security Design and Management**: Designed secure cloud infrastructure with IAM policies, network segmentation, and encryption.
- Leveraged PSM (Privileged Session Manager) to capture and record a secure session within CyberArk, facilitating isolated troubleshooting of a critical security incident.

- Proficient in building scalable, secure RESTful APIs on Azure using tools like Functions, App Service, and API Management. Experienced in consuming and integrating with enterprise APIs, leveraging C# and industry-standard design principles.
- Leverage Python programming with CyberArk SDK to automate the assignment of privileged accounts to users based on user online registered personal information data, ensuring efficient and secure access management.
- **Promote Self-Service and Self-Consumption**: **Foster an Industrialized IAM Landscape**: Establish standardized processes, templates, and workflows to enable self-service provisioning, deprovisioning, and access management.
- Leveraging hash maps to design and develop various types of Terraforms (AKS, EKS, GKE) and their data sources for extracting information from external systems and integrating it into Terraform configurations.
- Incorporated **DevSecOps** practices for secure infrastructure provisioning, vulnerability scanning, & automated deployments.
- **Threat Mitigation**: Monitored and mitigated threats using tools like Wireshark and Metasploit.
- **DevSecOps Collaboration**: Integrated security into CI/CD pipelines.
- **Incident Response Planning**: Developed incident response playbooks for cloud-based services.
- **Data Security and Management:** Designed and implemented AI-driven data solutions, pipelines, and infrastructure components using secure practices.
- **Threat Intelligence Analysis**: Investigated cyber threats, identified vulnerabilities, & recommended preventive measures.
- **Incident Handling**: Responded swiftly to security incidents, minimizing business impact.
- Setup and customized ServiceNow Enterprise-grade IT Service Management (ITSM) platform with robust features for incident, change, and problem management, pre-built integrations with IT infrastructure, security-focused design.

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Industry: Cloud Engineer/DevOps/Cyber Security Engineer/Software Test Engineer /SharePoint Consultant
Responsibilities:

- **Designed and Implemented Robust Security Architectures**: Network security (SIEM, Netskope), DevSecOps practices, secure AWS landing zones, multi-cloud development pipeline, secure cloud infrastructure.
- **Security Architecture Enhancement**: Collaborated on enhancing security architectures, conducted threat intelligence analysis, responded to incidents swiftly, reviewed code for vulnerabilities, and ensured compliance with standards.
- **Network Security via SIEM**: Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.
- **DevSecOps Security**: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
- Leverage Python programming with CyberArk SDK to automate the assignment of privileged accounts to workers compensation insurance and claims for schemes across Australia used for accessing sensitive data, systems, or applications related to claims processing, fraud investigation, or financial management. This automation process will ensure efficient and secure access management while ensuring strict compliance with data privacy regulations.
- #Determine privileged account assignments:account_assignments= determine_account_assignments(cv_expression_data)
- Utilized McAfee, Tenable, Splunk, and Secret Server to support operational Cyber Security Management programs of work.
- **Threat Mitigation**: Monitored and mitigated threats using tools like Wireshark and Metasploit.
- **Incident Response Planning**: Developed incident response playbooks for cloud-based services.
- **DevSecOps Integration**: Integrated CyberArk PSM security into GitHub Actions or GitLab CI/CD pipelines.
- Scrutinized code bases to eliminate common coding vulnerabilities and ensure secure application development and deployment. Deployed & managed advanced security tools to systematically identify & resolve security issues across systems & applications.Ensured adherence to industry-specific compliance standards & regulations in all security initiatives.
- Using Naive Bayes Classifier for Supervised Machine Learning to Predict Reparability Probability of Classified Items
- Utilized Splunk leading platform for log management, analysis, and security information and event management (SIEM), perform real-time monitoring, and conduct forensic investigations, including IDS, IPS, IAM and related.
- Configure McAfee and McAfee VPN to encrypts users' data and protects it from potential snoopers, ensuring that users' online activities and personal information remain secure even on untrusted networks.
- Setup and configured Tenable, Tenable.sc (formerly SecurityCenter) vulnerability management platform that helps organizations identify and prioritize security vulnerabilities across their infrastructure.
- Perform day-to-day monitoring and triaging critical alerts and coordinate with technical teams for incident response.
- Triage phishing emails, taking appropriate containment and remediation actions to prevent security incidents
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Cloud Engineer/DevOps/Cyber Security Engineer/Linux Systems Administrator/SharePt Consultant
Responsibilities:

- **Security Architecture Enhancement**: Collaborated with cross-functional teams to enhance security architectures for financial investment projects.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within Azure cloud architecture.

- **Network Security via SIEM**: Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.
- **Security Testing**: Performed penetration testing and vulnerability scanning.
- **Secure SDLC Implementation**: Advised development teams on secure coding practices.
- **NIST Best Practices**: Followed NIST guidelines for secure software development.
- Collaborating with IT Security for the Standard Operating Environment (SOE) security hardening in line with PSPF, ISM, ACSC, CIS and DTA Protected pattern guidelines.Stayed updated with industry security trends and emerging threats.
- Leveraged McAfee, Splunk, and Secret Server to enhance the effectiveness of Cyber Security Management programs.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Cloud Security Engineer/Infrastructure Engineer/Systems Administrator /DBA
Responsibilities:

- **Cloud Security Design**: Designed secure cloud infrastructure, including IAM policies, network segmentation, and encryption.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within Azure cloud architecture.
- **Network Security via SIEM**: Implemented a SIEM system to aggregate logs, streamline incident response, and monitor compliance.
- Configured a Netskope Cloud Private Access (NCPA) tenant and established secure connections between Azure and AWS environments to Netskope using virtual Private IPs or peering connections.
- **DevSecOps Security**: Extensive experience in code review, recognising code smells and implementing DevSecOps tooling such as SAST, DAST, RASP, RAST (e.g. IBM AppScan, Veracode, Synk, Coverity, SonarQube, Twistlock) and IaC security.
- **Network Security Enhancement**: Implemented network segmentation and intrusion detection systems.
- **Threat Hunting**: Proactively searched for vulnerabilities and potential threats.
- **Incident Handling**: Responded to security incidents, ensuring minimal impact.
- **Compliance Audits**: Conducted internal audits to meet industry standards.
- **Risk Assessment**: Evaluated emerging technologies for relevance and risk.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /Systems Administrator /DWH
Responsibilities:

- Supported ongoing operational Cyber Security Management programs of work, contributing to a strong security framework.
- Managed hardware and server administration, ensuring reliable infrastructure performance.
- Oversaw AD, DFS, ADFS, AAD, DNS, and DHCP services, maintaining their stability and security.
- Provided level 3 reactive support for incident/service request management, monitoring, and event management.
- Utilized McAfee, Tenable, Splunk, and Secret Server to support operational Cyber Security Management programs.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]

- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JSP, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI Multi-GPTs, Auto-Agents, GPT-3.5/4o, Q-Learning, Reflection, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB,etc.
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
- **Workflows**: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows

Senior Machine Learning Engineer – Woolis – Intelligen – 5/9/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior Machine Learning Engineer /Senior DevOps Engineer/Senior Data Engineer /Full Stack Developer(Java/.NET)/ Software Engineer /Python /AI/ML/DL/BI/SharePoint/D365 Developer /EDW/DBA -Japan(7.5yrs) and Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior ML Engineer/Snr DevOps/Data Engineer /Java/.NET/Python/AI/DL/ML /SP/D365 Developer/EDW
Responsibilities:

- Design, develop, implement, scale, and maintain Advanced Analytics solutions on GCP using Cloud Dataflow for data engineering, Vertex AI for ML model building and deployment, **Cloud Function** or **App Engine** or **Mulesoft Anypoint API** for API endpoints, & Kubeflow for scalable orchestration with Airflow, Docker, and Kubernetes for managing ML workloads.
- Collaborate with Data Scientists, ML Engineers, and Data Engineers to develop cutting-edge retail tools agilely.
- Design, develop, and maintain robust microservices-oriented architecture and REST APIs using Java or Python/Django for backend development and React/TypeScript for frontend development, to support advanced analytics solutions. Leverage GCP cloud and integration tools like MuleSoft and J2EE to ensure seamless data flow using RESTful web services.
- Develop Cloud based ML pipelines to solve a broad range of business problems.
- Grow the Engineering Guild capability through pioneering new tech and approaches to improve time to market, reliability and performance of Data Science applications.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Developed and deployed ML models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision: model = models.detection.fasterrcnn_resnet50_fpn(pretrained=True)
- Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.
- Utilized Harrison AI platform provided a common set of tools and processes for streamlining the entire AI development lifecycle for medical devices, including tasks like data preparation, model training, validation, deployment, and monitoring.
- Designed and maintained robust AI/ML pipelines using platform-specific tools:
  - Azure: Azure Machine Learning (AML), Azure OpenAI Service, Azure Databricks
  - AWS: AWS SageMaker, Amazon Rekognition
  - GCP: GCP Vertex AI, Cloud Natural Language API
- Develop medical-tailored LLMs (BERT, GPT-4, and other relevant models) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, & personalized medical insights & data-driven decision support.
- Utilized organoid intelligence (OI) to accelerate the development of brain organoids for neurological disease research.

- **Automation and CI/CD**: Develop, implement and maintain automated deployment pipelines for Data and Machine Learning pipelines, models, and related infrastructure using Ansible, GitLab CI/CD, Cloud Build, etc.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu Linux+ NVIDIA tools) for running containerized AI, HPC, & industry-specific workloads across DL training, inference, analytics, etc.
- Using the Harrison AI Platform to develop AI-powered medical devices which use ML algorithms to analyze data and make diagnostic or therapeutic decisions to deliver better care to patients of Aged Care Quality and Safety Commission.
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, RNN, ANN, GAN) based on dataset type, size, and desired result interpretability.
- **Design, develop, and maintain robust data pipelines and API orchestration using either:**
    - Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    - AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    - GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- **Architected and implemented a modern 3-layer lambda architecture on GCP for Advanced Analytics solutions**:
- **Batch layer**: Utilized Google Cloud Dataproc, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed BigQuery with its built-in Spark capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Cloud Pub/Sub as the unified service layer for real-time data ingestion and BigQuery for querying the data warehouse for advanced analytics and reporting.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- Utilized BigQuery for data querying, loading, exporting to Pandas DataFrame, and executing parameterized queries within the Python programming environment. query_job = client.query(query, job_config=job_config)

Dec. 2015 – Dec. 2017 Reserve Bank of Australia (RBA), Sydney
Gov. Central Bank: Snr Machine Learning Engineer /Snr DevOps/Data Engineer/AI/DL/ML /BI /Java/Python Developer/EDW
Responsibilities:

- Designed and implemented a hybrid cloud architecture, connecting AWS VPCs, GCP Cloud VPCs, and Azure VNets.
- Established encrypted tunnels for data transfer using Cloud VPN Gateways across all three platforms.
- Utilized infrastructure-as-code (IaC) tools like Terraform or Ansible to automate the provisioning and configuration of cloud resources, enabling optimized service operations and reducing manual errors.
- Design, develop, implement, scale, and maintain Advanced Analytics solutions on GCP using Cloud Dataflow for data engineering, Vertex AI for ML model building and deployment, **Cloud Function** or **App Engine** or **Mulesoft Anypoint API** for API endpoints, & Kubeflow for scalable orchestration with Airflow, Docker, and Kubernetes for managing ML workloads.
- Built secure & scalable GCP Landing Zones using IaC with Terraform or Cloud Build. Utilized GCP Cloud IAM for centralized governance and management of multiple GCP projects, enabling efficient parallel processing for various projects.
- Design, develop test and deploy security focused and AI-Powered ML applications, deep learning (DL) models and systems
- Based on bank business requirements, define most suitable ML Pipelines from Kubeflow, Mlflow, Dataflow MLOps , ZenML, Flyte, framework to create pipelines that orchestrate the ML workflow
- Design, develop, and maintain robust microservices-oriented architecture and REST APIs to support advanced analytics solutions, ensuring efficient data flow and scalable performance. This includes experience designing and developing Interactive Brokers (IB) OMS-like Trade WorkStation (TWS) applications using Java8/9/J2EE or IBPy (Python) for prime brokerage and prior trading within the FX, CFD, or fixed income, derivatives markets.
- Designed and maintained robust AI/ML pipelines using platform-specific tools:
    1. Azure: Azure Machine Learning (AML), Azure OpenAI Service, Azure Databricks
    2. AWS: AWS SageMaker, Amazon Rekognition
    3. GCP: GCP Vertex AI, Cloud Natural Language API
- Develop financial-tailored LLMs (BERT, GPT-3, and other relevant models) and integrated ML/DL algorithms for financial analysis with OpenCV, enabling accurate, compliant, & personalized financial insights & data-driven decision support.
- Monitored model performance, conducted A/B testing, & proactively identified and addressed model drift or potential issues.
- Built and implemented a **Medallion Architecture with a Data Lake in GCP** to efficiently process and analyze economic and financial data. **Batch Layer**: Utilized Google Cloud Dataproc for batch processing of large data volumes into the Bronze layer of the Data Lake. **Serving Layer**: Implemented BigQuery for near-real-time processing and analysis on the Silver layer of the Data Lake, enabling timely analysis and decision-making for critical economic indicators.
- **Real-time Layer**: Leveraged Cloud Pub/Sub for real-time data ingestion, ensuring adherence to regulatory compliance and data governance standards. Queried the Silver and the Gold layer of the Data Lake for advanced analytics and reporting.

- Utilized BigQuery for data querying, loading, exporting to Pandas DataFrame, and executing parameterized queries within the Python programming environment. query_job = client.query(query, job_config=job_config)
- **Integration Solution Design with GCP Cloud Build (CI/CD): Design Integration Flow:**
- Define data flow, transformations, and choose the orchestration platform (MuleSoft, Cloud Functions, or Hybrid).
- Used MuleSoft Anypoint Platform (including Anypoint API, Exchange, and DataWeave) for complex structured data transformations among disparate systems, while leveraging DBT for standardization and basic transformations.
- Built a high-performance Kubernetes cluster on Nvidia DGX-1 system with pre-installed DGX OS (optimized Ubuntu Linux+ NVIDIA tools) for running containerized AI, HPC, & industry-specific workloads across DL training, inference, analytics, etc.
- **CloudWatch and Third-Party Observability Expertise**: Implemented comprehensive observability solutions using CloudWatch, App Insights, Dynatrace, and Azure Log Analytics, leveraging metrics, logs, and dashboards. Utilized CloudWatch Insights for advanced log analysis and troubleshooting.
- **Designed, implemented and maintained robust CI/CD pipelines using platform-specific tools**:
    1. Azure: Azure DevOps
    2. AWS: AWS CodePipeline/CodeBuild/CodeDeploy
    3. GCP: Cloud Build/Cloud Deploy
- **Master of AI/ML deployment**: Built and deployed robust ML models (Flask/FastAPI) for web apps, managed cloud infra (AWS/Azure/GCP), secured pipelines with MLOps/DevSecOps.
- Design, develop, and maintain robust data pipeline and API orchestration using either:
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- **Big data wrangling**: Executed ML pipelines with Spark & Hadoop, leveraging AutoML tools for large-scale data processing (feature engineering, model selection, hyperparameter tuning).
- Built & orchestrate data pipelines with dbt, Cloud Dataflow, Cloud Pub/Sub, & Cloud Composer for efficient data movement and transformation. Then, normalized complex data in Cloud Spanner and Cloud SQL for optimal performance within GCP.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr ML/Operation Engineer /Snr DevOps/Data Engineer/Python/AI/DL/ML /SharePoint /K2 Developer /DBA
Responsibilities:

- ML Architect: Successfully designed, developed, and deployed AI/ML models to enhance international student engagement and drive sales. Implemented robust MLOps and DevSecOps practices for efficient and secure platform management. Contributed to the strategic planning and scoping of new AI/ML initiatives.
- Create, present and seek inputs and feedback on architecture, solution designs and foster alignment and support for solutions. Apply relevant testing frameworks to ensure the robustness of solutions
- Designed, built, and operated software systems to transform petabytes of data into actionable insights.
- Analyze student feedback and social media discussions using Cloud Natural Language API to identify trends, improve the enrollment process, and enhance student satisfaction.
- Security and automation expert: Secured and managed Linux systems, implemented best practices and automation, proactively mitigating threats and resolving complex issues.
- Developed and maintained AI /ML models to enhance international student's engagement and sales.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Master of AI/ML Deployment: Successfully built and deployed scalable ML models (using Flask, FastAPI, or REST APIs) for web applications. Proficient in managing cloud infrastructure (AWS, Azure, GCP) and ensuring secure pipelines through MLOps and DevSecOps practices. Communicate analyses, insights, solution designs and showcase working solutions
- Cloud Security Expert: Proactively managed and monitored cloud infrastructure (AWS) using tools such as SageMaker, Config, CloudTrail, Shield, and WAF to ensure robust security and automation. Experienced in setting up and optimizing CI/CD pipelines, leveraging shell scripting and unit/integration testing for efficient and secure software delivery.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Utilized Deep Learning CNNs to analyze medical images and improve diagnosis accuracy for specific diseases, contributing to a 20% increase in early detection rates.
- Deployed the infrastructure stacks for Platform (MLOps) related services as they were developed, using Terraform and Ansible. Developed, updated and improved upon a variety of Terraform modules for venture and internal consumption.
- Spearheaded the development and implementation of advanced machine learning models and algorithms, contributing to the Commission's position at the forefront of AI technology, from Physical AI to Generative AI to Interactive AI towards AGI.
- Design and develop ML models using Natural Language Processing (NLP), ML algorithms, OpenCV, MLOps, MLlib etc.

- Develop medical-tailored LLMs (BERT, GPT-3, Jurassic-1, Bloom) and integrated ML/DL algorithms for medical image analysis with OpenCV, enabling accurate, compliant, and personalized medical insights and decision support.
- **Model Deployment and Monitoring**: Deploy machine learning models either using fully managed AWS SageMaker, or high-performance and scalable TensorFlow Serving or Open-source Kubeflow run on Kubenetes on any major clouds.
- Implement and maintain monitoring solutions to track model performance, data drift, and system health.
- Conducted machine learning training and retraining using Python, scikit-learn, and TensorFlow.
- **Scalability and Optimization**: Optimize machine learning workflows for performance, scalability, and cost-effectiveness.
- **Security and Compliance**: Implement security best practices to safeguard machine learning infrastructure and data.
- Produced production-quality code deployed as Docker containers. Coded in a Linux environment using Git for source control. Implemented end-to-end solutions in a shared codebase.
- MLOps advocate: Implemented MLOps tools, prioritizing data security through encryption, access control, secure storage, audits, anonymization, etc.
- Automated tasks and scripts using Python, PowerShell, Bash scripts, Ansible, Puppet, Lambda, JavaScript and Typescript
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Employed Power BI, Tableau, or QlikView for real-time financial analysis.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Health Sector: Machine Learning & Operation Engineer /Snr DevOps/Data Engineer/SharePoint /D365 Consultant
Responsibilities:

- Designed, developed & deployed secure & efficient AI/ML applications, deep learning models and systems.
- Spearheaded the transformation and training of Data Science products and ML systems, ensuring optimal performance and adherence to industry best practices. Followed MLOps best practices for versioning, deployment, testing, and automation.
- Coded in a Linux environment using Git for source control. Produced production-quality code deployed as Docker containers. Implemented end-to-end solutions in a shared codebase.
- Manage support requests. Perform maintenance tasks such as software upgrades
- Utilized the AWS Deep Learning AMIs to build custom environments and workflows with MxNet as well as other frameworks including TensorFlow , PyTorch, Chainer, Keras, Caffe, Caffe2, and Microsoft Cognitive Toolkit.
- Built and deployed 20+ production-ready models using Random Forest and K-Means Clustering for automating medical claim processing, resulting in a 15% decrease in processing errors and a 10% reduction in administrative costs.
- Use IBM Cloud and Spark MLlib for distributed machine learning and manage code using Git.
- Implemented Infrastructure as Code (IAC) involves creating a CloudFormation template in YAML or JSON, parameterizing it for flexibility, utilizing version control for tracking changes, and automating deployment through CI/CD tools.
- Developed and deployed ML models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision): model = torch.hub.load('pytorch/vision:v0.6.0', 'mobilenet_v2', pretrained=True)
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
- Worked on projects involving web development (Spring Boot), data integration (Oracle Data Integrator) and business intelligence (Oracle BI Publisher). Skilled in creating impactful visualizations using Tableau and Power BI.
- Utilized expertise in AWS services such as EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance.
- Developed and trained machine learning models using Python, leveraging appropriate algorithms and tools. Conducted research and statistical analysis to optimize datasets and data representation methods. Extended existing ML libraries and frameworks to meet evolving business needs.
- Developed user interfaces with Flask, Django, or FastAPI, REST APIs, utilizing scikit-learn for model evaluation.
- Developed user interfaces and REST APIs: Created intuitive user interfaces using Flask, Django, or FastAPI, and implemented robust REST APIs for efficient data exchange.
- Implemented data automation solutions using R and Python. Designed OLAP Layer Cubes using SSAS and performed data profiling and cleansing.
- Used scikit-learn to create and evaluate models using various algorithms (logistic regression, Naive Bayes (NB) and Support Vector Machines (SVM) , random forest); Used Azure Machine Learning to deploy the model as a web service
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 Sunshine Coast Council (SCC), QLD Australia
Government Council: Machine Learning Engineer/ DevOps Engineer /Data Engineer/SharePoint Consultant

Responsibilities:

- Developed software in the Python ecosystem for complex applications.
- Worked on developing and training ML models.
- Conducted research on appropriate ML algorithms and tools according to business requirements.
- Developed appropriate datasets and data representation methods and performed statistical analysis.
- Trained and retrained systems when necessary whilst extending existing ML libraries and frameworks.
- Worked on data representation methods and contributed to effective data engineering processes.
- Proficient in Linux platform development, optimizing application performance.
- Leveraged containerization and Kubernetes for deployment scalability.
- Implemented unit testing and test-driven development for code quality.
- Engineered end-to-end data pipelines for diverse source data to warehouses.
- Used Jira and Confluence for effective project management and documentation.
- Conducted exploratory data analysis for evidence-based decision-making.
- Designed interactive Power BI dashboards for data visualization.
- Optimized SQL queries for efficient data extraction and transformation.
- Analyzed and enhanced OLAP tools for efficient data analysis.
- Analyse, review existing solutions to develop OLAP tools' drill-down and slice-and-dice functionality, including MOLAP, ROLAP, HOLAP to enable users to analyze data quickly by nesting the information in tabular or graphical formats.

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Data Engineer /BI Developer /Senior Systems Engineer /DWH /DBA
Responsibilities:

- Developed scheduled and ad hoc data visualizations aligning with business needs.
- Conducted data extraction, modeling, and statistical analysis for data-driven decisions.
- Designed DWH concepts (Inmon, Kimball), built data models (Power BI, Toad Data Modeler 6.x).
- Developed OLAP Cubes (SSAS, Cognos TM1) and visualized data (Power BI, Tableau, QlikView).
- Data Vault modeling for uniquely linked set of normalized tables that support one or more functional areas of business via its hybrid approach encompassing the best of breed among 3rd normal form (3NF), star schema and snowflake schema.
- Making reports either by SSRS, Crystal Report on Windows, Business Object BOXIR2 on HP-UX platform
- Perform data source analysis either via Oracle EBS BI, PL/SQL, SAP BI, SSAS, T-SQL, Sybase, MySQL etc.
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme, including 3NF,
- Designing Dashboards and Scordcards using SharePoint PerformancePoint Services 2010 and SSAS

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Data Engineer /Senior Systems Engineer /BI Consultant /DWH
Responsibilities:

- Prepared and analyzed large and diverse data sets to identify patterns and anomalies.
- Created aggregate data sets aligned with context for Insights Analysts and Visualization Specialists.
- Collaborated with stakeholders to fulfill data requirements and conducted data analysis using Python and machine learning.
- Managed the bank's infrastructure and applications using HP BCA, System Center 2010, and SPLUNK.
- Recommended suitable BI solutions based on clients' needs, considering various BI systems and tools.
- Developed complex SQL queries and procedures, enhancing an SQL Server 2008 R2 Analysis Services Cube.
- Conducted system demos for vendors and addressed technical queries regarding the project.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.

- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Sora, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Phi 3, Copilot, Google GEMINI, Metaverse Llama 3.1, Mistral
  - SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

InTune Engineer – The Hollard Insurance Company – The Aquarium – 2/9/2024
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior InTune Engineer/Senior Clouds/Linux Administrator/Senior DevOps Engineer/Senior Data Engineer /AI/ML/BI/SharePoint/Dynamics 365 Consultant /EDW –Japan (7.5yrs) & Australia (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Snr InTune Engineer/Snr Cloud/Systems Administrator/Snr DevOps/Data Engineer /AI/ML/SP/D365 Consultant
Responsibilities:

- Architected and Implemented hybrid cloud infrastructure architectures between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between on-prem & both cloud platforms (utilizing VPCs in AWS and VNets in Azure) using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- Federated Amazon Redshift access with Azure EntraID (formerly known as AAD), SSO and MFA in Azure Security Centre.
- Designed and implemented secure landing zones for cloud workloads using Azure Blueprints, focusing on Windows-based systems in M365 Environments (O365/MS Team/SharePoint 2019/Dynamics 365/Power BI/Power Automate & Power App)
- Successfully migrated device management tasks from SCCM to Microsoft Intune using Hybrid Azure AD Connect for user synchronization and Hybrid Azure AD Join for device management.
- Leveraged Intune Mobile Device Management (MDM) and Mobile Application Management (MAM) to effectively manage and secure mobile devices and applications within the MS Endpoint Manager admin center, including PowerShell script
- Integrated Intune MDM with Microsoft Endpoint Manager to deliver a unified platform for managing and securing devices across the organization, streamlining workflows and improving overall security posture.
- **Software deployment in a co-management environment**: Use Intune to deploy cloud-based applications (e.g., Office 365) and SCCM to deploy on-premises applications (e.g., legacy line-of-business apps).
- **Patch management**: Use Windows Autopatch with Intune to automate Windows 10/11 updates, and SCCM for server patches. Leverage PowerShell scripting for additional automation and customization as needed.
- **Device configuration**: Use Intune for mobile device management and SCCM for on-premises device configuration.

- **Autopilot Deployment Profile**: Create a deployment profile that specifies the desired configuration settings for the devices. Using Intune to enroll devices into Azure AD either using QR codes, bulk enrollment, or Autopilot deployment profiles.
- Train and support end-users and IT staff on Intune-related functionalities and features
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
    - **Azure: Azure DevOps**
    - **AWS: AWS CodePipeline/CodeBuild/CodeDeploy**
    - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to your specific needs. Or
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- **Deployed multi-cloud databases**: Utilized Azure Cosmos DB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using Azure Cognitive Search for native retrieval methods with RAG, libraries like Faiss or Preflexity, and Azure Cosmos DB connectors for pymongo-like access. Large language models (LLMs) like GPT-4, Sora, Mistral, and Gemini 1.5 can still be integrated for further analysis.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.
- Proven ability to manage, provide 3rd level support, & troubleshoot on-premises solutions like SharePoint 2019, Dynamics 365 CRM, & SQL Server 2022 on Windows Server 2022, along with extensive experience designing, implementing, and configuring Azure & Office 365 services (including Azure Files, D365 CRM, Power BI, Power Apps, Power Automate, SPoint & Exchange Online). This demonstrates a strong focus on system and procedure documentation for both environments.
- Provided day-to-day ICT technical support for AD, AAD, Microsoft 365, MFA/SSO, and Azure, ensuring smooth operations.
- Using InTune and Configuration Service Providers(CSP) for mobile device management (MDM) on Windows devices.
- Technical support for aged care business applications covering SharePoint 2019/Dynamics 365 CRM, TIBCO servers and its long run workflows linked SQL Databases, troubleshoot national users reported issues via Zendesk issue mgmt. systems
- Managed IT infrastructure & EDW data backup using Commvault Data Protection /Veeam Backup for Microsoft 365 v6.0/7.0
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Immersed experience adopting ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.
- Performing data transformation and integration tasks either using Python in Azure DevOps to leverage the power of Azure Pipelines to define CI/CD workflows or in Snowflake's Cloud Data Warehouse via snowflake-connector-python

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov. Central Bank: Snr InTune Engineer/Snr Clouds/Systems Administrator/Snr DevOps Engineer/ML/SP/BI Developer/EDW
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **Built and implemented Medallion Architecture with Data Lake in Azure**:
- Batch Layer: Utilize Azure Databricks, a scalable Apache Spark-based analytics service, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, and unrefined data.
- Serving Layer: Implement Azure Synapse Analytics (formerly Azure Data Lake Analytics) with its integrated Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- Real-time Layer: Leverage Azure Synapse Analytics as the unified service layer, combining Massively Parallel Processing (MPP) with native query support and data views for real-time analysis through Azure Stream Analytics. This layer can access both the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Design, develop, and maintain a robust ETL pipeline/API orchestration using either:
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, using either Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for stream processing developed custom ETL/API orchestrations which can apply to any cloud or on-premises application.
- Automate cloud infrastructure with Terraform and Packer for secure, standardized VM deployments. Leverage Ansible for post-provisioning configuration management & deployments on pre-built images, followed by integrated security scanning.

- Administer, support, maintain and monitor Windows Server fleet (VMWare/AWS/Azure/GCP) systems, & in addition provide web server system administration and technical support as required for the Digital Information Office (DIO) Business.
- Utilized MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between GCP infrastructure (including Apigee) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- Leveraged Intune Mobile Device Management (MDM) and Mobile Application Management (MAM) to effectively manage and secure mobile devices and applications within the Microsoft Endpoint Manager admin center.
- **Software deployment in a co-management environment**: Use Intune to deploy cloud-based applications (e.g., Office 365) and SCCM to deploy on-premises applications (e.g., legacy line-of-business apps).
- **Patch management**: Use Windows Autopatch with Intune to automate Windows 10 updates, and SCCM for server patches. Leverage PowerShell scripting for additional automation and customization as needed.
- **Device configuration**: Use Intune for mobile device management and SCCM for on-premises device configuration.
- Intune Engineer Key Achievements (Hybrid Join environment)
- Successfully migrated workflows from SCCM to Intune for hybrid Azure AD joined devices.
- Implemented and managed Intune configurations and policies for both domain-joined and Azure AD joined devices.
- Ensured seamless device management and synchronization between on-premises and cloud environments.
- Leveraged Intune's features to enhance security and improve device management processes.
- Successfully transitioned devices to hybrid Azure AD join while maintaining existing domain-joined infrastructure.
- Resolved policy conflicts and ensured compatibility between SCCM and Intune.
- Provided expert guidance on hybrid join and Intune best practices.
- Use Endpoint analytics to help identify the policies or the hardware issues that slow down the devices and improve end user experience. Deploy Microsoft 365 apps to the devices using Microsoft Intune.
- Connect Intune and Defender for Endpoint together and create policies to scan file, detect threats, and report threat levels. Configure CA policies to block access from non-compliant devices.
- Use Windows autopatch with Microsoft Intune to manage the patching and keep software up to date.
- Demonstrated expertise in managing, supporting, and troubleshooting SharePoint 2016, Dynamics CRM 2016, and SQL Server 2016 on Windows Server 2016, with a strong focus on system and procedure documentation.
- Provided essential day-to-day ICT support for AD, AAD, Microsoft 365, MFA/SSO, and Azure, ensuring smooth operations.
- Administer, support, troubleshoot SharePoint 2016, D365 CRM, SQL Server 2016 on Windows server 2016 OS platform.
- Developed and maintained ETL/ELT pipelines for Azure Synapse Analytics using Azure Data Factory/Synapse Pipelines, Spark Notebooks, SSIS, PolyBase, Stream Analytics
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Develop and optimize ETL processes using SQL and PySpark, ensuring seamless data integration and transformation.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: InTune Engineer/Snr Clouds/Systems Administrator/Snr DevOps Engineer /ML/SPoint /K2 Developer /DBA
Responsibilities:

- Architected and Implemented **hybrid cloud infrastructure architectures** between AWS and Azure, leveraging Netskope Cloud Private Access (NCPA) within Azure to enable secure and performant connections between both cloud environments using dedicated connections like Azure ExpressRoute or AWS Direct Connect.
- Administer and maintain Active Directory, both cloud and on-premise, with a strong focus on modern authentication technologies, Azure MFA, and single sign-on. Manage InTune/SCCM, identity services, including B2B and B2C tenants.
- Using InTune and Configuration Service Providers(CSP) for mobile device management (MDM) on Windows devices.
- Leveraged Intune Mobile Device Management (MDM) and Mobile Application Management (MAM) to effectively manage and secure mobile devices and applications within the Microsoft Endpoint Manager admin center.
- **Software deployment in a co-management environment**: Use Intune to deploy cloud-based applications (e.g., Office 365) and SCCM to deploy on-premises applications (e.g., legacy line-of-business apps).
- **Patch management**: Use Windows Autopatch with Intune to automate Windows 10 updates, and SCCM for server patches. Leverage PowerShell scripting for additional automation and customization as needed.
- **Device configuration**: Use Intune for mobile device management and SCCM for on-premises device configuration.
- **Image Creation**: Created a custom Windows image with the desired settings for hybrid deployment**.**
- **Hardware Preparation**: Ensure devices have the necessary hardware (e.g., TPM, UEFI).
- Successfully migrated on-premises workloads to Azure Cloud using Azure Migrate, Azure Site Recovery, Azure Import/Export, and Azure SQL.
- Demonstrated expertise in managing, supporting, and troubleshooting SharePoint 2013, Dynamics CRM 2016, and SQL Server 2016 on Windows Server 2016, with a strong focus on system and procedure documentation.
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins

- Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline, which will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Developed and maintained CI/CD pipelines for various applications using AWS/Azure DevOps, DevSecOps towards AIOps.
- **Automation and CI/CD**: Leveraged Kustomize & Kubespray, industry-standard tools, to streamline &customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Python programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- Proficiently set up and managed Microsoft Azure environments, including the establishment and management of MS secure landing zones via blueprints; Create VM either by New-AzVM in PowerShell or az vm create in CML
- Define and create an Azure Data Factory (ADF) pipeline that orchestrates the entire data pipeline. The pipeline will have activities for extracting data, transforming it, and loading it into Azure Synapse Analytics.
- Using SmartForms in K2 Designer to grab multiple line-of-business systems' databases (SharePoint, CRM, SQL Server, Oracle, SAP, IBM DB2, Cognos TM1, Sallesforce.com etc.) into a single form by SmartObjects for easy decision making.
- Using SSIS ETL Tools to extract complex data into DW staging area, transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Design and development of K2 workflows, forms and reports on a SharePoint platform and its backend SQL Server database.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Create and package custom workflows either by SP2013 or VS2013 or K2 Package and Deployment /K2 for Visual Studio Create SmartObject either by SmartObjects Service – Tester or via K2 Designer; Bind Service Objects to SmartObjects, etc.
- Python ML packages setup by $pip install numpy pandas matplotlib seaborn scikit-learn tensorflow keras on Anaconda 3
- Built predictive models using machine learning algorithms such as linear regression, logistic regression, decision trees and random forests using Scikit-learn and Keras. Developed predictive models to support data-driven decision-making.
- Applied natural language processing (NLP) techniques such as text preprocessing, sentiment analysis, and topic modeling using the NLTK (Natural Language Toolkit) and Gensim Python libraries.

Oct. 2012 – Sep.2013 Employers Mutual Limited (EML), Sydney, Australia
Healthcare Industry: Senior Clouds Engineer/DevOps Engineer /Senior Systems Administrator/SharePoint Consultant /DBA
Responsibilities:

- Administer, support, maintain and monitor Windows Server fleet (VMWare/AWS/Azure) systems, and in addition provide web server system administration and technical support as required for the Digital Information Office (DIO) Business.
- Proficiently managed Microsoft Windows 2012 server services, including Active Directory, both cloud and on-premise, RADIUS, trusts, group policies, remote access, DHCP, Enterprise CA, DNS, and network device enrolment services.
- Demonstrated expertise in managing, supporting, and troubleshooting SharePoint 2013, Dynamics 365 CRM, and SQL Server 2013 on Windows Server 2013, with a strong focus on system and procedure documentation.
- Implemented and Administered modern authentication technologies and InTune for efficient mobile management
- Successfully managed and maintained the Standard Operating Environment (SOE) for over 1000 workstations, ensuring system stability and optimal performance.
- Windows Server 2012 OS administration, providing support for services like DNS and DHCP, and contributing to network service maintenance. Contributed to the maintenance of enterprise CA and other network services
- Manage, support, troubleshoot SharePoint 2013, Dynamics CRM 2013, SQL Server 2012R2 on Windows server 2012R2 OS platform. Collaborated with IT Security to ensure SOE security hardening in compliance with industry standards.
- Offered 3rd level technical support through various channels, including phone, email, and in-person interactions, resolving hardware and software issues, while also installing and configuring software and hardware.
- Designed and developed fully automated CI/CD pipelines using Azure DevOps, Git, Docker, GitHub Actions, and Jenkins
- Implemented and maintained CI/CD pipelines using Git & Azure DevOps to automate build, test, and deployment processes
- Provisioned and managed Azure Kubernetes Service (AKS) clusters using Terraform and kubectl
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Utilized either dbt (data build tool), Fivetran or Dataform tools in the realm of modern data analytics and engineering.
- Secured Azure resources using Azure Active Directory, Azure Key Vault, Azure Policies and Azure Security Center
- Utilized bicep and ARM templates to provision and configure Azure resources: App Service, Functions, Logic Apps, Storage
- Multitasking with Unix/Linux server administrator and global firewall management, including install, configure and support VoIP IP Phones and Video conference meeting room setup for meeting with overseas branches management members.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: Systems Administrator (AD/SCCM/SCOM)/Systems Engineer/Data Engineer/SharePoint Consultant
Responsibilities:

- Provides technical support and system administration for the day-to-day operations of the organisation's ICT systems.
- Demonstrated expertise in managing, supporting, and troubleshooting SharePoint 2013, Dynamics 365 CRM, and SQL Server 2012 on Windows Server 2012, with a strong focus on system and procedure documentation.
- Provided essential day-to-day ICT support for AD, AAD, Microsoft 365, MFA/SSO, and Azure, ensuring smooth operations.
- Implemented and Administered modern authentication technologies and InTune for efficient mobile management
- Provided 3rd level desktop, systems, and network support, diagnosing and resolving complex technical issues.
- Manage Hyper-V VMs in Hybrid Cloud virtual infrastructure by System Center Virtual Machine Manager(SCVMM2012)
- Install and configure VMWare vCloud Directory 5.1 and Lab Manager Server interfacing Lab Manager Clients or REST APIs
- Using ConfigMrg2012 to manage software packages deployment & its change / release management policies & procedures.
- Database backup via Backup Exec 11D+, Backup Exec Intelligent Disaster Recovery (IDR) option, etc.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)
- Setting up a DR site either using NetApp MetroCluster failover solution or EMC VPLEX MetroCluster failover solution

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: Senior Systems Engineer /ICT System Administrator (AD/SCCM/SCOM)/DBA
Responsibilities:

- Conducted the installation, configuration, and troubleshooting of hardware, software, and networking equipment.
- Designing and architecting SCCM2010 for enterprise deployment, patch and update management solution at MLJ Windows Server 2008R2 /SQL Server 2008R2/SharePoint 2010 /Exchange2010/Office365/Lync2010 /SSRS platforms.
- Deploy a VSAN-enabled infrastructure to house numerous applications without concern for fabric resources event conflicts between these virtual environments by dividing a redundant physical SAN infrastructure into separate virtual SAN islands each with its own set of Fibre Channel fabric services.
- Created B-Series (Brocade) switch SAN zone with WWN via #zonecreate "zone1", "WWN1","WWN2"; Create a zoneset configuration by #cfgcreate "test_cfg", "zone1", "zone2"; Add a zone to a zone configuration via cfgadd, etc.
- Installed and configured a WebLogic domain with a frontend Apache Web Server acting as a reverse proxy. The backend integrates Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.
- Large scale Windows Server auto install by PXE script on Active Directory linked Altiris imaging server and auto install Windows desktop WinXP/Vista OS by MDT2008 /BDD2007 Update2 deployment best practice.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.
- Administer, support and troubleshoot SharePoint including various middleware and web server components as well as enterprise file transfer and web analytics tools, such as AD FS, SSO, DNS, DFS, IIS7.0~10, etc.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Systems Engineer /ICT System Administrator (AD/SCCM/SCOM) /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Installed and configured a WebLogic domain with a frontend Apache Web Server acting as a reverse proxy. The backend integrates Oracle Service Bus, Oracle Database 11gR2, and Access Manager for Oracle-deployed applications.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Systems Engineer /ICT Systems Administrator (AD/SCCM/SCOM) /Office Online Specialist /DWH
Responsibilities:

- Team leader, overseeing the design, development, and implementation of software systems, applications, infrastructure and related products on HP G3~G6 servers, HP&Dell Blade Servers, Dell and Citrix Desktops with HP/NetApp/Dell Storages.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.
- Designing and architecting SCCM2007R2 for enterprise deployment, patching for Operating Systems and Applications
- Planning, design and management of Active Directory with high level support to AD Migrations - within & between AD forests
- Proficiently handled hardware, software, and networking equipment installation and configuration. Provided effective support for Office Online Specialist functions, swiftly resolving technical issues while managing Active Directory and assisting in AD migrations between forests.
- Conducted installation, configuration, and testing of hardware and software solutions as well as routine maintenance tasks

- Create new SCCM .MSI application package either by AD linked Altiris imaging server or SCCM2007R2 Package Builder
- Interpret clients' SOE requirements and build SCCM infrastructure up from scratch to meet clients' specific SOE designs.
- 3000/5000/8500 EMC SAN/NetApp NAS servers, Cisco/Juniper switches, routers, Citrix Netscalers & F5 load balancers v9
- Building reports with SCCM 2007R2 using SC Orchestrator for document and maintain configuration & process information.
- Conduct research on emerging products, Services, protocols, standards and document them in iAdmin inventory.
- Setting up a wide range of Security technologies, such as MS FIM2007, Cisco ASA/Firewalls, Checkpoint NGX R65, VPN-1/firewall-1, Juniper NetScreen series, IDP/ISG/SRX/SA/SSG, SSL-VPN, IPSec-VPN, F5 App Firewall Manager etc.
- Working with the Microsoft's Azure cloud-based hosting services to host Office 365, Office Online etc. for the clients.
- Administration of Windows Server2003/2008R2, SQL Server2005/2008, AD, DNS, DHCP, Exchange2007, MOSS2007 etc.
- Administration of the SharePoint Farm and BizTalk2006(R2)/2209 configuration; support for out-of-the-box functionalities.
- Promoting & improving incident & problem management, change / release management policies & procedures by SCOM
- Expertise in configuration of SharePoint out of the box functionality, workflows, and administration via SharePoint Designer

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java/Kotlin, JSP, C#, VB, ASP.Net, TypeScript, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Copilot, Google Bard, GEMINI, MakerSuite; SQL Server Management Studio /BIDS, OBIEE11g~19c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; MuleSoft Anypoint Studio and Mule 4, BizTalk Server 1.0~2020, RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, etc.
- **Database**: SQL Server2005~2022, Azure SQL, Oracle9i/10gR2/11gR2/12c-19c, MySQL, PostgreSQL, Teradata, Exadata, AWS DynamoDB, MongoDB, GCP Cloud SQL, Cloud Firestore, Sybase,
- **Reporting**: SSMS, SSAS, SSIS, SSRS, BIDS, Azure Data Studio, Power BI, Tableau, SAP BusinessObjects, S/4 Hana, IBM Cognos~TM1, HP Trim 7.2, GraphQL/REST API
- **Clouds:** MS Azure/InTune, AWS Cloud, Google GCP, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum,
- **CMS:** SharePoint 2007~2019, Dynamics CRM3.0~D365, AX2009/2012, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, Hyper-V, SalesForce, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, ISO27001, ACSC Essential 8 Maturity Model, NIST, CIS and PCI-DSS frameworks.SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.

DevOps Engineer - ECS/EKS/K8s/GKE – Wipro Parramatta – 20/8/2024 – F.it- No response at all
Zhi Ming Zhang - CCNA, CompTIA Server+, MCSE, MCP+I; QANTAS English / Japanese Interpreter
Senior DevOps Engineer (Hybrid) /Integration Solution Architect /Software Engineer/Senior Data Engineer /Java/Salesforce/Mulesoft Developer /Python AI/ML/DL/BI/.NET/SP/D365 Developer /EDW -Japan (7.5yrs) and Aust. (24yrs)

PROFESSIONAL EXPERIENCE

Dec. 2017 – Present **Aged Care Quality Safety and Commission (ACQSC), Sydney**
Gov. Health: Senior DevOps Engineer/Snr Data Engineer/Python/Go/AI/ML/DL/Mulesoft/SharePoint/D365 Developer
Responsibilities:

- Designed and implemented a secure hybrid cloud architecture integrating AWS VPCs, GCP Cloud VPCs, and Azure VNets. Established encrypted site-to-site connections using Cloud VPN Gateways across all platforms. Leveraged Infrastructure as Code (IaC) principles to automate provisioning and configuration of network resources using Python (Boto3) with CloudFormation, Terraform, or Ansible. Utilized ECS Fargate for efficient and scalable container orchestration.
- Built secure and scalable AWS Landing Zones using Infrastructure as Code (IaC) with AWS CloudFormation templates or Terraform. Leveraged AWS Control Tower for centralized governance of multiple AWS accounts and implemented cross-account observability with Amazon CloudWatch.
- Build and maintain microservices applications deployed using AWS ECS/EKS and GCP's GKE to manage complex containerized environments. Navigate and enhance a larger ecosystem of evolving APIs and microservices using Amazon API Gateway, AWS AppSync, AWS Lambda, GCP Cloud Endpoints, Cloud Functions, or Mulesoft Anypoint Platform.
- Design and lead the migration of complex applications from Rancher/Docker on EC2 to ECS Fargate, optimizing performance and cost efficiency. Used CloudWatch Contributor Insights to analyze High-Cardinality Data & Traffic Trends.
- Built and managed complex ECS clusters and task definitions, ensuring seamless application deployment & management.
- Implemented robust CI/CD pipelines for automated builds, testing, and deployment to ECS Fargate.
- Established comprehensive monitoring and optimization strategies to maintain application health and performance.
- Successfully migrated critical applications with minimal downtime and disruption to business operations.
- **Declarative infrastructure**: Using Terraform configuration files (typically named main.tf or following a similar naming convention) to define the desired state of infrastructure.
- **Version control**: Using Git to store infrastructure configurations in a repository, while utilizing a .gitignore file to exclude sensitive data (e.g., .terraform.lock.hcl, terraform.tfstate) and enabling collaboration and change tracking.
- **Pull-based model**: Using Argo CD, the system actively fetches changes from Git and applies them to the infrastructure based on the defined app-of-apps.yaml configuration file.
- **Continuous reconciliation**: Using Terraform Cloud or other IaC platforms to continuously compare the desired state of infrastructure (defined in configuration files like terraform.tfvars for environment-specific variables) with the actual state and make necessary adjustments.
- Enhanced security posture, performance, and usability of a complex microservices ecosystem by implementing and maintaining CIS Controls through the strategic use of AWS WAF, Shield, and Inspector, along with GCP Cloud Armor, and third-party tools like vulnerability scanners(Nessus, OpenVAS) & performance monitoring platforms(New Relic, Dynatrace).
- Accelerated software delivery and infrastructure provisioning through implementing GitOps workflows and CI/CD pipelines.
- Implemented custom CloudWatch metrics using Boto3 to track critical application performance indicators, enabling data-driven decision-making and proactive issue resolution: cloudwatch = boto3.client('cloudwatch')
- Python or Go programming to design and implement AI-Driven Pipelines for Automate AWS IaC pipeline creation, prediction and optimization using SageMaker CloudFormation and boto3: cfn = boto3.client('cloudformation') …
- **Automation**:Utilized AWS CodeBuild, Git Actions, GitLab CI, or Jenkins for automated build, test, & deployment processes. Used Ansible, Chef, or Puppet for configuration management & infrastructure provisioning in cloud-agnostic environments.
- Built a high-performance Kubernetes cluster on DGX A100s system with pre-installed DGX OS (optimized Ubuntu + NVIDIA tools) for running containerized AI, HPC, and industry-specific workloads across DL training, inference, analytics, etc.
- **Designed and maintained robust CI/CD pipelines using platform-specific tools:**
  - **Azure: Azure DevOps**
  - **AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)**
  - **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- **Deployed multi-cloud databases**: Utilized either AWS DynamoDB or MongoDB for storing unstructured vector data, such as social media posts, videos, images, and audio.
- **Implemented Vector Search**: Facilitated efficient retrieval from vector databases using native retrieval methods with RAG, libraries like Preflexity, pymongo, and large language models (LLMs) such as GPT-4o, Sora, Mistral, and Gemini 1.5 Pro.
- Developed secure, local AI systems (PrivateGPT2.0, Ollama, Autogen Studio 2.0, etc.) for data analysis. The system leverages LangChain to create versatile AI agents and utilizes Pydantic for data validation. These agents assist Business Analysts and Data Analysts in efficient data analysis tasks, eliminating the need for an internet connection.

Dec. 2015 – Dec. 2017 **Reserve Bank of Australia (RBA), Sydney**
Gov Central Bank: Senior DevOps Consultant/Snr Data Engineer/Java/Mulesoft/Python/Go/AI/ML/BI/SP Developer /EDW
Responsibilities:

- **Built and implemented Medallion Architecture with Data Lake in GCP:**
- **Batch Layer:** Utilized Google Cloud Dataproc, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes into the Bronze layer of the Data Lake. This layer stores the initial, raw, & unrefined data.

- **Serving Layer:** Implemented BigQuery with its built-in Spark capabilities for near-real-time processing and analysis on the Silver layer of the Data Lake. This layer houses refined and transformed data suitable for initial analytical workloads.
- **Real-time Layer:** Leveraged Cloud Pub/Sub as the unified service layer for real-time data ingestion and BigQuery for querying the Silver and potentially the Gold layer of the Data Lake for advanced analytics and reporting.
- Utilized BigQuery for data querying, loading, exporting to Pandas DataFrame, and executing parameterized queries within the Python programming environment. query_job = client.query(query, job_config=job_config)
- Designed and developed Interactive Brokers(IB) OMS alike Trade WorkStation(TWS) applications either using Java8/9/ J2EE or IBPy(Python) for prime brokerage and prior trading within the FX, CFD or fixed income, derivatives markets
- Full-stack engineer skilled in React/TypeScript for frontend development, Java/APIs for backend development, and AWS cloud. Proficient in integration tools like MuleSoft and J2EE to ensure seamless data flow using RESTful web services.
- Designed and deployed highly available, scalable microservices architectures using container orchestration platforms and infrastructure as code principles.
- **CloudWatch Observability Expertise:** Implemented comprehensive CloudWatch observability solutions, including metrics, logs, and dashboards. Leveraged CloudWatch Insights for advanced log analysis and troubleshooting.
- Design a cloud-agnostic data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT, CI/CD tools like Airflow or Luigi). This ensures seamless portability between AWS, Azure, GCP, and even on-premises deployments, guaranteeing resilience against outages.
- Expand capabilities across the entire data pipeline lifecycle, encompassing the building, running, testing, and operating of data pipelines that ingest, filter, sort, join, and/or transform datasets on both GCP and SAP Data Services by utilization of Cloud Build for pipeline automation, GitOps for infrastructure-as-code management, and Argo CD for continuous delivery.
- **Automation and CI/CD**: Leveraged industry-standard tools like Helm and Kubespray to streamline & customize Kubernetes deployments across any cloud platform, offering a unified and efficient approach for managing containerized applications.
- Python programming to design and implement AI-Driven Pipelines for Automating GCP IaC pipeline creation, prediction, and optimization using GCP AI Platform/Vertex AI, Terraform, Cloud Build/Functions, and GCP client libraries.
- Designed and deployed scalable containerized applications on Kubernetes, leveraging CI/CD pipelines for automated deployments. Managed and optimized the Machine Learning Training Cluster based on Harrison Platform Nvidia DGX-1's.
- Successfully leveraged ITIL 4.0 frameworks, including Incident Management, Problem Management, and Continual Improvement, to optimize incident response processes and drive service excellence.
- Expertise in implementing and maintaining a robust suite of cloud security controls, including Identity and Access Management (IAM), data encryption, network security, vulnerability management, and data loss prevention (DLP). Ensured compliance with industry standards and regulations such as CIS, NIST, GDPR, and HIPAA.
- Design, develop, and maintain a robust ETL pipeline/API orchestration using either:
    1. Azure Synapse Analytics with its integration tools (Azure Data Factory/Synapse Pipelines) and deployment tool (Azure DevOps) for data processing in Azure.
    2. AWS Glue for data processing and integration with other AWS services to load data into target Aurora databases.
    3. GCP Cloud Dataflow or Cloud Data Fusion for data processing and integration and Google BigQuery for data warehousing and analytics
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- Leverage **Amazon Redshift Serverless** with its next-generation, AI-powered scaling and optimizations. Analyze data at any scale without managing infrastructure. This is achieved through machine learning (ML) models for forecasting and query resource prediction, enabling dynamic scaling for optimal performance.
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = cfn.get_template(StackName='my-iac-pipeline')['TemplateBody']
- Developed and deployed ML models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision: model = models.detection.fasterrcnn_resnet50_fpn(pretrained=True)
- Migrated on-premise applications to the Google Cloud Platform (GCP) cloud using Velostrata, Cloud Migrate for Compute Engine, Migrate for Anthos, and CloudEndure Migration to GKE environment.
- Worked with enterprise CI/CD capabilities and tools such as GoCD, GitLab, Code Build, and Cloud Formation to provide guidance on best practices and help enable major programs to deliver business value.
- Deploy and manage microservices-based applications using Docker CRE and Kubernetes on GCP GKE clusters.
- **Unleash insights from diverse data**: applying unsupervised ML to uncover patterns in unlabeled unstructured datasets, leveraging RStudio for reinforcement learning with supervised KNN, expertly selecting algorithms (KNN, CNN, RNN, ANN, GAN) based on dataset type, size, and desired result interpretability.

Oct. 2013 – Nov.2015 **University of Canberra (UC), ACT/SYD** (Working remotely from Sydney Home Office)
Gov. University: Snr DevOps Engineer/Integration Solution Architect/Java/.NET/Py/Salesforce/Mulesoft/SP/K2 Developer
Responsibilities:

- Integration Solution Architect to design Mulesoft integration framework for integrating the client's modern cloud-based software solutions, with the legacy on-site applications. Build and maintain all non-production and production environments.

- Leverage MuleSoft Anypoint Platform's hybrid deployment capabilities to design APIs that seamlessly integrate between AWS infrastructure (including API Gateway) and Azure IPaaS environments. This "write-once, deploy-anywhere" approach eliminates the need for Terraform or complex infrastructure management, offering greater agility and flexibility.
- Designed and implemented integration solutions using MuleSoft Anypoint Studio 7 and Mule 4 Platform to automate data exchange between disparate systems, improving data processing speed by 25%.
- **Develop Mule Flows**: Design flows in Anypoint Studio to act as intermediaries between systems.
- **Utilize Connectors**: Leverage pre-built connectors from Anypoint Exchange for AWS API Gateway, Azure IPaaS (choose specific service connector), BizTalk, SharePoint, Dynamics 365, and Nintet K2 Five workflows. Additionally, the platform integrates with data visualization tools like Power BI and Tableau for further data analysis.
- K2 BlackPearl (BP) Developer, working to tight timeframe for Affiliates SmartForms Workflow and PaidParking SmartForms Workflow deliverables across divisional IDM Projects for business processes, change management and documentation.
- Accelerated software delivery and infrastructure provisioning through implementing GitOps workflows and CI/CD pipelines.
- Established and maintained a DevOps culture, automating infrastructure and application deployment processes for increased efficiency and reliability.
- Configured CI/CD pipelines to automate code retrieval, testing, & deployment to staging orgs using `dfx force:source:deploy`. (Optional: additional deployment steps like code analysis and MuleSoft application deployments if applicable)
- **Streamlined Salesforce Development and Deployment**: Implemented a CICD pipeline using Salesforce DX tools (including the Salesforce CLI) for efficient code retrieval (e.g., `dfx force:source:pull`) from Salesforce orgs.
- **Automated Deployments with SFDX:** Minimized manual errors and accelerated releases.
- **Enhanced Code Quality**: Integrated unit testing frameworks with the CICD pipeline for continuous code quality checks.
- **MuleSoft Integration**: Built seamless Salesforce-MuleSoft integrations within CI/CD for end-to-end automation.
- Designed and maintained robust CI/CD pipelines using platform-specific tools:
    1. **Azure: Azure DevOps**
    2. **AWS: AWS DevOps (or AWS CodePipeline/CodeBuild/CodeDeploy)**
    3. **GCP: Cloud Build/Cloud Deploy**
- Alternatively, for highly customized CI/CD workflows that extend beyond the capabilities of platform-specific tools like GCP Composer, AWS CodePipeline, or Azure DevOps, consider using either:
- **Python with Pandas and Airflow** to offer flexibility for building intricate CI/CD pipelines tailored to the specific needs. Or
- **Automated Kubernetes Deployments:** Leveraged Kustomize & Kubespray for streamlined and customizable Kubernetes deployments across any cloud platform, offering efficient management of containerized applications.
- Proven ability to design and implement highly scalable, resilient, & cost-effective cloud-native applications on the Azure platform, leveraging technologies such as Azure Service Bus, Azure Event Hubs, Azure Container Service (ACS), and Azure DevOps.
- **Secured API Access for Student Enrollment**: Designed, secured, and managed APIs using MuleSoft API Manager to facilitate secure interaction between the student enrollment system and external partners. This ensured the privacy and security of students' data.
- **CI/CD and DevSecOps Automation:** Developed and maintained CI/CD pipelines (AWS/Azure DevOps) for various applications, employing DevSecOps practices for improved security.
- **Cloud-Agnostic Data Platform:** Designed a cloud-neutral data platform using Python, PySpark, and a metadata-driven ingestion framework (like DBT) for seamless portability across cloud providers and on-premises deployments.
- **Customizable ETL/API Orchestration (Alternative):** Developed custom ETL/API orchestrations using Python with Pandas and Airflow or Java with Spring Batch along with Kafka and Spark for any cloud or even on-premises environment.
- Developed and deployed ML models for image classification, object detection and face recognition using PyTorch and OpenCV(Open Source Computer Vision): model = torch.hub.load('pytorch/vision:v0.6.0', 'mobilenet_v2', pretrained=True)
- Crafted AI-driven pipelines in Python to automate AWS IaC pipeline creation, prediction, and optimization, leveraging SageMaker, CloudFormation, and boto3: template = update_template_based_on_predictions(template, predictions)

Oct. 2012 – Sep.2013 **Employers Mutual Limited (EML), Sydney, Australia**
Healthcare Sector: **DevOps Engineer/Data Engineer /Software Test Engineer/Linux Administrator**
Responsibilities:

- **Hybrid Cloud Architect:** Established secure connections between various cloud environments (AWS VPCs, GCP VPCs, Azure VNets) for data transfer. Utilized Cloud VPN Gateways for encrypted tunnels and infrastructure-as-code tools (Terraform/Ansible) for automated and consistent provisioning across all platforms.
- **Architected and implemented a modern 3-layer lambda architecture on AWS**:
- **Batch layer**: Utilized Amazon EMR, a managed service for running Apache Spark and Hadoop clusters, for batch processing of large data volumes.
- **Speed layer**: Employed Amazon Redshift with its built-in data processing capabilities for near-real-time processing and analysis.
- **Unified service layer**: Leveraged Amazon Kinesis as the unified service layer for real-time data ingestion and Amazon Redshift Spectrum for querying the data lake for advanced analytics and reporting.

- Designed and developed data solutions using AWS services like EC2, RDS, S3, PySpark/EMR, RedShift Spectrum, and Snowflake. Additionally, leveraged expertise in EC2, RDS, S3, DynamoDB, and Kinesis to optimize platform performance within the AWS ecosystem. Mentored and trained junior DevOps engineers, helping them develop their skills and expertise.
- Actively participate in Agile meetings (daily stand-ups, retrospectives, sprint planning), assist in developing scalable software practices, uphold SRE practices (measurement, operational excellence, release engineering), and collaborate with product engineering teams for quality integration and functionality.
- Design, Develop, Review, and Implement CI/CD pipelines for containerized scalable and resilient applications using Docker.
- Developed and maintained automation scripts using Python, Go, Bash to streamline development & operational workflows.
- **Logging and monitoring**: Configure logging and monitoring to track user activity, security events, and system performance.
- Using Splunk for log management, analysis, perform real-time monitoring, and conduct forensic investigations.
- Applied technical expertise in monitoring, observability, performance management, risk management, troubleshooting, and service recoverability. Executed and reported on non-functional testing, contributing to system improvements.
- **Automation Expertise**: automates tests for both front-end and back-end using various tools (Selenium, Postman, RestAssured etc.). Testing within AWS cloud architecture.
- **Security**: Secured secrets with HashiCorp Vault or AWS Secrets Manager and implemented network segmentation and firewalls. Integrated security scanning tools throughout development.
- **Leverage AWS services for cloud-native design patterns**:
- **Highly scalable and fault-tolerant monitoring**: Employed a comprehensive monitoring strategy leveraging services like Amazon CloudWatch, Splunk, ELK Stack (Elasticsearch, Logstash, and Kibana), and Prometheus to monitor and observe resources, track metrics, logs, and events from applications and infrastructure.
- Set up Kubernetes Cluster by creating a Kubernetes cluster either using GKE, EKS or AKS or a local deployment tool like Minikube. Ensure the cluster has sufficient resources (CPU, memory, storage) to accommodate the ML workloads.
- **Optimized CI/CD pipelines with DevOps tools**: Used tools like Jenkins, GitLab CI, Bamboo, alongside cloud platforms (Azure/AWS) to create fast, efficient, and reliable software delivery pipelines.
- Select a suitable MLOps framework or platform that integrates with Kubernetes, such as Kubeflow Pipelines, Mlflow, Seldon Core or Argo Workflows. Followed MLOps best practices for versioning, deployment, testing, and automation.
- **Docker Images**: Proficient in building and running Docker containers in AWS and GCP Hybrid Cloud using Docker commands and managed services like Amazon Elastic Container Registry (ECR) and Google Container Registry (GCR): `docker build`, `docker push`, and `docker run`, including dockerfile in Python.
- **Linux Server Provisioning**: Utilized Ansible-Playbook to automate the creation of EC2 instances, network configuration, package installation, and application deployment. Executed the following command: Bash ansible-playbookec2_instance.yml,configure_router.yml,network_config.yml,install_packages.yml,deploy_app.yml
- Proficient in Linux system administration, including kernel optimization (sysctl, vmstat), performance tuning (iostat, sar), and security hardening (iptables, fail2ban).
- Windows Server Configuration: Employ Chef, initiated with knife bootstrap windows_server_ip chef-server_url, for managing Active Directory, group policies, and software installations.
- Database Provisioning: Leverage Puppet to automate database creation (e.g., mysql_database { 'mydatabase': ensure => present, }) and user management.

Oct. 2011 – Sep. 2012 **Sunshine Coast Council(SCC), QLD Australia**
Government Council: **DevOps Engineer/Data Engineer /Software Test Engineer/SharePoint Consultant /DBA**
Responsibilities:

- Architected and built scalable cloud-native web applications using AWS services and Python.
- Utilize AWS Cloud Development Kit (CDK) for infrastructure provisioning and management
- Develop serverless applications using AWS Lambda. Work with Amazon DynamoDB for data storage and retrieval
- Implement APIs and integrate with various systems using AWS API Gateway
- Deploy and maintain web applications on Amazon Web Services (AWS) cloud platform
- **Agile & Data Expertise:** Experienced working in agile environments and with data engineering practices.
- **Security and Compliance**: **Implement security best practices**: Regularly patch systems and applications.
- **GCP Stack Proficiency:** Possess strong understanding of GCP, Cloud Composer, DBT, and Python scripting languages).
- Managed infrastructure as code (IaC) with Terraform, integrated tools into CI/CD pipelines with Jenkins and GitLab CI/CD, developed REST APIs with Python and Java, monitored AWS cloud resources with EC2 instances and S3 buckets using tools like CloudWatch, and implemented large-scale logging, monitoring, and security measures.
- Installed, configured, and maintained Windows servers, Unix and Linux servers for various business applications
- Utilized Apache Tomcat, NGINX /NGINX Plus and JBoss for application/web servers

Aug.2009 - Oct. 2011 **Manulife Financial Corporation (MLJ), Tokyo**
Finance/Insurance Domain: **DevOps Engineer/Senior Systems Engineer /Unix/Linux Systems Administrator/DWH/DBA**
Responsibilities:

- Spearheaded the improvement of CI/CD pipelines, leading to a significant reduction in software feature delivery time.
- Ensured robust security measures were in place to safeguard the expanding infrastructure.

- Utilized Terraform to implement Infrastructure as Code (IAC), enhancing deployment processes and scalability.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Level 2/3 technical support for all Microsoft infrastructures like AD, DNS, DHCP, Exchange2010, Office 365, SharePoint etc.
- Deployed the VMM Self-Service Portal in System Center 2010 for MLJ employees to minimize IT Helpdesk running cost.

Aug. 2008 to Jul. 2009 **ING Life Insurance Company, LTD, Tokyo**
Banking/Insurance Domain: Project Manager /Senior Systems Engineer /Unix/Linux Systems Administrator /DWH
Responsibilities:

- Project Manager to replace existing insurance web application from IBM AS/400 and **LAMP** to **WISA** based ING Link project to drop down the IT operation cost and link it to Sales Force Automation (SFA) cloud for best marketing result.
- Systems planning, implementing and supporting all core IT systems including Hyper-V, VMware ESXi & vSphere 5.x, HP/EMC/Dell SAN Storage, Windows Server 2008R2, security principles, and general software management best practices.
- Administered and monitored Linux-based systems, ensuring smooth operation of critical applications.
- Provided technical support and troubleshooting assistance to end-users for their Office Online issues.

Jul. 2004 to Apr. 2008 **Microsoft Japan, Tokyo**
Software Vendor: Software Engineer /BI Developer /Data Engineer /Senior Systems Engineer /DWH
Responsibilities:

- Led a team in planning, designing, and developing large-scale, complex enterprise-level packaged software for BI solutions using Microsoft BI tools (SSAS, SSIS, SSRS). Utilized Managed Extensions for C++ (MEX) for SSAS, C#, VB.NET, and T-SQL for SSIS, and C#, VB.NET, and RDL for SSRS programming.
- Design data warehouse architecture with hardware infrastructure landscape, define schemes, Fact and Relational Dimensions tables, including Extended Star Schema (hierarchies, attributes and text), Wormholes, Constellation and Galaxy schemas where Multiple Fact Tables Sharing Dimensions using Multidimensional eXpressions (MDX) and **BIML script**.
- Using SSIS ETL Tools to extract complex data into DW staging area transform and load cleansed data as Master Data either for SSAS to produce Analysis Services Cubes for business reporting by SSRS or Data Mats to serve business apps.
- Managed and maintained company databases in iAdmin, ensuring data accuracy and integrity.
- Built and deployed ML models using k-nearest neighbors algorithm (k-NN), non-parametric supervised learning method and Python for customer churn prediction and fraud detection, achieving 15% improvement in accuracy.
- Developed SQL queries for data extraction, transformation, and reporting purposes.
- Create and deploy reports and ETL packages via SSRS and SSIS in SQL Server Business Intelligence Development Studio platform, including debug, monitor and troubleshoot MS BI Solutions
- Develop data warehouse and business reporting solutions to meet stakeholder requirements via SSIS, SSAS and SSRS
- Defining a Target Warehouse Module, create Target User, Target Module and Target scheme.
- Involving UAT, TEST in a QA Development environment by writing PowerShell scripts to automate processes

EDUCATION / CERTIFICATION

- Bachelor of Automobile Technology, Shanghai University, China [1987]
- Associate Diploma of Business Computing, (C/C++, SQL and COBOL). NSW Business College, Sydney, Australia [1995]

TECHNICAL SKILLS

- **Programming Language :** Java, JavaScript, C#, VB/VBA, Ajax, ASP.net, Action Script, ASP.NET(Web Forms, MVC2~6), XML, T-SQL, PL/SQL, SQL*PlusADO, HTML5/DHTML/XHTML, WSDL, SOAP, CSS, LINQ, Silverlight, JavaScript, MS Access
- **OS**: Windows, Unix/Linux/BSD, Ubuntu 16~21,Solaris8-10, HP-UX8-11, RHEL3~9, SuSe7-10, CenOS 5.0, AIX5.3/6.1, etc
- **AI, ML, DL, BI IDE**: Harrison Platform on Nvidia DGX-1 systems ~ Nvidia DGX A100s systems; OpenAI, ChatGPT-4o, Sora, Multi-GPTs, Auto-Agents, LM-Studio, Azure Open AI ML Services, Phi 3, Copilot, Google GEMINI, Metaverse Llama 3.1, Mistral SQL Server Management Studio /BIDS, OBIEE11g~12c, OWB10gR2~11R2, Business Object, Cognos 8~11, TM1 10.2; Visual Studio 2005~2022, VS Code, Eclipse, NetBeans, Flex Builder, Oracle SQL Developer; RStudio, R 3.5~4.3, Python 3.7~3.11, PSPP, JMP, JASP, EViews, StatCrunch, IBM SPSS, SAS, Stata, MATLAB, SalesForce, MuleSoft Anypoint 7 and Mule 4.0
- **Database**: Oracle9i/10gR2/11gR2/12c, Teradata, Exadata, SQL Server2005~2022, Azure SQL, Power BI, Sybase, MySQL
- **Clouds:** MS Azure, AWS Cloud, Apache Spark, PySpark, Amazon Athena, Redshift/Redshift Spectrum, Google GCP
- **CMS:** SharePoint 2007~2022, CRM3.0~D365, AX2009/2012, HP Trim 7.2, Sitecore 6.x/7.0, Kentico, Umbraco4.x/6 etc.
- **Server**: Wintel Servers, Solaris, Linux Servers, GlassFish, WAS/MQ , Apache/Tomcat, IIS 6/7/8.5/10, LAMP, WISA
- **Virtualization:** Citrix Xen 5.0/5.5, VMWare ESX 3.5~5 / ESXi5.5, IBM PowerVM, TM1, Hyper-V, Sun, Dell VIS.
- **Security:** MS Defender Suite, FortiGate Firewalls, FortiClient, MFA, SSL-VPN, IPSec-VPN, Public Key Infrastructure (PKI), Digital Signing based on RSA Data Security Inc, Public Key Cryptographic Standards (PKCS#1), Hash functions SHA-1/MD5
- **Framework**: .Net 2.0~6.0/Core, Apache Tomcat 6~10.1, JQuery, jqGrid, AvePoint, K2 Black Perl, AvePoint DocAve
  Workflows: K2 Workflow, Nintex 2010 Workflow and SharePoint 2007 ~ 2022 Workflows.